

**INSTITUTE OF ENGINEERING  
PULCHOWK CAMPUS**

**Department of Electrical Engineering**



**A Project Report on  
Industrial Power Distribution and Illumination**

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Sincerely,

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## 1. Introduction:

This report presents the design and development of an electrical power distribution system and Illumination design for Babarmahal Administrative Plaza Project as part of the *Industrial Power Distribution and Illumination* course. The primary objective of the project is to design an efficient and reliable power distribution and illumination system using computer-aided design software, AutoCAD. The project involves several key steps, beginning with the calculation of the required number of light fixtures and power sockets for each flat. These components are then strategically placed within the building layout using AutoCAD. Subsequently, Sub Distribution Boards (SDBs) are designed, and appropriately rated Miniature Circuit Breakers (MCBs) are selected. The design process continued with the development of the Main Distribution Board (MDB) and the determination of suitable cable sizes for connections between the SDBs and MDB. Finally, a transformer with the appropriate rating is selected to ensure a stable and efficient power supply.

## 2. About: Administrative Plaza Babarmahal

The Babarmahal Administrative Plaza project in Nepal is a visionary initiative aimed at accommodating various government offices affiliated under different ministries of Government of Nepal. An 18-story state-of-the-art building, situated on a site spanning 30,021 square meters, is designed to provide high-quality and convenient services to the public and various concerned bodies, thereby optimizing administrative processes and public service delivery.

For this project report, seven floors of AutoCAD schematic, including a basement is considered with each group member responsible for design of two floors.

### 3. Light Sub-Circuit Tables:

| LIGHT SUB-CIRCUIT SPECIFICATION OF BASEMENT |            |    |    |           |              |    |       |                 |                |
|---------------------------------------------|------------|----|----|-----------|--------------|----|-------|-----------------|----------------|
| Light Sub-Circuit                           | Wattage(W) |    |    |           |              |    | Total | Current (I) (A) | MCB Rating (A) |
|                                             | 10         | 18 | 28 | 40(En-Ex) | 40 (parking) | 75 |       |                 |                |
| LC1                                         | 0          | 0  | 6  | 14        | 0            | 0  | 728   | 3.956521739     | 6              |
| LC2                                         | 0          | 0  | 0  | 0         | 19           | 0  | 760   | 4.130434783     | 6              |
| LC3                                         | 0          | 0  | 0  | 0         | 14           | 0  | 560   | 3.043478261     | 6              |
| LC4                                         | 6          | 2  | 0  | 0         | 10           | 0  | 496   | 2.695652174     | 6              |
| LC5                                         | 7          | 0  | 0  | 0         | 16           | 0  | 710   | 3.858695652     | 6              |
| LC6                                         | 5          | 2  | 0  | 0         | 13           | 0  | 606   | 3.293478261     | 6              |
| LC7                                         | 0          | 0  | 0  | 0         | 16           | 0  | 640   | 3.47826087      | 6              |
| LC8                                         | 9          | 0  | 0  | 0         | 14           | 0  | 650   | 3.532608696     | 6              |
| LC9                                         | 0          | 0  | 0  | 0         | 16           | 0  | 640   | 3.47826087      | 6              |
| LC10                                        | 6          | 2  | 0  | 0         | 12           | 0  | 576   | 3.130434783     | 6              |
| LC11                                        | 7          | 0  | 0  | 0         | 16           | 0  | 710   | 3.858695652     | 6              |
| LC12                                        | 0          | 0  | 6  | 0         | 14           | 0  | 728   | 3.956521739     | 6              |
| LC13                                        | 0          | 0  | 0  | 0         | 17           | 0  | 680   | 3.695652174     | 6              |
| LC14                                        | 0          | 0  | 6  | 0         | 13           | 0  | 688   | 3.739130435     | 6              |
| LC15                                        | 0          | 0  | 0  | 0         | 14           | 0  | 560   | 3.043478261     | 6              |
| LC16                                        | 0          | 0  | 0  | 0         | 14           | 0  | 560   | 3.043478261     | 6              |
| LC17                                        | 0          | 0  | 0  | 0         | 14           | 0  | 560   | 3.043478261     | 6              |
| LC18                                        | 0          | 0  | 0  | 0         | 14           | 0  | 560   | 3.043478261     | 6              |
| LC19                                        | 0          | 0  | 0  | 0         | 14           | 0  | 560   | 3.043478261     | 6              |
| LC20                                        | 0          | 0  | 0  | 0         | 14           | 0  | 560   | 3.043478261     | 6              |
| LC21                                        | 0          | 0  | 0  | 0         | 14           | 0  | 560   | 3.043478261     | 6              |

| LIGHT SUB-CIRCUIT SPECIFICATION OF GROUND FLOOR |            |    |    |    |    |       |                 |                |
|-------------------------------------------------|------------|----|----|----|----|-------|-----------------|----------------|
| Light Sub-Circuit                               | Wattage(W) |    |    |    |    | Total | Current (I) (A) | MCB Rating (A) |
|                                                 | 10         | 18 | 20 | 28 | 42 |       |                 |                |
| LC1                                             | 0          | 0  | 0  | 0  | 18 | 756   | 4.109           | 6              |
| LC2                                             | 0          | 0  | 0  | 0  | 14 | 588   | 3.196           | 6              |
| LC3                                             | 0          | 0  | 0  | 0  | 16 | 672   | 3.652           | 6              |
| LC4                                             | 0          | 0  | 0  | 0  | 12 | 504   | 2.739           | 6              |
| LC5                                             | 20         | 0  | 0  | 16 | 0  | 648   | 3.522           | 6              |
| LC6                                             | 0          | 0  | 30 | 0  | 0  | 600   | 3.261           | 6              |
| LC7                                             | 4          | 0  | 36 | 0  | 0  | 760   | 4.130           | 6              |
| LC8                                             | 16         | 1  | 6  | 0  | 0  | 298   | 1.620           | 6              |
| LC9                                             | 0          | 0  | 0  | 0  | 12 | 504   | 2.739           | 6              |
| LC10                                            | 0          | 0  | 0  | 0  | 12 | 504   | 2.739           | 6              |
| LC11                                            | 0          | 0  | 0  | 0  | 16 | 672   | 3.652           | 6              |
| LC12                                            | 0          | 0  | 0  | 0  | 18 | 756   | 4.109           | 6              |
| LC13                                            | 0          | 0  | 0  | 0  | 18 | 756   | 4.109           | 6              |
| LC14                                            | 20         | 0  | 24 | 0  | 0  | 680   | 3.696           | 6              |
| LC15                                            | 20         | 0  | 26 | 0  | 0  | 720   | 3.913           | 6              |
| LC16                                            | 0          | 0  | 0  | 0  | 16 | 672   | 3.652           | 6              |
| LC17                                            | 0          | 0  | 0  | 0  | 18 | 756   | 4.109           | 6              |
| LC18                                            | 0          | 0  | 0  | 0  | 18 | 756   | 4.109           | 6              |
| LC19                                            | 0          | 0  | 0  | 0  | 16 | 672   | 3.652           | 6              |
| LC20                                            | 0          | 0  | 0  | 0  | 16 | 672   | 3.652           | 6              |
| LC21                                            | 20         | 0  | 26 | 0  | 0  | 720   | 3.913           | 6              |
| LC22                                            | 4          | 0  | 36 | 0  | 0  | 760   | 4.130           | 6              |
| LC23                                            | 16         | 1  | 6  | 0  | 0  | 298   | 1.620           | 6              |
| LC24                                            | 0          | 0  | 0  | 0  | 12 | 504   | 2.739           | 6              |
| LC25                                            | 0          | 0  | 0  | 0  | 12 | 504   | 2.739           | 6              |
| LC26                                            | 0          | 0  | 0  | 0  | 16 | 672   | 3.652           | 6              |
| LC27                                            | 0          | 0  | 0  | 0  | 12 | 504   | 2.739           | 6              |
| LC28                                            | 0          | 0  | 0  | 0  | 12 | 504   | 2.739           | 6              |
| LC29                                            | 0          | 0  | 0  | 0  | 16 | 672   | 3.652           | 6              |
| LC30                                            | 0          | 0  | 0  | 0  | 16 | 672   | 3.652           | 6              |
| LC31                                            | 0          | 0  | 0  | 0  | 12 | 504   | 2.739           | 6              |
| LC32                                            | 0          | 0  | 0  | 0  | 12 | 504   | 2.739           | 6              |

| LIGHT SUB-CIRCUIT SPECIFICATION OF FIRST FLOOR |            |    |    |    |    |    |    |       |                 |                |
|------------------------------------------------|------------|----|----|----|----|----|----|-------|-----------------|----------------|
| Light Sub-Circuit                              | Wattage(W) |    |    |    |    |    |    | Total | Current (I) (A) | MCB Rating (A) |
|                                                | 10         | 15 | 18 | 20 | 28 | 34 | 42 |       |                 |                |
| LC1                                            | 1          | 0  | 0  | 0  | 0  | 23 | 0  | 792   | 4.304           | 6              |
| LC2                                            | 1          | 0  | 0  | 0  | 0  | 19 | 0  | 656   | 3.565           | 6              |
| LC3                                            | 0          | 0  | 0  | 0  | 0  | 16 | 0  | 544   | 2.957           | 6              |
| LC4                                            | 1          | 0  | 0  | 0  | 0  | 23 | 0  | 792   | 4.304           | 6              |
| LC5                                            | 4          | 0  | 0  | 0  | 0  | 10 | 0  | 380   | 2.065           | 6              |
| LC6                                            | 16         | 0  | 0  | 19 | 0  | 0  | 0  | 540   | 2.935           | 6              |
| LC7                                            | 1          | 0  | 0  | 0  | 0  | 15 | 0  | 520   | 2.826           | 6              |
| LC8                                            | 0          | 0  | 0  | 0  | 0  | 12 | 0  | 408   | 2.217           | 6              |
| LC9                                            | 4          | 0  | 0  | 34 | 0  | 0  | 0  | 720   | 3.913           | 6              |
| LC10                                           | 16         | 0  | 1  | 7  | 0  | 10 | 0  | 658   | 3.576           | 6              |
| LC11                                           | 1          | 0  | 0  | 0  | 0  | 17 | 0  | 588   | 3.196           | 6              |
| LC12                                           | 0          | 0  | 0  | 0  | 0  | 13 | 0  | 442   | 2.402           | 6              |
| LC13                                           | 0          | 0  | 0  | 0  | 0  | 17 | 0  | 578   | 3.141           | 6              |
| LC14                                           | 0          | 0  | 0  | 0  | 9  | 11 | 0  | 626   | 3.402           | 6              |
| LC15                                           | 20         | 0  | 0  | 24 | 0  | 0  | 0  | 680   | 3.696           | 6              |
| LC16                                           | 20         | 0  | 0  | 26 | 0  | 0  | 0  | 720   | 3.913           | 6              |
| LC17                                           | 0          | 0  | 0  | 0  | 0  | 0  | 16 | 672   | 3.652           | 6              |
| LC18                                           | 0          | 0  | 0  | 0  | 0  | 0  | 18 | 756   | 4.109           | 6              |
| LC19                                           | 0          | 0  | 0  | 0  | 0  | 0  | 18 | 756   | 4.109           | 6              |
| LC20                                           | 0          | 0  | 0  | 0  | 16 | 0  | 0  | 448   | 2.435           | 6              |
| LC21                                           | 0          | 0  | 0  | 0  | 0  | 0  | 16 | 672   | 3.652           | 6              |
| LC22                                           | 20         | 0  | 0  | 21 | 0  | 0  | 4  | 788   | 4.283           | 6              |
| LC23                                           | 4          | 0  | 0  | 36 | 0  | 0  | 0  | 760   | 4.130           | 6              |
| LC24                                           | 0          | 0  | 0  | 0  | 0  | 0  | 16 | 672   | 3.652           | 6              |
| LC25                                           | 0          | 0  | 0  | 0  | 0  | 0  | 12 | 504   | 2.739           | 6              |
| LC26                                           | 0          | 0  | 0  | 0  | 0  | 0  | 12 | 504   | 2.739           | 6              |
| LC27                                           | 16         | 0  | 1  | 0  | 7  | 0  | 0  | 374   | 2.033           | 6              |
| LC28                                           | 1          | 0  | 0  | 0  | 0  | 17 | 0  | 588   | 3.196           | 6              |
| LC29                                           | 0          | 0  | 0  | 0  | 0  | 15 | 0  | 510   | 2.772           | 6              |
| LC30                                           | 2          | 0  | 0  | 0  | 10 | 8  | 0  | 572   | 3.109           | 6              |
| LC31                                           | 11         | 0  | 0  | 0  | 10 | 9  | 0  | 696   | 3.783           | 6              |
| LC32                                           | 0          | 0  | 0  | 0  | 0  | 14 | 0  | 476   | 2.587           | 6              |
| LC33                                           | 0          | 0  | 0  | 0  | 0  | 13 | 0  | 442   | 2.402           | 6              |

| LIGHT SUB-CIRCUIT SPECIFICATION OF FLOOR 2 |            |    |    |    |    |    |       |                    |                   |
|--------------------------------------------|------------|----|----|----|----|----|-------|--------------------|-------------------|
| Light Sub-Circuit                          | Wattage(W) |    |    |    |    |    | Total | Current (I)<br>(A) | MCB Rating<br>(A) |
|                                            | 10         | 12 | 15 | 20 | 34 | 75 |       |                    |                   |
| LC1                                        | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| LC2                                        | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| LC3                                        | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| LC4                                        | 0          | 0  | 0  | 0  | 15 | 0  | 510   | 2.77173913         | 6                 |
| LC5                                        | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| LC6                                        | 0          | 0  | 0  | 0  | 15 | 0  | 510   | 2.77173913         | 6                 |
| LC7                                        | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| LC8                                        | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| LC9                                        | 0          | 0  | 0  | 0  | 12 | 0  | 408   | 2.217391304        | 6                 |
| LC10                                       | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| LC11                                       | 0          | 0  | 0  | 0  | 12 | 0  | 408   | 2.217391304        | 6                 |
| LC12                                       | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| LC13                                       | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| LC14                                       | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| LC15                                       | 0          | 0  | 0  | 0  | 11 | 0  | 374   | 2.032608696        | 6                 |
| LC16                                       | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| LC17                                       | 16         | 4  | 0  | 10 | 0  | 0  | 408   | 2.217391304        | 6                 |
| LC18                                       | 0          | 4  | 0  | 29 | 2  | 0  | 696   | 3.782608696        | 6                 |
| LC19                                       | 16         | 4  | 0  | 9  | 0  | 0  | 388   | 2.108695652        | 6                 |
| LC20                                       | 0          | 4  | 0  | 11 | 2  | 0  | 336   | 1.826086957        | 6                 |
| LC21                                       | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| LC22                                       | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| LC23                                       | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| LC24                                       | 0          | 0  | 0  | 0  | 12 | 0  | 408   | 2.217391304        | 6                 |
| LC25                                       | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| LC26                                       | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| LC27                                       | 0          | 0  | 0  | 0  | 12 | 0  | 408   | 2.217391304        | 6                 |
| LC28                                       | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| LC29                                       | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| LC30                                       | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| LC31                                       | 0          | 0  | 0  | 0  | 11 | 0  | 374   | 2.032608696        | 6                 |
| LC32                                       | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| LC33                                       | 16         | 4  | 0  | 10 | 0  | 0  | 408   | 2.217391304        | 6                 |
| LC34                                       | 16         | 4  | 0  | 9  | 0  | 0  | 388   | 2.108695652        | 6                 |
| LC35                                       | 0          | 4  | 0  | 11 | 2  | 0  | 336   | 1.826086957        | 6                 |
| LC36                                       | 0          | 4  | 0  | 29 | 2  | 0  | 696   | 3.782608696        | 6                 |
| LC37                                       | 0          | 4  | 21 | 20 | 0  | 0  | 763   | 4.14673913         | 6                 |
| LC38                                       | 0          | 4  | 21 | 20 | 0  | 0  | 763   | 4.14673913         | 6                 |
| LC39                                       | 16         | 2  | 0  | 5  | 0  | 0  | 284   | 1.543478261        | 6                 |
| LC40                                       | 16         | 2  | 0  | 4  | 0  | 0  | 264   | 1.434782609        | 6                 |
| LC41                                       | 0          | 0  | 0  | 0  | 0  | 2  | 150   | 0.815217391        | 6                 |
| LC42                                       | 0          | 0  | 0  | 0  | 0  | 2  | 150   | 0.815217391        | 6                 |

| LIGHT SUB-CIRCUIT SPECIFICATION OF FLOOR 3 |            |    |    |    |    |    |       |                    |                   |
|--------------------------------------------|------------|----|----|----|----|----|-------|--------------------|-------------------|
| Light Sub-Circuit                          | Wattage(W) |    |    |    |    |    | Total | Current (I)<br>(A) | MCB Rating<br>(A) |
|                                            | 10         | 12 | 15 | 20 | 34 | 75 |       |                    |                   |
| 1L                                         | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| 2L                                         | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| 3L                                         | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| 4L                                         | 0          | 0  | 0  | 0  | 15 | 0  | 510   | 2.77173913         | 6                 |
| 5L                                         | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| 6L                                         | 0          | 0  | 0  | 0  | 15 | 0  | 510   | 2.77173913         | 6                 |
| 7L                                         | 1          | 0  | 0  | 0  | 23 | 0  | 792   | 4.304347826        | 6                 |
| 8L                                         | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| 9L                                         | 0          | 0  | 0  | 0  | 19 | 0  | 646   | 3.510869565        | 6                 |
| 10L                                        | 1          | 0  | 0  | 0  | 20 | 0  | 690   | 3.75               | 6                 |
| 11L                                        | 0          | 0  | 0  | 0  | 23 | 0  | 782   | 4.25               | 6                 |
| 12L                                        | 1          | 0  | 0  | 0  | 23 | 0  | 792   | 4.304347826        | 6                 |
| 13L                                        | 0          | 0  | 0  | 0  | 19 | 0  | 646   | 3.510869565        | 6                 |
| 14L                                        | 1          | 0  | 0  | 0  | 20 | 0  | 690   | 3.75               | 6                 |
| 15L                                        | 0          | 0  | 21 | 0  | 0  | 0  | 315   | 1.711956522        | 6                 |
| 16L                                        | 0          | 0  | 21 | 0  | 0  | 0  | 315   | 1.711956522        | 6                 |
| 17L                                        | 16         | 4  | 0  | 6  | 1  | 0  | 362   | 1.967391304        | 6                 |
| 18L                                        | 0          | 4  | 0  | 29 | 2  | 0  | 696   | 3.782608696        | 6                 |
| 19L                                        | 0          | 0  | 12 | 0  | 0  | 0  | 180   | 0.97826087         | 6                 |
| 20L                                        | 0          | 4  | 0  | 17 | 2  | 0  | 456   | 2.47826087         | 6                 |
| 21L                                        | 16         | 4  | 0  | 6  | 0  | 0  | 328   | 1.782608696        | 6                 |
| 22L                                        | 16         | 2  | 0  | 5  | 0  | 0  | 284   | 1.543478261        | 6                 |
| 23L                                        | 16         | 2  | 0  | 4  | 0  | 0  | 264   | 1.434782609        | 6                 |
| 24L                                        | 16         | 4  | 0  | 6  | 1  | 0  | 362   | 1.967391304        | 6                 |
| 25L                                        | 0          | 4  | 0  | 29 | 2  | 0  | 696   | 3.782608696        | 6                 |
| 26L                                        | 0          | 4  | 0  | 17 | 2  | 0  | 456   | 2.47826087         | 6                 |
| 27L                                        | 16         | 4  | 0  | 6  | 0  | 0  | 328   | 1.782608696        | 6                 |
| 28L                                        | 0          | 0  | 12 | 0  | 0  | 0  | 180   | 0.97826087         | 6                 |
| 29L                                        | 0          | 0  | 0  | 0  | 0  | 10 | 750   | 4.076086957        | 6                 |
| 30L                                        | 0          | 0  | 0  | 0  | 0  | 10 | 750   | 4.076086957        | 6                 |

| LIGHT SUB-CIRCUIT SPECIFICATION OF FLOOR 4 |            |    |    |    |    |    |       |                    |                   |
|--------------------------------------------|------------|----|----|----|----|----|-------|--------------------|-------------------|
| Light Sub-Circuit                          | Wattage(W) |    |    |    |    |    | Total | Current (I)<br>(A) | MCB Rating<br>(A) |
|                                            | 10         | 12 | 15 | 20 | 34 | 75 |       |                    |                   |
| 1L                                         | 0          | 1  | 0  | 0  | 20 | 0  | 692   | 3.760869565        | 6                 |
| 2L                                         | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| 3L                                         | 5          | 19 | 0  | 22 | 2  | 0  | 786   | 4.27173913         | 6                 |
| 4L                                         | 20         | 20 | 0  | 13 | 0  | 0  | 700   | 3.804347826        | 6                 |
| 5L                                         | 0          | 2  | 0  | 0  | 16 | 0  | 568   | 3.086956522        | 6                 |
| 6L                                         | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| 7L                                         | 0          | 0  | 0  | 6  | 11 | 0  | 494   | 2.684782609        | 6                 |
| 8L                                         | 0          | 0  | 0  | 0  | 0  | 7  | 525   | 2.85326087         | 6                 |
| 9L                                         | 0          | 0  | 0  | 0  | 0  | 7  | 525   | 2.85326087         | 6                 |
| 10L                                        | 0          | 0  | 0  | 0  | 0  | 7  | 525   | 2.85326087         | 6                 |
| 11L                                        | 0          | 0  | 0  | 0  | 0  | 7  | 525   | 2.85326087         | 6                 |
| 12L                                        | 4          | 19 | 0  | 0  | 12 | 0  | 676   | 3.673913043        | 6                 |
| 13L                                        | 5          | 0  | 0  | 0  | 20 | 0  | 730   | 3.967391304        | 6                 |
| 14L                                        | 5          | 0  | 0  | 0  | 20 | 0  | 730   | 3.967391304        | 6                 |
| 15L                                        | 4          | 21 | 0  | 0  | 12 | 0  | 700   | 3.804347826        | 6                 |
| 16L                                        | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| 17L                                        | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| 18L                                        | 0          | 0  | 0  | 0  | 16 | 0  | 544   | 2.956521739        | 6                 |
| 19L                                        | 1          | 0  | 0  | 12 | 2  | 0  | 318   | 1.72826087         | 6                 |
| 20L                                        | 2          | 0  | 0  | 28 | 0  | 0  | 580   | 3.152173913        | 6                 |
| 21L                                        | 0          | 1  | 0  | 0  | 20 | 0  | 692   | 3.760869565        | 6                 |
| 22L                                        | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| 23L                                        | 5          | 19 | 0  | 22 | 2  | 0  | 786   | 4.27173913         | 6                 |
| 24L                                        | 20         | 20 | 0  | 13 | 0  | 0  | 700   | 3.804347826        | 6                 |
| 25L                                        | 0          | 2  | 0  | 0  | 16 | 0  | 568   | 3.086956522        | 6                 |
| 26L                                        | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| 27L                                        | 0          | 0  | 0  | 6  | 11 | 0  | 494   | 2.684782609        | 6                 |
| 28L                                        | 0          | 0  | 0  | 0  | 0  | 7  | 525   | 2.85326087         | 6                 |
| 29L                                        | 0          | 0  | 0  | 0  | 0  | 7  | 525   | 2.85326087         | 6                 |
| 30L                                        | 0          | 0  | 0  | 0  | 0  | 7  | 525   | 2.85326087         | 6                 |
| 31L                                        | 0          | 0  | 0  | 0  | 0  | 7  | 525   | 2.85326087         | 6                 |

| LIGHT SUB-CIRCUIT SPECIFICATION OF FIFTH FLOOR 5 |            |    |    |    |    |    |       |                    |                   |
|--------------------------------------------------|------------|----|----|----|----|----|-------|--------------------|-------------------|
| Light Sub-Circuit                                | Wattage(W) |    |    |    |    |    | Total | Current (I)<br>(A) | MCB Rating<br>(A) |
|                                                  | 10         | 12 | 15 | 20 | 34 | 75 |       |                    |                   |
| LC1                                              | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| LC2                                              | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| LC3                                              | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| LC4                                              | 0          | 0  | 0  | 0  | 15 | 0  | 510   | 2.77173913         | 6                 |
| LC5                                              | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| LC6                                              | 0          | 0  | 0  | 0  | 15 | 0  | 510   | 2.77173913         | 6                 |
| LC7                                              | 0          | 4  | 21 | 12 | 0  | 0  | 603   | 3.277173913        | 6                 |
| LC8                                              | 16         | 6  | 0  | 21 | 0  | 0  | 652   | 3.543478261        | 6                 |
| LC9                                              | 0          | 4  | 12 | 18 | 2  | 0  | 656   | 3.565217391        | 6                 |
| LC10                                             | 16         | 4  | 0  | 6  | 12 | 0  | 736   | 4                  | 6                 |
| LC11                                             | 0          | 1  | 0  | 0  | 20 | 0  | 692   | 3.760869565        | 6                 |
| LC12                                             | 0          | 0  | 0  | 0  | 14 | 1  | 551   | 2.994565217        | 6                 |
| LC13                                             | 0          | 4  | 0  | 0  | 10 | 0  | 388   | 2.108695652        | 6                 |
| LC14                                             | 16         | 8  | 0  | 23 | 2  | 0  | 784   | 4.260869565        | 6                 |
| LC15                                             | 0          | 0  | 0  | 0  | 17 | 0  | 578   | 3.141304348        | 6                 |
| LC16                                             | 0          | 1  | 0  | 0  | 21 | 0  | 726   | 3.945652174        | 6                 |
| LC17                                             | 0          | 1  | 0  | 0  | 21 | 0  | 726   | 3.945652174        | 6                 |
| LC18                                             | 0          | 0  | 0  | 0  | 17 | 0  | 578   | 3.141304348        | 6                 |
| LC19                                             | 16         | 4  | 0  | 6  | 12 | 0  | 736   | 4                  | 6                 |
| LC20                                             | 0          | 1  | 0  | 0  | 20 | 0  | 692   | 3.760869565        | 6                 |
| LC21                                             | 0          | 0  | 0  | 0  | 14 | 1  | 551   | 2.994565217        | 6                 |
| LC22                                             | 0          | 4  | 0  | 0  | 10 | 0  | 388   | 2.108695652        | 6                 |
| LC23                                             | 0          | 0  | 21 | 16 | 0  | 0  | 635   | 3.451086957        | 6                 |
| LC24                                             | 16         | 4  | 0  | 10 | 2  | 0  | 476   | 2.586956522        | 6                 |
| LC25                                             | 0          | 4  | 12 | 14 | 2  | 0  | 576   | 3.130434783        | 6                 |

| LIGHT SUB-CIRCUIT SPECIFICATION OF FLOOR 6 |            |    |    |    |    |    |       |                    |                   |
|--------------------------------------------|------------|----|----|----|----|----|-------|--------------------|-------------------|
| Light Sub-Circuit                          | Wattage(W) |    |    |    |    |    | Total | Current (I)<br>(A) | MCB Rating<br>(A) |
|                                            | 10         | 12 | 15 | 20 | 34 | 75 |       |                    |                   |
| LC1                                        | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| LC2                                        | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| LC3                                        | 0          | 0  | 0  | 0  | 22 | 0  | 748   | 4.065217391        | 6                 |
| LC4                                        | 0          | 0  | 0  | 0  | 15 | 0  | 510   | 2.77173913         | 6                 |
| LC5                                        | 0          | 0  | 0  | 0  | 18 | 0  | 612   | 3.326086957        | 6                 |
| LC6                                        | 0          | 0  | 0  | 0  | 15 | 0  | 510   | 2.77173913         | 6                 |
| LC7                                        | 0          | 4  | 0  | 12 | 2  | 0  | 356   | 1.934782609        | 6                 |
| LC8                                        | 0          | 0  | 0  | 0  | 15 | 0  | 510   | 2.77173913         | 6                 |
| LC9                                        | 7          | 1  | 0  | 0  | 21 | 0  | 796   | 4.326086957        | 6                 |
| LC10                                       | 6          | 10 | 0  | 15 | 2  | 0  | 548   | 2.97826087         | 6                 |
| LC11                                       | 16         | 0  | 0  | 0  | 0  | 6  | 610   | 3.315217391        | 6                 |
| LC12                                       | 0          | 4  | 0  | 0  | 15 | 0  | 558   | 3.032608696        | 6                 |
| LC13                                       | 0          | 4  | 0  | 12 | 2  | 0  | 356   | 1.934782609        | 6                 |
| LC14                                       | 6          | 10 | 0  | 15 | 2  | 3  | 773   | 4.201086957        | 6                 |
| LC15                                       | 0          | 10 | 0  | 16 | 0  | 0  | 440   | 2.391304348        | 6                 |
| LC16                                       | 0          | 6  | 0  | 12 | 0  | 0  | 312   | 1.695652174        | 6                 |
| LC17                                       | 16         | 0  | 0  | 0  | 0  | 6  | 610   | 3.315217391        | 6                 |
| LC18                                       | 7          | 1  | 0  | 0  | 21 | 0  | 796   | 4.326086957        | 6                 |

## 4. Illumination Specification:

### a. Basement:

| BASEMENT ILLUMINATION SPECIFICATIONS |                                      |                        |                                                 |                               |                |                |                         |                         |                         |                                    |             |        |
|--------------------------------------|--------------------------------------|------------------------|-------------------------------------------------|-------------------------------|----------------|----------------|-------------------------|-------------------------|-------------------------|------------------------------------|-------------|--------|
| Serial No.                           | Description                          | Dimensions             | Required Illumination (E <sub>req</sub> ) (lux) | Lumen <sub>s</sub> Output (φ) | μ <sub>m</sub> | μ <sub>u</sub> | N <sub>calculated</sub> | N <sub>considered</sub> | E <sub>considered</sub> | Catalogue: Havells Office Lighting |             |        |
|                                      | Designation                          | Area (m <sup>2</sup> ) |                                                 |                               |                |                |                         |                         |                         | Model                              | Wattage (W) | Pg.no. |
| 1                                    | Guard Room 1 (right bottom)          | 34.04                  | 300                                             | 2700                          | 0.75           | 0.85           | 5.93                    | 6                       | 303.39                  | VENUSNEOHE4XIPLR28-36WLED830MOD    | 28          | 41     |
| 2                                    | Guard Room 2 (right top)             | 34.63                  | 300                                             | 2700                          | 0.75           | 0.85           | 6.04                    | 6                       | 298.22                  | VENUSNEOHE4XIPLR28-36WLED830MOD    | 28          | 41     |
| 3                                    | Guard Room 3 (left)                  | 33.88                  | 300                                             | 2700                          | 0.75           | 0.85           | 5.91                    | 6                       | 304.83                  | VENUSNEOHE4XIPLR28-36WLED830MOD    | 28          | 41     |
| 4                                    | Staircase 1 U3 (left area - bottom)  | 16.00                  | 150                                             | 900                           | 0.75           | 0.85           | 4.18                    | 4                       | 143.44                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 5                                    | Staircase 2 U3 (left area - top)     | 16.00                  | 150                                             | 900                           | 0.75           | 0.85           | 4.18                    | 4                       | 143.44                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 6                                    | Staircase 3 U3 (middle area - left)  | 15.27                  | 150                                             | 900                           | 0.75           | 0.85           | 3.99                    | 4                       | 150.29                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 7                                    | Staircase 4 U3 (middle area - right) | 15.27                  | 150                                             | 900                           | 0.75           | 0.85           | 3.99                    | 4                       | 150.29                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 8                                    | Staircase 5 U3 (right area - top)    | 15.98                  | 150                                             | 900                           | 0.75           | 0.85           | 4.18                    | 4                       | 143.62                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 9                                    | Staircase 6 U3 (right area - bottom) | 15.98                  | 150                                             | 900                           | 0.75           | 0.85           | 4.18                    | 4                       | 143.62                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 10                                   | Basement parking bottom              | 6479.00                | 70                                              | 4000                          | 0.75           | 0.85           | 177.85                  | 176                     | 69.27                   | CONSTA1144LP40WLED830SPMMAWH       | 40          | 27     |
| 11                                   | Basement parking top                 | 4097.17                | 70                                              | 4000                          | 0.75           | 0.85           | 112.47                  | 111                     | 69.08                   | CONSTA1144LP40WLED830SPMMAWH       | 40          | 27     |
| 12                                   | Store 1 (left area)                  | 14.90                  | 150                                             | 1800                          | 0.75           | 0.85           | 1.95                    | 2                       | 154.03                  | VENUSNEO1-10V2X2PLR34WLED840MOD    | 18          | 37     |
| 13                                   | Store 2 (middle area)                | 14.21                  | 150                                             | 1800                          | 0.75           | 0.85           | 1.86                    | 2                       | 161.51                  | VENUSNEO1-10V2X2PLR34WLED840MOD    | 18          | 37     |
| 14                                   | Store 3 (right area)                 | 14.90                  | 150                                             | 1800                          | 0.75           | 0.85           | 1.95                    | 2                       | 154.03                  | VENUSNEO1-10V2X2PLR34WLED840MOD    | 18          | 37     |
| 15                                   | Lobby 1 (left area - bottom)         | 11.16                  | 125                                             | 900                           | 0.75           | 0.85           | 2.43                    | 2                       | 112.84                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 16                                   | Lobby 2 (left area - top)            | 16.25                  | 125                                             | 900                           | 0.75           | 0.85           | 3.54                    | 3                       | 115.92                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 17                                   | Lobby 3 (middle area - freight)      | 5.47                   | 125                                             | 900                           | 0.75           | 0.85           | 1.19                    | 1                       | 114.89                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 18                                   | Lobby 4 (middle area - lift)         | 27.58                  | 125                                             | 900                           | 0.75           | 0.85           | 6.01                    | 5                       | 114.02                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 19                                   | Lobby 5 (right area - bottom)        | 11.39                  | 125                                             | 900                           | 0.75           | 0.85           | 2.48                    | 2                       | 110.75                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 20                                   | Lobby 6 (right area - top)           | 15.33                  | 125                                             | 900                           | 0.75           | 0.85           | 2.34                    | 3                       | 128.28                  | INTEGRAGRANDDLR10WLED830S          | 10          | 73     |
| 21                                   | Entry - Exit                         | 409.58                 | 90                                              | 4000                          | 0.75           | 0.85           | 14.46                   | 14                      | 87.16                   | STOUTPLUS1200BS40WLED857SPCWH      | 40          | 71     |

b. Ground Floor:

| S. N | Description                            | Area (m <sup>2</sup> ) | Required illumination level (E) (lux) | Required lumens output (Φ) | μm   | μu   | Ncalc | N considered | E considered | Havells Office Lighting Catalogue               |      |         |
|------|----------------------------------------|------------------------|---------------------------------------|----------------------------|------|------|-------|--------------|--------------|-------------------------------------------------|------|---------|
|      |                                        |                        |                                       |                            |      |      |       |              |              | Model                                           | Watt | Pg. No. |
| 1    | Gym/Health Club                        | 293.50                 | 250                                   | 3600                       | 0.75 | 0.85 | 31.97 | 32           | 250.22       | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42   | 37 (1)  |
| 2    | Yoga/Meditation Hall                   | 358                    | 250                                   | 3600                       | 0.75 | 0.85 | 39.00 | 40           | 256.42       | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42   | 37 (1)  |
| 3    | Clinic 1 (bottom)                      | 14.48                  | 300                                   | 4000                       | 0.75 | 0.85 | 1.70  | 2            | 352.21       | VENUSNEOTW2X2PLR42WLEDTWMOD                     | 42   | 39      |
| 4    | Clinic 2                               | 16.38                  | 300                                   | 4000                       | 0.75 | 0.85 | 1.93  | 2            | 311.36       | VENUSNEOTW2X2PLR42WLEDTWMOD                     | 42   | 39      |
| 5    | Clinic 3                               | 16.38                  | 300                                   | 4000                       | 0.75 | 0.85 | 1.93  | 2            | 311.36       | VENUSNEOTW2X2PLR42WLEDTWMOD                     | 42   | 39      |
| 6    | Clinic 4                               | 16.38                  | 300                                   | 4000                       | 0.75 | 0.85 | 1.93  | 2            | 311.36       | VENUSNEOTW2X2PLR42WLEDTWMOD                     | 42   | 39      |
| 7    | Clinic Desk Area                       | 16.15                  | 300                                   | 4000                       | 0.75 | 0.85 | 1.90  | 2            | 315.79       | VENUSNEOTW2X2PLR42WLEDTWMOD                     | 42   | 39      |
| 8    | Clinic 5                               | 16.36                  | 300                                   | 4000                       | 0.75 | 0.85 | 1.92  | 2            | 311.74       | VENUSNEOTW2X2PLR42WLEDTWMOD                     | 42   | 39      |
| 9    | Clinic 6                               | 16.39                  | 300                                   | 4000                       | 0.75 | 0.85 | 1.93  | 2            | 311.17       | VENUSNEOTW2X2PLR42WLEDTWMOD                     | 42   | 39      |
| 10   | Clinic 7                               | 18.11                  | 300                                   | 4000                       | 0.75 | 0.85 | 2.13  | 2            | 281.61       | VENUSNEOTW2X2PLR42WLEDTWMOD                     | 42   | 39      |
| 11   | Lab                                    | 108.25                 | 275                                   | 4000                       | 0.75 | 0.85 | 11.67 | 12           | 282.68       | VENUSNEOTW2X2PLR42WLEDTWMOD                     | 42   | 39      |
| 12   | Waiting area (Clinic & Lab)            | 140.51                 | 200                                   | 2550                       | 0.75 | 0.85 | 17.29 | 16           | 185.11       | VENUSNEO1-10V2X2PLR34WLED840MOD+1LEDDRIVER36V28 | 28   | 37 (1)  |
| 13   | Mart                                   | 315.57                 | 250                                   | 3600                       | 0.75 | 0.85 | 34.38 | 36           | 261.81       | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42   | 37 (1)  |
| 14   | Bank                                   | 445                    | 275                                   | 3780                       | 0.75 | 0.85 | 50.78 | 52           | 281.59       | VENUSNEO1-10V2X2PLR34WLED857MOD                 | 42   | 37 (2)  |
| 15   | Information Office                     | 145.875                | 250                                   | 3600                       | 0.75 | 0.85 | 15.89 | 16           | 251.72       | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42   | 37 (1)  |
| 16   | Logistics Support                      | 71.75                  | 250                                   | 3600                       | 0.75 | 0.85 | 7.82  | 8            | 255.89       | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42   | 37 (1)  |
| 17   | Courier Office                         | 74.25                  | 250                                   | 3600                       | 0.75 | 0.85 | 8.09  | 8            | 247.27       | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42   | 37 (1)  |
| 18   | Library                                | 348.25                 | 250                                   | 3600                       | 0.75 | 0.85 | 37.94 | 40           | 263.60       | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42   | 37 (1)  |
| 19   | Book Shop                              | 160.25                 | 225                                   | 3600                       | 0.75 | 0.85 | 15.71 | 16           | 229.14       | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42   | 37 (1)  |
| 20   | Reproduction Office                    | 212.45                 | 250                                   | 3600                       | 0.75 | 0.85 | 23.14 | 24           | 259.26       | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42   | 37 (1)  |
| 21   | Provident Fund Office - Life Insurance | 358.58                 | 250                                   | 3600                       | 0.75 | 0.85 | 39.06 | 40           | 256.01       | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42   | 37 (1)  |
| 22   | Staircase 1 L2 (left area - bottom)    | 7.84                   | 150                                   | 900                        | 0.75 | 0.85 | 2.05  | 2            | 146.36       | INTEGRAGRANDDL10WLED830S                        | 10   | 73 (1)  |
| 23   | Staircase 2 L3 (left area - bottom)    | 15.98                  | 150                                   | 900                        | 0.75 | 0.85 | 4.18  | 4            | 143.62       | INTEGRAGRANDDL10WLED830S                        | 10   | 73 (1)  |

|    |                                       |        |     |      |      |      |       |    |        |                                 |    |        |
|----|---------------------------------------|--------|-----|------|------|------|-------|----|--------|---------------------------------|----|--------|
| 24 | Staircase 3 L3 (left area - top)      | 15.98  | 150 | 900  | 0.75 | 0.85 | 4.18  | 4  | 143.62 | INTEGRAGRANDDL R10WLED830S      | 10 | 73 (I) |
| 25 | Staircase 4 L2 (left area - top)      | 7.84   | 150 | 900  | 0.75 | 0.85 | 2.05  | 2  | 146.36 | INTEGRAGRANDDL R10WLED830S      | 10 | 73 (I) |
| 26 | Staircase 6 L2 (left terrace - top)   | 11.80  | 100 | 900  | 0.75 | 0.85 | 2.06  | 2  | 97.25  | INTEGRAGRANDDL R10WLED830S      | 10 | 73 (I) |
| 27 | Staircase 8 L2 (right terrace - top)  | 11.80  | 100 | 900  | 0.75 | 0.85 | 2.06  | 2  | 97.25  | INTEGRAGRANDDL R10WLED830S      | 10 | 73 (I) |
| 28 | Staircase 9 L3 (lobby area - left)    | 15.27  | 150 | 900  | 0.75 | 0.85 | 3.99  | 4  | 150.29 | INTEGRAGRANDDL R10WLED830S      | 10 | 73 (I) |
| 29 | Staircase 10 L3 (lobby area - right)  | 15.27  | 150 | 900  | 0.75 | 0.85 | 3.99  | 4  | 150.29 | INTEGRAGRANDDL R10WLED830S      | 10 | 73 (I) |
| 30 | Staircase 11 L2 (right area - bottom) | 7.84   | 150 | 900  | 0.75 | 0.85 | 2.05  | 2  | 146.36 | INTEGRAGRANDDL R10WLED830S      | 10 | 73 (I) |
| 31 | Staircase 12 L3 (right area - bottom) | 15.98  | 150 | 900  | 0.75 | 0.85 | 4.18  | 4  | 143.62 | INTEGRAGRANDDL R10WLED830S      | 10 | 73 (I) |
| 32 | Staircase 13 L3 (right area - top)    | 15.98  | 150 | 900  | 0.75 | 0.85 | 4.18  | 4  | 143.62 | INTEGRAGRANDDL R10WLED830S      | 10 | 73 (I) |
| 33 | Staircase 14 L2 (right area - top)    | 7.84   | 150 | 900  | 0.75 | 0.85 | 2.05  | 2  | 146.36 | INTEGRAGRANDDL R10WLED830S      | 10 | 73 (I) |
| 34 | Store 1 (left area)                   | 6.98   | 150 | 1800 | 0.75 | 0.85 | 0.91  | 1  | 164.40 | VENUSNEO1-10V2X2PLR34WLED840MOD | 18 | 37 (I) |
| 36 | Store 2 (right area)                  | 6.98   | 150 | 1800 | 0.75 | 0.85 | 0.91  | 1  | 164.40 | VENUSNEO1-10V2X2PLR34WLED840MOD | 18 | 37 (I) |
| 37 | Washroom 1 (left area - top)          | 49.8   | 150 | 810  | 0.75 | 0.85 | 14.47 | 14 | 145.17 | INTEGRANEODLR10WLED830S         | 10 | 71 (I) |
| 38 | Washroom 2 (left area - bottom)       | 49.8   | 150 | 810  | 0.75 | 0.85 | 14.47 | 14 | 145.17 | INTEGRANEODLR10WLED830S         | 10 | 71 (I) |
| 39 | Washroom 3 (middle area - left)       | 49.8   | 150 | 810  | 0.75 | 0.85 | 14.47 | 14 | 145.17 | INTEGRANEODLR10WLED830S         | 10 | 71 (I) |
| 40 | Washroom 4 (middle area - right)      | 49.8   | 150 | 810  | 0.75 | 0.85 | 14.47 | 14 | 145.17 | INTEGRANEODLR10WLED830S         | 10 | 71 (I) |
| 41 | Washroom 5 (right area - top)         | 49.8   | 150 | 810  | 0.75 | 0.85 | 14.47 | 14 | 145.17 | INTEGRANEODLR10WLED830S         | 10 | 71 (I) |
| 42 | Washroom 6 (right area - bottom)      | 49.8   | 150 | 810  | 0.75 | 0.85 | 14.47 | 14 | 145.17 | INTEGRANEODLR10WLED830S         | 10 | 71 (I) |
| 43 | Corridor 1 (whole left area)          | 575.36 | 150 | 1920 | 0.75 | 0.85 | 70.51 | 72 | 153.17 | PLUTO2X1PIR20WLED840S           | 20 | 53 (I) |
| 44 | Corridor 2 (whole middle area)        | 374.56 | 150 | 1920 | 0.75 | 0.85 | 45.90 | 50 | 163.39 | PLUTO2X1PIR20WLED840S           | 20 | 53 (I) |
| 45 | Corridor 3 (whole right area)         | 545.26 | 150 | 1920 | 0.75 | 0.85 | 66.82 | 68 | 152.65 | PLUTO2X1PIR20WLED840S           | 20 | 53 (I) |

c. First Floor:

| S. N | Description                                       | Area (m <sup>2</sup> ) | Required illumination level (E) (lux) | Required lumens output (Φ) | μm   | μu   | Ncalc | N considered | E considered | Model                                            | Watt | Pg. No. |
|------|---------------------------------------------------|------------------------|---------------------------------------|----------------------------|------|------|-------|--------------|--------------|--------------------------------------------------|------|---------|
| 1    | Office Area 1 (left area - left)                  | 732.99                 | 250                                   | 3000                       | 0.75 | 0.85 | 95.82 | 101          | 263.53       | VENUSNEO1-10V2X2PLR34WLED840MOD                  | 34   | 37 (1)  |
| 2    | Office Area 2 (left area - right bottom)          | 355.82                 | 250                                   | 3000                       | 0.75 | 0.85 | 46.51 | 47           | 252.62       | VENUSNEO1-10V2X2PLR34WLED840MOD                  | 34   | 37 (1)  |
| 3    | Office Area 3 (left area - right top)             | 206.51                 | 250                                   | 3000                       | 0.75 | 0.85 | 26.99 | 27           | 250.05       | VENUSNEO1-10V2X2PLR34WLED840MOD                  | 34   | 37 (1)  |
| 4    | Office Area 4 (middle)                            | 77.97                  | 250                                   | 3000                       | 0.75 | 0.85 | 10.19 | 11           | 269.82       | VENUSNEO1-10V2X2PLR34WLED840MOD                  | 34   | 37 (1)  |
| 5    | Office Area 5 (right area - bottom)               | 280.00                 | 250                                   | 3000                       | 0.75 | 0.85 | 36.60 | 40           | 273.21       | VENUSNEO1-10V2X2PLR34WLED840MOD                  | 34   | 37 (1)  |
| 6    | Office Area 6 (right area - top)                  | 275.01                 | 250                                   | 3000                       | 0.75 | 0.85 | 35.95 | 38           | 264.26       | VENUSNEO1-10V2X2PLR34WLED840MOD                  | 34   | 37 (1)  |
| 7    | Waiting Lobby 1 (right area - bottom)             | 72.85                  | 200                                   | 2550                       | 0.75 | 0.85 | 8.96  | 10           | 223.15       | VENUSNEO1-10V2X2PLR34WLED840MOD+H-LEDDRIVER36V28 | 28   | 37 (1)  |
| 8    | Waiting Lobby 2 (right area - top)                | 72.85                  | 200                                   | 2550                       | 0.75 | 0.85 | 8.96  | 10           | 223.15       | VENUSNEO1-10V2X2PLR34WLED840MOD+H-LEDDRIVER36V28 | 28   | 37 (1)  |
| 9    | Office Washroom 1 (left area -left bottom)        | 3.39                   | 150                                   | 810                        | 0.75 | 0.85 | 0.98  | 1            | 152.32       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 10   | Office Washroom 2 (left area - left middle)       | 2.53                   | 150                                   | 810                        | 0.75 | 0.85 | 0.73  | 1            | 204.10       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 11   | Office Washroom 3 (left area - left top)          | 3.39                   | 150                                   | 810                        | 0.75 | 0.85 | 0.98  | 1            | 152.32       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 12   | Office Washroom 4 (left area - right bottom)      | 3.39                   | 150                                   | 810                        | 0.75 | 0.85 | 0.98  | 1            | 152.32       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 13   | Office Washroom 5 (left area - right top)         | 3.39                   | 150                                   | 810                        | 0.75 | 0.85 | 0.98  | 1            | 152.32       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 14   | Office Washroom 6 (right area - bottom chamber 1) | 2.62                   | 150                                   | 810                        | 0.75 | 0.85 | 0.76  | 1            | 197.09       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 15   | Office Washroom 7 (right area - bottom chamber 2) | 2.14                   | 150                                   | 810                        | 0.75 | 0.85 | 0.62  | 1            | 241.30       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 16   | Office Washroom 8 (right area - bottom chamber 3) | 2.13                   | 150                                   | 810                        | 0.75 | 0.85 | 0.62  | 1            | 242.43       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 17   | Office Washroom 9 (right area - top chamber 1)    | 2.62                   | 150                                   | 810                        | 0.75 | 0.85 | 0.76  | 1            | 197.09       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 18   | Office Washroom 10 (right area - top chamber 2)   | 2.14                   | 150                                   | 810                        | 0.75 | 0.85 | 0.62  | 1            | 241.30       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 19   | Office Washroom 11 (right area - top chamber 3)   | 2.13                   | 150                                   | 810                        | 0.75 | 0.85 | 0.62  | 1            | 242.43       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 20   | Washroom 1 (left area - top)                      | 49.8                   | 150                                   | 810                        | 0.75 | 0.85 | 14.47 | 14           | 145.17       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 21   | Washroom 2 (left area - bottom)                   | 49.8                   | 150                                   | 810                        | 0.75 | 0.85 | 14.47 | 14           | 145.17       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 22   | Washroom 3 (middle area - left)                   | 49.8                   | 150                                   | 810                        | 0.75 | 0.85 | 14.47 | 14           | 145.17       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 23   | Washroom 4 (middle area - right)                  | 49.8                   | 150                                   | 810                        | 0.75 | 0.85 | 14.47 | 14           | 145.17       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 24   | Washroom 5 (right area - top)                     | 49.8                   | 150                                   | 810                        | 0.75 | 0.85 | 14.47 | 14           | 145.17       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 25   | Washroom 6 (right area - bottom)                  | 49.8                   | 150                                   | 810                        | 0.75 | 0.85 | 14.47 | 14           | 145.17       | INTEGRANEODLRI0WLED830S                          | 10   | 71 (1)  |
| 26   | Staircase 1 L2 (left area - bottom)               | 7.84                   | 150                                   | 900                        | 0.75 | 0.85 | 2.05  | 2            | 146.36       | INTEGRAGRANDDLRI0WLED830S                        | 10   | 73 (1)  |

|    |                                       |        |     |      |      |      |       |    |        |                                                 |    |           |
|----|---------------------------------------|--------|-----|------|------|------|-------|----|--------|-------------------------------------------------|----|-----------|
| 27 | Staircase 2 L3 (left area - bottom)   | 15.98  | 150 | 900  | 0.75 | 0.85 | 4.18  | 4  | 143.62 | INTEGRAGRANDDLRL10WLED830S                      | 10 | 73<br>(1) |
| 28 | Staircase 3 L3 (left area - top)      | 15.98  | 150 | 900  | 0.75 | 0.85 | 4.18  | 4  | 143.62 | INTEGRAGRANDDLRL10WLED830S                      | 10 | 73<br>(1) |
| 29 | Staircase 4 L2 (left area - top)      | 7.84   | 150 | 900  | 0.75 | 0.85 | 2.05  | 2  | 146.36 | INTEGRAGRANDDLRL10WLED830S                      | 10 | 73<br>(1) |
| 30 | Staircase 5 L2 (middle area - left)   | 11.80  | 100 | 900  | 0.75 | 0.85 | 2.06  | 2  | 97.25  | INTEGRAGRANDDLRL10WLED830S                      | 10 | 73<br>(1) |
| 31 | Staircase 6 L2 (middle area - right)  | 11.80  | 100 | 900  | 0.75 | 0.85 | 2.06  | 2  | 97.25  | INTEGRAGRANDDLRL10WLED830S                      | 10 | 73<br>(1) |
| 32 | Staircase 7 L3 (lobby area - left)    | 15.27  | 150 | 900  | 0.75 | 0.85 | 3.99  | 4  | 150.29 | INTEGRAGRANDDLRL10WLED830S                      | 10 | 73<br>(1) |
| 33 | Staircase 8 L3 (lobby area - right)   | 15.27  | 150 | 900  | 0.75 | 0.85 | 3.99  | 4  | 150.29 | INTEGRAGRANDDLRL10WLED830S                      | 10 | 73<br>(1) |
| 34 | Staircase 9 L2 (right area - bottom)  | 7.84   | 150 | 900  | 0.75 | 0.85 | 2.05  | 2  | 146.36 | INTEGRAGRANDDLRL10WLED830S                      | 10 | 73<br>(1) |
| 35 | Staircase 10 L3 (right area - bottom) | 15.98  | 150 | 900  | 0.75 | 0.85 | 4.18  | 4  | 143.62 | INTEGRAGRANDDLRL10WLED830S                      | 10 | 73<br>(1) |
| 36 | Staircase 11 L3 (right area - top)    | 15.98  | 150 | 900  | 0.75 | 0.85 | 4.18  | 4  | 143.62 | INTEGRAGRANDDLRL10WLED830S                      | 10 | 73<br>(1) |
| 37 | Staircase 12 L2 (right area - top)    | 7.84   | 150 | 900  | 0.75 | 0.85 | 2.05  | 2  | 146.36 | INTEGRAGRANDDLRL10WLED830S                      | 10 | 73<br>(1) |
| 38 | Store 1 (left area - near lift)       | 6.98   | 150 | 1800 | 0.75 | 0.85 | 0.91  | 1  | 164.40 | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 18 | 37<br>(1) |
| 39 | Store Sakha (middle area - left)      | 73.48  | 200 | 2550 | 0.75 | 0.85 | 9.04  | 9  | 199.11 | VENUSNEO1-10V2X2PLR34WLED840MOD+LED DRIVER36V28 | 28 | 37<br>(1) |
| 40 | Store 2 (middle area - top right)     | 160.5  | 150 | 2550 | 0.75 | 0.85 | 14.81 | 16 | 162.06 | VENUSNEO1-10V2X2PLR34WLED840MOD+LED DRIVER36V29 | 28 | 37<br>(1) |
| 41 | Store 3 (rightmost middle)            | 35.64  | 175 | 1180 | 0.75 | 0.85 | 8.29  | 8  | 168.86 | PLUTOSURFACE1X1PLS15 WLED857S                   | 15 | 61<br>(2) |
| 42 | Store 4 (right area - near lift)      | 6.98   | 150 | 1800 | 0.75 | 0.85 | 0.91  | 1  | 164.40 | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 18 | 37<br>(1) |
| 43 | Bank                                  | 445    | 275 | 3780 | 0.75 | 0.85 | 50.78 | 52 | 281.59 | VENUSNEO1-10V2X2PLR34WLED857MOD                 | 42 | 37<br>(2) |
| 44 | Library                               | 148.26 | 250 | 3600 | 0.75 | 0.85 | 16.15 | 16 | 247.67 | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42 | 37<br>(1) |
| 45 | IT Section                            | 37.98  | 250 | 3600 | 0.75 | 0.85 | 4.14  | 4  | 241.71 | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42 | 37<br>(1) |
| 46 | Sakha Area (right area - bottom left) | 358.58 | 250 | 3600 | 0.75 | 0.85 | 39.06 | 40 | 256.01 | VENUSNEO1-10V2X2PLR34WLED840MOD                 | 42 | 37<br>(1) |
| 47 | Corridor 1 (whole left area)          | 458.23 | 150 | 1920 | 0.75 | 0.85 | 56.16 | 59 | 157.60 | PLUTO2X1PLR20WLED840S                           | 20 | 53<br>(1) |
| 48 | Corridor 2 (whole middle area)        | 374.53 | 150 | 1920 | 0.75 | 0.85 | 45.90 | 50 | 163.40 | PLUTO2X1PLR20WLED840S                           | 20 | 53<br>(1) |
| 49 | Corridor 3 (whole right area)         | 492.37 | 150 | 1920 | 0.75 | 0.85 | 60.34 | 63 | 156.61 | PLUTO2X1PLR20WLED840S                           | 20 | 53<br>(1) |

d. Second Floor:

| SECOND FLOOR ILLUMINATION SPECIFICATIONS |              |        |                        |                                                 |                    |      |      |                         |                         |                                    |             |        |       |
|------------------------------------------|--------------|--------|------------------------|-------------------------------------------------|--------------------|------|------|-------------------------|-------------------------|------------------------------------|-------------|--------|-------|
| Serial No.                               | Description  |        | Dimensions             | Required Illumination (E <sub>req</sub> ) (lux) | Lumen's Output (φ) | μm   | μu   | N <sub>calculated</sub> | N <sub>considered</sub> | Catalogue: Havells Office Lighting |             |        |       |
|                                          | Designation  | Code   | Area (m <sup>2</sup> ) |                                                 |                    |      |      |                         |                         | Model                              | Wattage (W) | Pg.no. | Scale |
| 1                                        | Left Hall 1  | 2FLH1  | 397.832                | 300                                             | 3400               | 0.75 | 0.85 | 55.0632526              | 57                      | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34          | 37     | -     |
| 2                                        | Left hall 3  | 2FLH3  | 397.832                | 300                                             | 3400               | 0.75 | 0.85 | 55.0632526              | 57                      | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34          | 37     | -     |
| 3                                        | Left hall 4  | 2FLH4  | 397.832                | 300                                             | 3400               | 0.75 | 0.85 | 55.0632526              | 57                      | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34          | 37     | -     |
| 4                                        | Right hall 1 | 2RHH1  | 397.832                | 300                                             | 3400               | 0.75 | 0.85 | 55.0632526              | 57                      | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34          | 37     | -     |
| 5                                        | Right hall 3 | 2RHH3  | 397.832                | 300                                             | 3400               | 0.75 | 0.85 | 55.0632526              | 57                      | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34          | 37     | -     |
| 6                                        | Right hall 4 | 2RHH4  | 397.832                | 300                                             | 3400               | 0.75 | 0.85 | 55.0632526              | 57                      | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34          | 37     | -     |
| 7                                        | Right hall 2 | 2RHH2  | 332.603                | 300                                             | 3400               | 0.75 | 0.85 | 46.0350173              | 46                      | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34          | 37     | -     |
| 8                                        | Left hall 2  | 2FLH2  | 332.603                | 300                                             | 3400               | 0.75 | 0.85 | 46.0350173              | 46                      | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34          | 37     | -     |
| 9                                        | Restaurant   | 2BMR   | 484.964                | 275                                             | 3400               | 0.75 | 0.85 | 61.5294579              | 58                      | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34          | 37     | -     |
| 10                                       | Food Court   | 2TFMF  | 372.225                | 275                                             | 3400               | 0.75 | 0.85 | 47.22577855             | 48                      | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34          | 37     | -     |
| 11                                       | Store-1      | 2MLSto | 36.05                  | 175                                             | 1180               | 0.75 | 0.85 | 8.386507145             | 9                       | PLUTOSURFACEIXIPLS15 WLED857S      | 15          | 61     | -     |
| 12                                       | Store-2      | 2MRSto | 36.05                  | 175                                             | 1180               | 0.75 | 0.85 | 8.386507145             | 9                       | PLUTOSURFACEIXIPLS15 WLED857S      | 15          | 61     | -     |
| 13                                       | Pantry-1     | 2FMLO  | 38.54                  | 250                                             | 1180               | 0.75 | 0.85 | 12.80824194             | 12                      | PLUTOSURFACEIXIPLS15 WLED857S      | 15          | 61     | -     |
| 14                                       | Pantry-2     | 2FMRO  | 38.54                  | 250                                             | 1180               | 0.75 | 0.85 | 12.80824194             | 12                      | PLUTOSURFACEIXIPLS15 WLED857S      | 15          | 61     | -     |
| 15                                       | Corridor-2   | 2MRC   | 371.158                | 150                                             | 1920               | 0.75 | 0.85 | 45.48504902             | 48                      | PLUTO2X1PIR20WLED840S              | 20          | 53     | -     |
| 16                                       | Corridor-1   | 2RLC   | 432.072                | 150                                             | 1920               | 0.75 | 0.85 | 52.95                   | 59                      | PLUTO2X1PIR20WLED840S              | 20          | 53     | -     |
| 17                                       | Corridor-3   | 2RRC   | 432.072                | 150                                             | 1920               | 0.75 | 0.85 | 52.95                   | 59                      | PLUTO2X1PIR20WLED840S              | 20          | 53     | -     |
| 18                                       | Staircase-1  | 2FTLS  | 18.325                 | 125                                             | 900                | 0.75 | 0.85 | 3.992374728             | 4                       | INTEGRAGRANDDLRL12WLED840S         | 12          | 71     | -     |
| 19                                       | Staircase-2  | 2FTBS  | 18.325                 | 125                                             | 900                | 0.75 | 0.85 | 3.992374728             | 4                       | INTEGRAGRANDDLRL12WLED840S         | 12          | 71     | -     |
| 20                                       | Staircase-3  | 2FTRS  | 18.325                 | 125                                             | 900                | 0.75 | 0.85 | 3.992374728             | 4                       | INTEGRAGRANDDLRL12WLED840S         | 12          | 71     | -     |
| 21                                       | Staircase-4  | 2FTBS  | 18.325                 | 125                                             | 900                | 0.75 | 0.85 | 3.992374728             | 4                       | INTEGRAGRANDDLRL12WLED840S         | 12          | 71     | -     |
| 22                                       | Staircase-5  | 2FTLS  | 20.02                  | 125                                             | 900                | 0.75 | 0.85 | 4.361655773             | 4                       | INTEGRAGRANDDLRL12WLED840S         | 12          | 71     | -     |
| 23                                       | Staircase-6  | 2FTBS  | 20.02                  | 125                                             | 900                | 0.75 | 0.85 | 4.361655773             | 4                       | INTEGRAGRANDDLRL12WLED840S         | 12          | 71     | -     |
| 24                                       | Staircase-7  | 2FTRS  | 20.02                  | 125                                             | 900                | 0.75 | 0.85 | 4.361655773             | 4                       | INTEGRAGRANDDLRL12WLED840S         | 12          | 71     | -     |
| 25                                       | Staircase-8  | 2FTBS  | 20.02                  | 125                                             | 900                | 0.75 | 0.85 | 4.361655773             | 4                       | INTEGRAGRANDDLRL12WLED840S         | 12          | 71     | -     |
| 26                                       | Staircase-9  | 2FMRS  | 20.02                  | 125                                             | 900                | 0.75 | 0.85 | 4.361655773             | 4                       | INTEGRAGRANDDLRL12WLED840S         | 12          | 71     | -     |

|    |                 |       |        |     |      |      |      |             |    |            |                                       |    |    |        |
|----|-----------------|-------|--------|-----|------|------|------|-------------|----|------------|---------------------------------------|----|----|--------|
| 27 | Staircase-10    | 2fMLS | 20.02  | 125 | 900  | 0.75 | 0.85 | 4.361655773 | 4  | 114.635365 | INTEGRAGRANDDLRL12WLED840S            | 12 | 71 | -      |
| 28 | Staircase-11    | 2fMLS | 18.325 | 125 | 900  | 0.75 | 0.85 | 3.992374728 | 4  | 125.238745 | INTEGRAGRANDDLRL12WLED840S            | 12 | 71 | -      |
| 29 | Staircase-12    | 2fMRS | 18.325 | 125 | 900  | 0.75 | 0.85 | 3.992374728 | 4  | 125.238745 | INTEGRAGRANDDLRL12WLED840S            | 12 | 71 | -      |
| 30 | Washroom-1      | 2fMLW | 49.138 | 150 | 810  | 0.75 | 0.85 | 14.27392883 | 16 | 168.138711 | INTEGRANEODLRL10WLED830S              | 10 | 71 | 1 by 2 |
| 31 | Washroom-2      | 2fMRW | 49.138 | 150 | 810  | 0.75 | 0.85 | 14.27392883 | 16 | 168.138711 | INTEGRANEODLRL10WLED830S              | 10 | 71 | 1 by 2 |
| 32 | Washroom-3      | 2fTLW | 49.138 | 150 | 810  | 0.75 | 0.85 | 14.27392883 | 16 | 168.138711 | INTEGRANEODLRL10WLED830S              | 10 | 71 | 1 by 2 |
| 33 | Washroom-4      | 2fTRW | 49.138 | 150 | 810  | 0.75 | 0.85 | 14.27392883 | 16 | 168.138711 | INTEGRANEODLRL10WLED830S              | 10 | 71 | 1 by 2 |
| 34 | Washroom-5      | 2fBLW | 49.138 | 150 | 810  | 0.75 | 0.85 | 14.27392883 | 16 | 168.138711 | INTEGRANEODLRL10WLED830S              | 10 | 71 | 1 by 2 |
| 35 | Washroom-6      | 2fBRW | 49.138 | 150 | 810  | 0.75 | 0.85 | 14.27392883 | 16 | 168.138711 | INTEGRANEODLRL10WLED830S              | 10 | 71 | 1 by 2 |
| 36 | Baluster area-1 | 2fLB  | 32.77  | 250 | 3400 | 0.75 | 0.85 | 3.779700115 | 4  | 264.571254 | VENUSNEO1-10V2X2PLR34WLED840MOD       | 34 | 37 | -      |
| 37 | Baluster area-2 | 2fRB  | 32.77  | 250 | 3400 | 0.75 | 0.85 | 3.779700115 | 4  | 264.571254 | VENUSNEO1-10V2X2PLR34WLED840MOD       | 34 | 37 | -      |
| 38 | Terrace         | 2fMLT | 66.46  | 150 | 9000 | 0.75 | 0.85 | 1.73751634  | 2  | 172.660247 | HEXAHWCWPA75WLED850ASYMTGSWS<br>CIP66 | 75 | 27 |        |
| 39 | Terrace         | 2fMRT | 66.46  | 150 | 9000 | 0.75 | 0.85 | 1.73751634  | 2  | 172.660247 | HEXAHWCWPA75WLED850ASYMTGSWS<br>CIP67 | 75 | 27 |        |

e. Third Floor:

| THIRD FLOOR ILLUMINATION SPECIFICATIONS |                   |         |                        |                                                  |                    |      |      |                         |                          |                         |                                    |             |        |        |
|-----------------------------------------|-------------------|---------|------------------------|--------------------------------------------------|--------------------|------|------|-------------------------|--------------------------|-------------------------|------------------------------------|-------------|--------|--------|
| Serial No.                              | Description       |         | Dimensions             | Required Illuminati on (E <sub>req</sub> ) (lux) | Lumen s Output (φ) | μm   | μu   | N <sub>calculated</sub> | N <sub>consider ed</sub> | E <sub>considered</sub> | Catalogue: Havells Office Lighting |             |        |        |
|                                         | Designation       | Code    | Area (m <sup>2</sup> ) |                                                  |                    |      |      |                         |                          |                         | Model                              | Wattage (W) | Pg.no. | Scale  |
| 1                                       | Office Area-1     | 3FBLO   | 376.915                | 250                                              | 3400               | 0.75 | 0.85 | 43.4734717              | 46                       | 264.529138              | VENUSNEOI - 10V2X2PLR34WLED840MOD  | 34          | 37     | -      |
| 2                                       | Office Washroom-1 | 3FBLO W | 3.46                   | 150                                              | 900                | 0.75 | 0.85 | 0.90457516              | 1                        | 165.823699              | INTEGRAGRANDDL R12WLED840S         | 12          | 71     | -      |
| 3                                       | Corridor-1        | 3FLC    | 445.391                | 150                                              | 1920               | 0.75 | 0.85 | 54.5822304              | 58                       | 159.392534              | PLUTO2X1PL R20WLED840S             | 20          | 53     | -      |
| 4                                       | Pantry-1          | 3FLP    | 38.55                  | 250                                              | 1180               | 0.75 | 0.85 | 12.8115653              | 12                       | 234.163424              | PLUTOSURFACE1X1PLS15 WLED857S      | 15          | 61     | -      |
| 5                                       | Office Area-2     | 3FTLO   | 302.8                  | 250                                              | 3400               | 0.75 | 0.85 | 34.9250288              | 38                       | 272.011229              | VENUSNEOI - 10V2X2PLR34WLED840MOD  | 34          | 37     | -      |
| 6                                       | Office Washroom-2 | 3FTLO W | 3.906                  | 150                                              | 900                | 0.75 | 0.85 | 1.02117647              | 1                        | 146.889401              | INTEGRAGRANDDL R12WLED840S         | 12          | 71     | -      |
| 7                                       | Washroom-1        | 3FBLW   | 48.67                  | 150                                              | 810                | 0.75 | 0.85 | 14.1379811              | 16                       | 169.755496              | INTEGRANEODL R10WLED830S           | 10          | 71     | 1 by 2 |
| 8                                       | Washroom-2        | 3FTLW   | 48.67                  | 150                                              | 810                | 0.75 | 0.85 | 14.1379811              | 16                       | 169.755496              | INTEGRANEODL R10WLED830S           | 10          | 71     | 1 by 2 |
| 9                                       | Staircase-1       | 3FTLS   | 18.325                 | 125                                              | 900                | 0.75 | 0.85 | 3.99237473              | 4                        | 125.238745              | INTEGRAGRANDDL R12WLED840S         | 12          | 71     | -      |
| 10                                      | Staircase-2       | 3FBLS   | 18.325                 | 125                                              | 900                | 0.75 | 0.85 | 3.99237473              | 4                        | 125.238745              | INTEGRAGRANDDL R12WLED840S         | 12          | 71     | -      |
| 11                                      | Staircase-3       | 3FTLS   | 20.02                  | 125                                              | 900                | 0.75 | 0.85 | 4.36165577              | 4                        | 114.635365              | INTEGRAGRANDDL R12WLED840S         | 12          | 71     | -      |
| 12                                      | Staircase-4       | 3FBLS   | 20.02                  | 125                                              | 900                | 0.75 | 0.85 | 4.36165577              | 4                        | 114.635365              | INTEGRAGRANDDL R12WLED840S         | 12          | 71     | -      |
| 13                                      | Office-1          | 3BMO    | 484.964                | 275                                              | 3400               | 0.75 | 0.85 | 61.5294579              | 58                       | 259.225427              | VENUSNEOI - 10V2X2PLR34WLED840MOD  | 34          | 37     | -      |
| 14                                      | Office-2          | 3TMO    | 372.225                | 275                                              | 3400               | 0.75 | 0.85 | 47.2257785              | 48                       | 279.508362              | VENUSNEOI - 10V2X2PLR34WLED840MOD  | 34          | 37     | -      |
| 15                                      | Pantry-2          | 3MLP    | 38.54                  | 250                                              | 1180               | 0.75 | 0.85 | 12.8082419              | 12                       | 234.224183              | PLUTOSURFACE1X1PLS15 WLED857S      | 15          | 61     | -      |
| 16                                      | Pantry-3          | 3MRP    | 38.54                  | 250                                              | 1180               | 0.75 | 0.85 | 12.8082419              | 12                       | 234.224183              | PLUTOSURFACE1X1PLS15 WLED857S      | 15          | 61     | -      |
| 17                                      | Store-1           | 3MLSt o | 36.05                  | 175                                              | 1180               | 0.75 | 0.85 | 8.38650715              | 9                        | 187.801664              | PLUTOSURFACE1X1PLS15 WLED857S      | 15          | 61     | -      |
| 18                                      | Store-2           | 3MRSt o | 36.05                  | 175                                              | 1180               | 0.75 | 0.85 | 8.38650715              | 9                        | 187.801664              | PLUTOSURFACE1X1PLS15 WLED857S      | 15          | 61     | -      |
| 19                                      | Corridor-2        | 3MC     | 371.158                | 150                                              | 1920               | 0.75 | 0.85 | 45.485049               | 48                       | 158.293772              | PLUTO2X1PL R20WLED840S             | 20          | 53     | -      |
| 20                                      | Washroom-3        | 3MLW    | 49.138                 | 150                                              | 810                | 0.75 | 0.85 | 14.2739288              | 16                       | 168.138711              | INTEGRANEODL R10WLED830S           | 10          | 71     | 1 by 2 |
| 21                                      | Washroom-4        | 3MRW    | 49.138                 | 150                                              | 810                | 0.75 | 0.85 | 14.2739288              | 16                       | 168.138711              | INTEGRANEODL R10WLED830S           | 10          | 71     | 1 by 2 |
| 22                                      | Staircase-5       | 3MBL S  | 18.325                 | 150                                              | 900                | 0.75 | 0.85 | 4.79084967              | 4                        | 125.238745              | INTEGRAGRANDDL R12WLED840S         | 12          | 71     | -      |
| 23                                      | Staircase-6       | 3MBR S  | 18.325                 | 150                                              | 900                | 0.75 | 0.85 | 4.79084967              | 4                        | 125.238745              | INTEGRAGRANDDL R12WLED840S         | 12          | 71     | -      |
| 24                                      | Staircase-7       | 3MTL S  | 20.02                  | 150                                              | 900                | 0.75 | 0.85 | 5.23398693              | 4                        | 114.635365              | INTEGRAGRANDDL R12WLED840S         | 12          | 71     | -      |
| 25                                      | Staircase-8       | 3MTR S  | 20.02                  | 150                                              | 900                | 0.75 | 0.85 | 5.23398693              | 4                        | 114.635365              | INTEGRAGRANDDL R12WLED840S         | 12          | 71     | -      |
| 26                                      | Office Area-3     | 3BRO    | 376.915                | 250                                              | 3400               | 0.75 | 0.85 | 43.4734717              | 46                       | 264.529138              | VENUSNEOI - 10V2X2PLR34WLED840MOD  | 34          | 37     | -      |

|    |                   |         |         |     |      |      |      |            |    |            |                                    |    |    |        |
|----|-------------------|---------|---------|-----|------|------|------|------------|----|------------|------------------------------------|----|----|--------|
| 27 | Office Washroom-3 | 3fBLO W | 3.46    | 150 | 900  | 0.75 | 0.85 | 0.90457516 | 1  | 165.823699 | INTEGRAGRANDDDLR10WLED830S         | 10 | 73 | -      |
| 28 | Corridor-3        | 3fLC    | 445.391 | 150 | 1920 | 0.75 | 0.85 | 54.5822304 | 58 | 159.392534 | PLUTOX2P1R20WLED840S               | 20 | 53 | -      |
| 29 | Pantry-4          | 3fRP    | 38.54   | 250 | 1180 | 0.75 | 0.85 | 12.8082419 | 12 | 234.224183 | PLUTOSURFACE1X1PLS15 WLED857S      | 15 | 61 | -      |
| 30 | Office Area-4     | 3fTRO   | 302.8   | 250 | 3400 | 0.75 | 0.85 | 34.9250288 | 38 | 272.011229 | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34 | 37 | -      |
| 31 | Office Washroom-4 | 3fTRO W | 3.906   | 150 | 900  | 0.75 | 0.85 | 1.02117647 | 1  | 146.889401 | INTEGRAGRANDDDLR12WLED840S         | 12 | 71 | -      |
| 32 | Washroom-5        | 3fBRW   | 48.67   | 150 | 810  | 0.75 | 0.85 | 14.1379811 | 16 | 169.755496 | INTEGRANEODLR10WLED830S            | 10 | 71 | 1 by 2 |
| 33 | Washroom-6        | 3fTRW   | 48.67   | 150 | 810  | 0.75 | 0.85 | 14.1379811 | 16 | 169.755496 | INTEGRANEODLR10WLED830S            | 10 | 71 | 1 by 2 |
| 34 | Staircase-9       | 3fTRS   | 22.325  | 150 | 900  | 0.75 | 0.85 | 5.83660131 | 6  | 154.199328 | INTEGRAGRANDDDLR12WLED840S         | 12 | 71 | -      |
| 35 | Staircase-10      | 3fBRS   | 22.325  | 150 | 900  | 0.75 | 0.85 | 5.83660131 | 6  | 154.199328 | INTEGRAGRANDDDLR12WLED840S         | 12 | 71 | -      |
| 36 | Staircase-11      | 3fTRS   | 24.02   | 150 | 900  | 0.75 | 0.85 | 6.27973856 | 6  | 143.318068 | INTEGRAGRANDDDLR12WLED840S         | 12 | 71 | -      |
| 37 | Staircase-12      | 3fBRS   | 24.02   | 150 | 900  | 0.75 | 0.85 | 6.27973856 | 6  | 143.318068 | INTEGRAGRANDDDLR12WLED840S         | 12 | 71 | -      |
| 38 | Office store-1    | 3fBLOS  | 16.23   | 150 | 3400 | 0.75 | 0.85 | 1.12318339 | 1  | 133.548983 | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34 | 37 | -      |
| 39 | Office store-2    | 3fTROS  | 16.23   | 150 | 3400 | 0.75 | 0.85 | 1.12318339 | 1  | 133.548983 | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34 | 37 | -      |
| 40 | Blauster area-1   | 3fLB    | 32.77   | 250 | 3400 | 0.75 | 0.85 | 3.77970012 | 4  | 264.571254 | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34 | 37 | -      |
| 41 | Blauster area-2   | 3fRB    | 32.77   | 250 | 3400 | 0.75 | 0.85 | 3.77970012 | 4  | 264.571254 | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34 | 37 | -      |
| 42 | Office store-3    | 3fBROS  | 16.23   | 150 | 3400 | 0.75 | 0.85 | 1.12318339 | 1  | 133.548983 | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34 | 37 | -      |
| 43 | Office store-4    | 3fTLOS  | 16.23   | 150 | 3400 | 0.75 | 0.85 | 1.12318339 | 1  | 133.548983 | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34 | 37 | -      |
| 44 | Terrace-1         | 3fTRT   | 341.25  | 150 | 9000 | 0.75 | 0.85 | 8.92156863 | 10 | 168.131868 | HEXAHWCWPA75WLED850ASYMTGSWS CIP65 | 75 | 27 | -      |
| 45 | Terrace-2         | 3fTLT   | 341.25  | 150 | 9000 | 0.75 | 0.85 | 8.92156863 | 10 | 168.131868 | HEXAHWCWPA75WLED850ASYMTGSWS CIP66 | 75 | 27 | -      |

f. Fourth Floor:

| FOURTH FLOOR ILLUMINATION SPECIFICATIONS |                    |            |                        |                                                 |                    |      |      |                         |                          |                         |                                      |             |        |       |
|------------------------------------------|--------------------|------------|------------------------|-------------------------------------------------|--------------------|------|------|-------------------------|--------------------------|-------------------------|--------------------------------------|-------------|--------|-------|
| Serial No.                               | Description        |            | Dimensions             | Required Illumination (E <sub>req</sub> ) (lux) | Lumen s Output (φ) | μm   | μu   | N <sub>calculated</sub> | N <sub>consider ed</sub> | E <sub>considered</sub> | Catalogue: Havells Office Lighting   |             |        |       |
|                                          | Designation        | Code       | Area (m <sup>2</sup> ) |                                                 |                    |      |      |                         |                          |                         | Model                                | Wattage (W) | Pg.no. | Scale |
| 1                                        | Office Area-1      | 4FBLO      | 396.34                 | 250                                             | 3400               | 0.75 | 0.85 | 45.7139562              | 44                       | 240.626735              | VENUSNEOI-10V2X2PLR34WLED840MOD      | 34          | 37     | -     |
| 2                                        | Toilet in office 1 | 4FBLCCT    | 3.687                  | 175                                             | 900                | 0.75 | 0.85 | 1.12457516              | 1                        | 155.614321              | INTEGRAGRANDDDLR10WLED830S           | 10          | 73     | -     |
| 3                                        | staircase 1        | 4FBLS      | 25.34                  | 125                                             | 1200               | 0.75 | 0.85 | 4.14052288              | 5                        | 150.947119              | INTEGRAGRANDDDLR12WLED840S           | 12          | 71     | -     |
| 4                                        | washroom 1         | 4FBLW      | 56.312                 | 175                                             | 900                | 0.75 | 0.85 | 17.1757734              | 19                       | 193.586625              | INTEGRAGRANDDDLR10WLED830S           | 10          | 73     | -     |
| 5                                        | Corridor 1         | 4FBLC      | 25.2                   | 175                                             | 1800               | 0.75 | 0.85 | 3.84313725              | 4                        | 182.142857              | PLUTO2X1PLR20WLED840S                | 20          | 53     | -     |
| 6                                        | Staircase 2        | 4BBS       | 45.55                  | 125                                             | 1200               | 0.75 | 0.85 | 7.44281046              | 7                        | 117.563117              | INTEGRAGRANDDDLR12WLED840S           | 12          | 71     | -     |
| 7                                        | Corridor-2         | 4FCLC      | 148.754                | 150                                             | 1800               | 0.75 | 0.85 | 19.4449673              | 18                       | 138.853409              | PLUTO2X1PLR20WLED840S                | 20          | 53     | -     |
| 8                                        | Corridor-3         | 4FCRC      | 93.5                   | 175                                             | 1800               | 0.75 | 0.85 | 14.2592593              | 13                       | 159.545455              | PLUTO2X1PLR20WLED840S                | 20          | 53     | -     |
| 9                                        | Staircase-3        | 4FULS      | 54.29                  | 125                                             | 1200               | 0.75 | 0.85 | 8.87091503              | 8                        | 112.727943              | INTEGRAGRANDDDLR12WLED840S           | 12          | 71     | -     |
| 10                                       | Washroom-2         | 4FULW      | 58.626                 | 175                                             | 900                | 0.75 | 0.85 | 17.8815686              | 20                       | 195.732269              | INTEGRAGRANDDDLR10WLED830S           | 10          | 73     | -     |
| 11                                       | Staircase-4        | 4FUCS      | 34.34                  | 125                                             | 1200               | 0.75 | 0.85 | 5.61111111              | 5                        | 111.386139              | INTEGRAGRANDDDLR10WLED830S           | 12          | 71     | -     |
| 12                                       | Office Area-2      | 4FURO      | 410.022                | 250                                             | 3400               | 0.75 | 0.85 | 47.2920415              | 49                       | 259.028784              | VENUSNEOI-10V2X2PLR34WLED840MOD      | 34          | 37     | -     |
| 13                                       | Corridor-4         | 4FCRC      | 40.322                 | 175                                             | 1800               | 0.75 | 0.85 | 6.14932462              | 6                        | 170.750459              | PLUTO2X1PLR20WLED840S                | 20          | 53     | -     |
| 14                                       | Toilet in office 2 | 4FCRT      | 3.15                   | 175                                             | 900                | 0.75 | 0.85 | 0.96078431              | 1                        | 182.142857              | INTEGRAGRANDDDLR12WLED830S           | 10          | 73     | -     |
| 15                                       | Toilet in office 3 | 4FURC<br>T | 3.172                  | 175                                             | 900                | 0.75 | 0.85 | 0.96749455              | 1                        | 180.879571              | INTEGRAGRANDDDLR12WLED830S           | 10          | 73     | -     |
| 16                                       | Terrace-1          | 4FULTe     | 355.43                 | 200                                             | 9000               | 0.75 | 0.85 | 12.3897168              | 14                       | 225.993867              | HEXAHWCWA75WLED850ASYMTGSWS<br>CIP66 | 75          | 27     | -     |
| 17                                       | Terrace-2          | 4FBRTe     | 355.43                 | 200                                             | 9000               | 0.75 | 0.85 | 12.3897168              | 14                       | 225.993867              | HEXAHWCWA75WLED850ASYMTGSWS<br>CIP65 | 75          | 27     | -     |
| 18                                       | Office Area-3      | 4FBRO      | 396.34                 | 250                                             | 3400               | 0.75 | 0.85 | 45.7139562              | 44                       | 240.626735              | VENUSNEOI-10V2X2PLR34WLED840MOD      | 34          | 37     | -     |
| 19                                       | Toilet in office 4 | 4FBRCT     | 3.687                  | 175                                             | 900                | 0.75 | 0.85 | 1.12457516              | 1                        | 155.614321              | INTEGRAGRANDDDLR12WLED830S           | 10          | 73     | -     |
| 20                                       | Staircase-5        | 4BBS       | 25.34                  | 125                                             | 1200               | 0.75 | 0.85 | 4.14052288              | 5                        | 150.947119              | INTEGRAGRANDDDLR10WLED840S           | 12          | 71     | -     |
| 21                                       | Washroom-3         | 4FBRW      | 56.312                 | 175                                             | 900                | 0.75 | 0.85 | 17.1757734              | 19                       | 193.586625              | INTEGRAGRANDDDLR12WLED830S           | 10          | 73     | -     |
| 22                                       | Corridor-5         | 4FBRC      | 25.2                   | 175                                             | 1800               | 0.75 | 0.85 | 3.84313725              | 4                        | 182.142857              | PLUTO2X1PLR20WLED840S                | 20          | 53     | -     |
| 23                                       | Staircase-6        | 4BBS       | 45.55                  | 125                                             | 1200               | 0.75 | 0.85 | 7.44281046              | 7                        | 117.563117              | INTEGRAGRANDDDLR10WLED840S           | 12          | 71     | -     |
| 24                                       | Corridor-6         | 4FCRC      | 148.754                | 175                                             | 1800               | 0.75 | 0.85 | 22.6857952              | 18                       | 138.853409              | PLUTO2X1PLR20WLED840S                | 20          | 53     | -     |
| 25                                       | Corridor-7         | 4FCLC      | 93.5                   | 175                                             | 1800               | 0.75 | 0.85 | 14.2592593              | 13                       | 159.545455              | PLUTO2X1PLR20WLED840S                | 20          | 53     | -     |

|    |                    |            |         |     |      |      |      |            |    |            |                                      |    |    |   |
|----|--------------------|------------|---------|-----|------|------|------|------------|----|------------|--------------------------------------|----|----|---|
| 26 | Staircase 7        | 4fURS      | 54.29   | 125 | 1200 | 0.75 | 0.85 | 8.87091503 | 8  | 112.727943 | INTEGRAGRANDDDLR10WLED830S           | 10 | 73 | - |
| 27 | Washroom-4         | 4fURW      | 58.626  | 175 | 900  | 0.75 | 0.85 | 17.8815686 | 20 | 195.732269 | INTEGRAGRANDDDLR12WLED830S           | 10 | 73 | - |
| 28 | Staircase 8        | 4fUCS      | 34.34   | 125 | 1200 | 0.75 | 0.85 | 5.61111111 | 5  | 111.386139 | INTEGRAGRANDDDLR10WLED840S           | 12 | 71 | - |
| 29 | Office Area-4      | 4fULO      | 410.022 | 250 | 3400 | 0.75 | 0.85 | 47.2920415 | 49 | 259.028784 | VENUSNEO1-<br>10V2X2PLR34WLED840MOD  | 34 | 37 | - |
| 30 | Corridor-8         | 4fCLC      | 40.322  | 175 | 1800 | 0.75 | 0.85 | 6.14932462 | 6  | 170.750459 | PLUTO2X1PLR20WLED840S                | 20 | 53 | - |
| 31 | Toilet in office 5 | 4fCLT      | 3.15    | 175 | 900  | 0.75 | 0.85 | 0.96078431 | 1  | 182.142857 | INTEGRAGRANDDDLR12WLED830S           | 10 | 73 | - |
| 32 | Toilet in office 6 | 4fULCT     | 3.172   | 175 | 900  | 0.75 | 0.85 | 0.96749455 | 1  | 180.879571 | INTEGRAGRANDDDLR12WLED830S           | 10 | 73 | - |
| 33 | Terrace 3          | 4fURTe     | 355.43  | 150 | 9000 | 0.75 | 0.85 | 9.29228758 | 14 | 225.993867 | HEXAHCWPA75WLED850ASYMTGSWS<br>CIP66 | 75 | 27 | - |
| 34 | Terrace 4          | 4fBLTe     | 355.43  | 150 | 9000 | 0.75 | 0.85 | 9.29228758 | 14 | 225.993867 | HEXAHCWPA75WLED850ASYMTGSWS<br>CIP65 | 75 | 27 | - |
| 35 | Office Area-5      | 4fBCO      | 471.67  | 250 | 3400 | 0.75 | 0.85 | 54.4025375 | 52 | 238.959442 | VENUSNEO1-<br>10V2X2PLR34WLED840MOD  | 34 | 37 | - |
| 36 | Office Area-6      | 4fUCO      | 363.21  | 250 | 3400 | 0.75 | 0.85 | 41.8927336 | 40 | 238.704881 | VENUSNEO1-<br>10V2X2PLR34WLED840MOD  | 34 | 37 | - |
| 37 | Staircase-9        | 4fMUL<br>S | 32.09   | 125 | 1200 | 0.75 | 0.85 | 5.24346405 | 5  | 119.196011 | INTEGRAGRANDDDLR10WLED840S           | 12 | 71 | - |
| 38 | Staircase 10       | 4fMUR<br>S | 32.09   | 125 | 1200 | 0.75 | 0.85 | 5.24346405 | 5  | 119.196011 | INTEGRAGRANDDDLR10WLED840S           | 12 | 71 | - |
| 39 | Staircase 11       | 4fMBL<br>S | 26.175  | 125 | 1200 | 0.75 | 0.85 | 4.27696078 | 4  | 116.905444 | INTEGRAGRANDDDLR10WLED840S           | 12 | 71 | - |
| 40 | Staircase 12       | 4fMBR<br>S | 26.175  | 125 | 1200 | 0.75 | 0.85 | 4.27696078 | 4  | 116.905444 | INTEGRAGRANDDDLR10WLED840S           | 12 | 71 | - |
| 41 | Washroom 5         | 4fMCL<br>W | 58.7    | 175 | 900  | 0.75 | 0.85 | 17.9041394 | 19 | 185.711244 | INTEGRAGRANDDDLR12WLED830S           | 10 | 73 | - |
| 42 | Washroom 6         | 4fMCR<br>W | 58.7    | 175 | 900  | 0.75 | 0.85 | 17.9041394 | 19 | 185.711244 | INTEGRAGRANDDDLR12WLED830S           | 10 | 73 | - |
| 43 | Office area-7      | 4fMCL<br>O | 107.83  | 250 | 3400 | 0.75 | 0.85 | 12.4371396 | 12 | 241.21302  | VENUSNEO1-<br>10V2X2PLR34WLED840MOD  | 34 | 37 | - |
| 44 | Office area 8      | 4fMCR<br>O | 107.83  | 250 | 3400 | 0.75 | 0.85 | 12.4371396 | 12 | 241.21302  | VENUSNEO1-<br>10V2X2PLR34WLED840MOD  | 34 | 37 | - |
| 45 | Corridor-9         | 4fMCC      | 260     | 175 | 1800 | 0.75 | 0.85 | 39.6514161 | 38 | 167.711538 | PLUTO2X1PLR20WLED840S                | 20 | 53 | - |
| 46 | Lift Lobby 1       | 4fMCL<br>L | 8.09    | 150 | 1800 | 0.75 | 0.85 | 1.05751634 | 1  | 141.84178  | PLUTO2X1PLR20WLED840S                | 10 | 73 | - |
| 47 | Lift Lobby 2       | 4fMCR<br>L | 14.182  | 175 | 1800 | 0.75 | 0.85 | 2.15520697 | 2  | 161.824848 | PLUTO2X1PLR20WLED840S                | 10 | 73 | - |
| 48 | DB Room            | 4fMCL<br>R | 16.52   | 250 | 3400 | 0.75 | 0.85 | 1.90542099 | 2  | 262.409201 | VENUSNEO1-<br>10V2X2PLR34WLED840MOD  | 34 | 37 | - |

g. Fifth Floor:

| FIFTH FLOOR ILLUMINATION SPECIFICATIONS |                   |         |                        |                                                 |                    |      |      |                         |                         |                         |                                    |             |        |        |
|-----------------------------------------|-------------------|---------|------------------------|-------------------------------------------------|--------------------|------|------|-------------------------|-------------------------|-------------------------|------------------------------------|-------------|--------|--------|
| Serial No.                              | Description       |         | Dimensions             | Required Illumination (E <sub>req</sub> ) (lux) | Lumen's Output (φ) | μm   | μu   | N <sub>calculated</sub> | N <sub>considered</sub> | E <sub>considered</sub> | Catalogue: Havells Office Lighting |             |        |        |
|                                         | Designation       | Code    | Area (m <sup>2</sup> ) |                                                 |                    |      |      |                         |                         |                         | Model                              | Wattage (W) | Pg.no. | Scale  |
| 1                                       | Office Area-1     | 3fBLO   | 450                    | 275                                             | 3400               | 0.75 | 0.85 | 57.09343                | 56                      | 269.7333                | VENUSNEOI-10V2X2PLR34WLED840MOD    | 34          | 37     | -      |
| 2                                       | Office Washroom-1 | 3fBLOW  | 3.46                   | 150                                             | 900                | 0.75 | 0.85 | 0.904575                | 1                       | 159.1749                | INTEGRAGRANDDLR12WLED840S          | 12          | 71     | -      |
| 3                                       | Corridor-1        | 3fLC    | 446                    | 150                                             | 1920               | 0.75 | 0.85 | 54.65686                | 58                      | 159.1749                | PLUTO2X1PLR20WLED840S              | 20          | 53     | -      |
| 4                                       | Pantry-1          | 3fLP    | 38.55                  | 250                                             | 1180               | 0.75 | 0.85 | 12.81157                | 12                      | 234.1634                | PLUTOSURFACEIXIPLS15 WLED857S      | 15          | 61     | -      |
| 5                                       | Office Area-2     | 3fTLO   | 300                    | 250                                             | 3400               | 0.75 | 0.85 | 34.60208                | 35                      | 252.875                 | VENUSNEOI-10V2X2PLR34WLED840MOD    | 34          | 37     | -      |
| 6                                       | Office Washroom-2 | 3fTLOW  | 3.906                  | 150                                             | 900                | 0.75 | 0.85 | 1.021176                | 1                       | 146.8894                | INTEGRAGRANDDLR12WLED840S          | 12          | 71     | -      |
| 7                                       | Washroom-1        | 3fBLW   | 49                     | 150                                             | 810                | 0.75 | 0.85 | 14.23384                | 15                      | 158.074                 | INTEGRANEODLR10WLED830S            | 10          | 71     | 1 by 2 |
| 8                                       | Washroom-2        | 3fTLW   | 49                     | 150                                             | 810                | 0.75 | 0.85 | 14.23384                | 16                      | 168.6122                | INTEGRANEODLR10WLED830S            | 10          | 71     | 1 by 2 |
| 9                                       | Staircase-1       | 3fTLS   | 19                     | 125                                             | 900                | 0.75 | 0.85 | 4.139434                | 4                       | 120.7895                | INTEGRAGRANDDLR12WLED840S          | 12          | 71     | -      |
| 10                                      | Staircase-2       | 3fBLS   | 19                     | 125                                             | 900                | 0.75 | 0.85 | 4.139434                | 4                       | 120.7895                | INTEGRAGRANDDLR12WLED840S          | 12          | 71     | -      |
| 11                                      | Staircase-3       | 3fTLS   | 20.02                  | 125                                             | 900                | 0.75 | 0.85 | 4.361656                | 4                       | 114.6354                | INTEGRAGRANDDLR12WLED840S          | 12          | 71     | -      |
| 12                                      | Staircase-4       | 3fBLS   | 20.02                  | 125                                             | 900                | 0.75 | 0.85 | 4.361656                | 4                       | 114.6354                | INTEGRAGRANDDLR12WLED840S          | 12          | 71     | -      |
| 13                                      | Office-1          | 3fBMO   | 484.964                | 275                                             | 3400               | 0.75 | 0.85 | 61.52946                | 58                      | 259.2254                | VENUSNEOI-10V2X2PLR34WLED840MOD    | 34          | 37     | -      |
| 14                                      | Office-2          | 3fTMO   | 372.225                | 275                                             | 3400               | 0.75 | 0.85 | 47.22578                | 47                      | 273.6853                | VENUSNEOI-10V2X2PLR34WLED840MOD    | 34          | 37     | -      |
| 15                                      | Pantry-2          | 3fMLP   | 38.54                  | 250                                             | 1180               | 0.75 | 0.85 | 12.80824                | 12                      | 234.2242                | PLUTOSURFACEIXIPLS15 WLED857S      | 15          | 61     | -      |
| 16                                      | Pantry-3          | 3fMRP   | 38.54                  | 250                                             | 1180               | 0.75 | 0.85 | 12.80824                | 12                      | 234.2242                | PLUTOSURFACEIXIPLS15 WLED857S      | 15          | 61     | -      |
| 17                                      | Store-1           | 3fMLSto | 36.05                  | 175                                             | 1180               | 0.75 | 0.85 | 8.386507                | 9                       | 187.8017                | PLUTOSURFACEIXIPLS15 WLED857S      | 15          | 61     | -      |
| 18                                      | Store-2           | 3fMRSo  | 36.05                  | 175                                             | 1180               | 0.75 | 0.85 | 8.386507                | 9                       | 187.8017                | PLUTOSURFACEIXIPLS15 WLED857S      | 15          | 61     | -      |
| 19                                      | Corridor-2        | 3fMC    | 371.158                | 150                                             | 1920               | 0.75 | 0.85 | 45.48505                | 48                      | 158.2938                | PLUTO2X1PLR20WLED840S              | 20          | 53     | -      |
| 20                                      | Washroom-3        | 3fMLW   | 49.138                 | 150                                             | 810                | 0.75 | 0.85 | 14.27393                | 16                      | 168.1387                | INTEGRANEODLR10WLED830S            | 10          | 71     | 1 by 2 |
| 21                                      | Washroom-4        | 3fMRW   | 49.138                 | 150                                             | 810                | 0.75 | 0.85 | 14.27393                | 16                      | 168.1387                | INTEGRANEODLR10WLED830S            | 10          | 71     | 1 by 2 |
| 22                                      | Staircase-5       | 3fMBLS  | 18.325                 | 150                                             | 900                | 0.75 | 0.85 | 4.79085                 | 4                       | 125.2387                | INTEGRAGRANDDLR12WLED840S          | 12          | 71     | -      |
| 23                                      | Staircase-6       | 3fMBRS  | 18.325                 | 150                                             | 900                | 0.75 | 0.85 | 4.79085                 | 4                       | 125.2387                | INTEGRAGRANDDLR12WLED840S          | 12          | 71     | -      |
| 24                                      | Staircase-7       | 3fMTLS  | 20.02                  | 150                                             | 900                | 0.75 | 0.85 | 5.233987                | 4                       | 114.6354                | INTEGRAGRANDDLR12WLED840S          | 12          | 71     | -      |
| 25                                      | Staircase-8       | 3fMTRS  | 20.02                  | 150                                             | 900                | 0.75 | 0.85 | 5.233987                | 4                       | 114.6354                | INTEGRAGRANDDLR12WLED840S          | 12          | 71     | -      |
| 26                                      | Office Area-3     | 3fBRO   | 376.915                | 250                                             | 3400               | 0.75 | 0.85 | 43.47347                | 46                      | 264.5291                | VENUSNEOI-10V2X2PLR34WLED840MOD    | 34          | 37     | -      |

|    |                   |            |         |     |      |      |      |          |    |          |                                       |    |    |        |
|----|-------------------|------------|---------|-----|------|------|------|----------|----|----------|---------------------------------------|----|----|--------|
| 27 | Office Washroom-3 | 3fBLO<br>W | 3.46    | 150 | 900  | 0.75 | 0.85 | 0.904575 | 1  | 165.8237 | INTEGRAGRANDDLR10WLED830S             | 10 | 73 | -      |
| 28 | Corridor-3        | 3fLC       | 445.391 | 150 | 1920 | 0.75 | 0.85 | 54.58223 | 58 | 159.3925 | PLUTO2X1PLR20WLED840S                 | 20 | 53 | -      |
| 29 | Pantry-4          | 3fRP       | 38.54   | 250 | 1180 | 0.75 | 0.85 | 12.80824 | 12 | 234.2242 | PLUTOSURFACE1X1PLS15 WLED857S         | 15 | 61 | -      |
| 30 | Office Area-4     | 3fTRO      | 302.8   | 250 | 3400 | 0.75 | 0.85 | 34.92503 | 38 | 272.0112 | VENUSNEO1-<br>10V2X2PLR34WLED840MOD   | 34 | 37 | -      |
| 31 | Office Washroom-4 | 3fTRO<br>W | 3.906   | 150 | 900  | 0.75 | 0.85 | 1.021176 | 1  | 146.8894 | INTEGRAGRANDDLR12WLED840S             | 12 | 71 | -      |
| 32 | Washroom-5        | 3fBRW      | 48.67   | 150 | 810  | 0.75 | 0.85 | 14.13798 | 16 | 169.7555 | INTEGRANEODLR10WLED830S               | 10 | 71 | 1 by 2 |
| 33 | Washroom-6        | 3fTRW      | 48.67   | 150 | 810  | 0.75 | 0.85 | 14.13798 | 16 | 169.7555 | INTEGRANEODLR10WLED830S               | 10 | 71 | 1 by 2 |
| 34 | Staircase-9       | 3fTRS      | 22.325  | 150 | 900  | 0.75 | 0.85 | 5.836601 | 6  | 154.1993 | INTEGRAGRANDDLR12WLED840S             | 12 | 71 | -      |
| 35 | Staircase-10      | 3fBRS      | 22.325  | 150 | 900  | 0.75 | 0.85 | 5.836601 | 6  | 154.1993 | INTEGRAGRANDDLR12WLED840S             | 12 | 71 | -      |
| 36 | Staircase-11      | 3fTRS      | 24.02   | 150 | 900  | 0.75 | 0.85 | 6.279739 | 6  | 143.3181 | INTEGRAGRANDDLR12WLED840S             | 12 | 71 | -      |
| 37 | Staircase-12      | 3fBRS      | 24.02   | 150 | 900  | 0.75 | 0.85 | 6.279739 | 6  | 143.3181 | INTEGRAGRANDDLR12WLED840S             | 12 | 71 | -      |
| 38 | Office store-1    | 3fBLOS     | 16.23   | 150 | 3400 | 0.75 | 0.85 | 1.123183 | 1  | 133.549  | VENUSNEO1-<br>10V2X2PLR34WLED840MOD   | 34 | 37 | -      |
| 39 | Office store-2    | 3fTROS     | 16.23   | 150 | 3400 | 0.75 | 0.85 | 1.123183 | 1  | 133.549  | VENUSNEO1-<br>10V2X2PLR34WLED840MOD   | 34 | 37 | -      |
| 40 | Blauster area-1   | 3fLB       | 32.77   | 250 | 3400 | 0.75 | 0.85 | 3.7797   | 4  | 264.5713 | VENUSNEO1-<br>10V2X2PLR34WLED840MOD   | 34 | 37 | -      |
| 41 | Blauster area-2   | 3fRB       | 32.77   | 250 | 3400 | 0.75 | 0.85 | 3.7797   | 4  | 264.5713 | VENUSNEO1-<br>10V2X2PLR34WLED840MOD   | 34 | 37 | -      |
| 42 | Office store-3    | 3fBROS     | 16.23   | 150 | 3400 | 0.75 | 0.85 | 1.123183 | 1  | 133.549  | VENUSNEO1-<br>10V2X2PLR34WLED840MOD   | 34 | 37 | -      |
| 43 | Office store-4    | 3fTLOS     | 16.23   | 150 | 3400 | 0.75 | 0.85 | 1.123183 | 1  | 133.549  | VENUSNEO1-<br>10V2X2PLR34WLED840MOD   | 34 | 37 | -      |
| 44 | Terrace-1         | 3fTRT      | 341.25  | 150 | 9000 | 0.75 | 0.85 | 8.921569 | 10 | 168.1319 | HEXAHCWPA75WLED850ASYMTGWSWS<br>CIP65 | 75 | 27 | -      |
| 45 | Terrace-2         | 5fTLT      | 30.7    | 150 | 9000 | 0.75 | 0.85 | 0.802614 | 1  | 186.8893 | HEXAHCWPA75WLED850ASYMTGWSWS<br>CIP66 | 75 | 27 | -      |

h. Sixth Floor:

| Sixth Floor Illumination Specifications |                      |        |                        |                                                 |                   |                |                |                         |                         |                         |                                    |            |        |       |
|-----------------------------------------|----------------------|--------|------------------------|-------------------------------------------------|-------------------|----------------|----------------|-------------------------|-------------------------|-------------------------|------------------------------------|------------|--------|-------|
| Serial No.                              | Description          |        | Dimensions             | Required Illumination (E <sub>req</sub> ) (lux) | Lumens Output (φ) | μ <sub>m</sub> | μ <sub>a</sub> | N <sub>calculated</sub> | N <sub>considered</sub> | E <sub>considered</sub> | Catalogue: Havells Office Lighting |            |        |       |
|                                         | Designation          | Code   | Area (m <sup>2</sup> ) |                                                 |                   |                |                |                         |                         |                         | Model                              | Wattage(W) | Pg.no. | Scale |
| 1                                       | Office Area-1        | 6FBLO  | 185                    | 250                                             | 3400              | 0.75           | 0.85           | 21.33794694             | 21                      | 246.040541              | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34         | 37     | -     |
| 2                                       | Office Area-2        | 6FBRO  | 185                    | 250                                             | 3400              | 0.75           | 0.85           | 21.33794694             | 21                      | 246.040541              | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34         | 37     | -     |
| 3                                       | Office Area-3        | 6FTRO  | 138                    | 250                                             | 3400              | 0.75           | 0.85           | 15.91695502             | 15                      | 235.597826              | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34         | 37     | -     |
| 4                                       | Office Area-4        | 6FTLO  | 138                    | 250                                             | 3400              | 0.75           | 0.85           | 15.91695502             | 15                      | 235.597826              | VENUSNEO1-10V2X2PLR34WLED840MOD    | 34         | 37     | -     |
| 5                                       | left RIGHT washroom  | 6FLRW  | 22.953                 | 150                                             | 900               | 0.75           | 0.85           | 6.000784314             | 7                       | 174.977127              | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     | -     |
| 6                                       | Office Washroom-1    | 6FLOW  | 3.46                   | 150                                             | 900               | 0.75           | 0.85           | 0.904575163             | 1                       | 165.823699              | INTEGRAGRANDDLRI0WLED830S          | 12         | 73     | -     |
| 7                                       | Office Washroom-2    | 6FROW  | 3.46                   | 150                                             | 900               | 0.75           | 0.85           | 0.904575163             | 1                       | 165.823699              | INTEGRAGRANDDLRI0WLED830S          | 12         | 73     | -     |
| 8                                       | left LEFT washroom   | 6ELLW  | 22.953                 | 150                                             | 900               | 0.75           | 0.85           | 6.000784314             | 7                       | 174.977127              | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     | -     |
| 9                                       | right RIGHT washroom | 6FRRW  | 22.953                 | 150                                             | 900               | 0.75           | 0.85           | 6.000784314             | 7                       | 174.977127              | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     | -     |
| 10                                      | right LEFT washroom  | 6FRLW  | 22.953                 | 150                                             | 900               | 0.75           | 0.85           | 6.000784314             | 7                       | 174.977127              | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     | -     |
| 11                                      | Staircase-1          | 6FTLLS | 15.98                  | 125                                             | 900               | 0.75           | 0.85           | 3.481481481             | 4                       | 143.617021              | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     | -     |
| 12                                      | Staircase-2          | 6FLBS  | 20                     | 125                                             | 900               | 0.75           | 0.85           | 4.357298475             | 4                       | 114.75                  | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     |       |
| 13                                      | Staircase-3          | 6FLRS  | 20                     | 125                                             | 900               | 0.75           | 0.85           | 4.357298475             | 4                       | 114.75                  | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     |       |
| 14                                      | Staircase-4          | 6FLTS  | 14.3                   | 125                                             | 900               | 0.75           | 0.85           | 3.11546841              | 4                       | 160.48951               | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     |       |
| 15                                      | Staircase-5          | 6FTRRS | 15.98                  | 125                                             | 900               | 0.75           | 0.85           | 3.481481481             | 4                       | 143.617021              | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     | -     |
| 16                                      | Staircase-6          | 6FRBS  | 20                     | 125                                             | 900               | 0.75           | 0.85           | 4.357298475             | 4                       | 114.75                  | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     |       |
| 17                                      | Staircase-7          | 6FRLS  | 26                     | 125                                             | 900               | 0.75           | 0.85           | 5.664488017             | 4                       | 88.2692308              | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     |       |
| 18                                      | Staircase-8          | 6FRTS  | 14.3                   | 125                                             | 900               | 0.75           | 0.85           | 3.11546841              | 4                       | 160.48951               | INTEGRAGRANDDLRI2WLED840S          | 12         | 71     |       |
| 19                                      | Corridor-1           | 6FLC   | 238                    | 150                                             | 1920              | 0.75           | 0.85           | 29.16666667             | 31                      | 159.428571              | PLUTO2X1PLR20WLED840S              | 20         | 53     | -     |

|    |                       |        |         |     |      |      |      |             |    |            |                                  |    |    |        |
|----|-----------------------|--------|---------|-----|------|------|------|-------------|----|------------|----------------------------------|----|----|--------|
| 20 | Corridor-2            | 6FRC   | 238     | 150 | 1920 | 0.75 | 0.85 | 29.16666667 | 31 | 159.428571 | PLUTO2X1PLR20WLED840S            | 20 | 53 | -      |
| 21 | Office-1              | 6FBMO  | 484.964 | 275 | 3400 | 0.75 | 0.85 | 61.5294579  | 58 | 259.225427 | VENUSNEO1-10V2X2PLR34WLED840MOD  | 34 | 37 | -      |
| 22 | Office-2              | 6FTMO  | 372.225 | 275 | 3400 | 0.75 | 0.85 | 47.22577855 | 48 | 279.508362 | VENUSNEO1-10V2X2PLR34WLED840MOD  | 34 | 37 | -      |
| 23 | Washroom              | 6FMLW  | 54.2    | 150 | 810  | 0.75 | 0.85 | 15.74437182 | 16 | 152.435424 | INTEGRANEODLR10WLED830S          | 10 | 71 | 1 by 2 |
| 24 | Washroom              | 6FMRW  | 54.2    | 150 | 810  | 0.75 | 0.85 | 15.74437182 | 16 | 152.435424 | INTEGRANEODLR10WLED830S          | 10 | 71 | 1 by 2 |
| 25 | Terrace-1             | 6FBLT  | 109.1   | 150 | 9000 | 0.75 | 0.85 | 2.852287582 | 3  | 157.768103 | HEXAHCWPA75WLED850ASYMTGSWSCIP65 | 75 | 27 | -      |
| 26 | Terrace-2             | 6FMLT  | 293.16  | 150 | 9000 | 0.75 | 0.85 | 7.664313725 | 8  | 156.569791 | HEXAHCWPA75WLED850ASYMTGSWSCIP66 | 75 | 27 | -      |
| 27 | Terrace-3             | 6FBRT  | 109.1   | 150 | 9000 | 0.75 | 0.85 | 2.852287582 | 3  | 157.768103 | HEXAHCWPA75WLED850ASYMTGSWSCIP65 | 75 | 27 | -      |
| 28 | Terrace-4             | 6FMRT  | 293.16  | 150 | 9000 | 0.75 | 0.85 | 7.664313725 | 8  | 156.569791 | HEXAHCWPA75WLED850ASYMTGSWSCIP66 | 75 | 27 | -      |
| 29 | middle RIGHT washroom | 6FMRW  | 54      | 150 | 900  | 0.75 | 0.85 | 14.11764706 | 16 | 170        | INTEGRAGRANDDLRI2WLED840S        | 12 | 71 | -      |
| 30 | middle LEFT washroom  | 6FMLW  | 54      | 150 | 900  | 0.75 | 0.85 | 14.11764706 | 16 | 170        | INTEGRAGRANDDLRI2WLED840S        | 12 | 71 | -      |
| 31 | Staircase-9           | 6FTMLS | 15.98   | 125 | 900  | 0.75 | 0.85 | 3.481481481 | 4  | 143.617021 | INTEGRAGRANDDLRI2WLED840S        | 12 | 71 | -      |
| 32 | Staircase-10          | 6FMLBS | 20      | 125 | 900  | 0.75 | 0.85 | 4.357298475 | 4  | 114.75     | INTEGRAGRANDDLRI2WLED840S        | 12 | 71 |        |
| 33 | Staircase-11          | 6FMLRS | 20      | 125 | 900  | 0.75 | 0.85 | 4.357298475 | 4  | 114.75     | INTEGRAGRANDDLRI2WLED840S        | 12 | 71 |        |
| 34 | Staircase-12          | 6FMLTS | 14.3    | 125 | 900  | 0.75 | 0.85 | 3.11546841  | 4  | 160.48951  | INTEGRAGRANDDLRI2WLED840S        | 12 | 71 |        |
| 35 | Corridor-3            | 6FMC   | 193.83  | 150 | 1920 | 0.75 | 0.85 | 23.75367647 | 28 | 176.814735 | PLUTO2X1PLR20WLED840S            | 20 | 53 | -      |

## 5. SDB Calculation Table:

a. Basement:

| Details of Basement SDB-1(Below Parking left side) |                      |                    |                   |                                       |               |                                      |                                              |            |
|----------------------------------------------------|----------------------|--------------------|-------------------|---------------------------------------|---------------|--------------------------------------|----------------------------------------------|------------|
| Light and Power Circuit                            | Wattage Per Phase(W) |                    |                   | Total Connected Load                  | Demand Factor | Demand Load                          | Current(I)                                   | MCB Rating |
|                                                    | R                    | Y                  | B                 |                                       |               |                                      |                                              |            |
| LC1                                                | 728                  | 0                  | 0                 | 728                                   | 1             | 728                                  | 3.956521739                                  | 6          |
| LC2                                                | 0                    | 760                | 0                 | 760                                   | 1             | 760                                  | 4.130434783                                  | 6          |
| LC3                                                | 0                    | 0                  | 560               | 560                                   | 1             | 560                                  | 3.043478261                                  | 6          |
| LC4                                                | 496                  | 0                  | 0                 | 496                                   | 1             | 496                                  | 2.695652174                                  | 6          |
| LC5                                                | 0                    | 710                | 0                 | 710                                   | 1             | 710                                  | 3.858695652                                  | 6          |
| PC1                                                | 3000                 | 0                  | 0                 | 3000                                  | 0.5           | 1500                                 | 8.152173913                                  | 16         |
| PC2                                                | 0                    | 8000               | 0                 | 8000                                  | 1             | 8000                                 | 43.47826087                                  | 63         |
| PC3                                                | 0                    | 0                  | 8000              | 8000                                  | 1             | 8000                                 | 43.47826087                                  | 63         |
| PC4                                                | 8000                 | 0                  | 0                 | 8000                                  | 1             | 8000                                 | 43.47826087                                  | 63         |
| PC5                                                | 0                    | 8000               | 0                 | 8000                                  | 1             | 8000                                 | 43.47826087                                  | 63         |
|                                                    | $\Sigma R = 12224$   | $\Sigma Y = 17470$ | $\Sigma B = 8560$ | $\Sigma$ Total Connected Load = 38254 | 0.95          | $\Sigma$ Demand Load = 36754         | $\Sigma I$ (per phase) = 199.75              |            |
|                                                    | 80A RCCB             | 125A RCCB          | 63A RCCB          |                                       |               | $\Sigma$ Demand Load (KW) = 36.754   | $\Sigma I$ (per phase calculated) = 199.75 A |            |
|                                                    | Spare LC 23          | Spare LC 24        | Spare LC 22       |                                       |               | $\Sigma$ Demand Load (KVA) = 45.9425 | I (MCCB Rating) = 79.574 A                   |            |
|                                                    | Spare PC 27,30       | Spare PC 28,31     | Spare PC 26,29    |                                       |               |                                      | 10 Way TPN                                   |            |

| Details of Basement Floor SDB-2(Below parking middle section) |                      |                    |                   |                                       |               |                                    |                                             |            |
|---------------------------------------------------------------|----------------------|--------------------|-------------------|---------------------------------------|---------------|------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                                       | Wattage Per Phase(W) |                    |                   | Total Connected Load                  | Demand Factor | Demand Load                        | Current(I)                                  | MCB Rating |
|                                                               | R                    | Y                  | B                 |                                       |               |                                    |                                             |            |
| LC6                                                           | 606                  | 0                  | 0                 | 606                                   | 1             | 606                                | 3.293478261                                 | 6          |
| LC7                                                           | 0                    | 640                | 0                 | 640                                   | 1             | 640                                | 3.47826087                                  | 6          |
| LC8                                                           | 0                    | 0                  | 650               | 650                                   | 1             | 650                                | 3.532608696                                 | 6          |
| LC9                                                           | 640                  | 0                  | 0                 | 640                                   | 1             | 640                                | 3.47826087                                  | 6          |
| PC6                                                           | 8000                 | 0                  | 0                 | 8000                                  | 1             | 8000                               | 43.47826087                                 | 63         |
| PC7                                                           | 0                    | 8000               | 0                 | 8000                                  | 1             | 8000                               | 43.47826087                                 | 63         |
| PC8                                                           | 0                    | 0                  | 8000              | 8000                                  | 1             | 8000                               | 43.47826087                                 | 63         |
| PC9                                                           | 8000                 | 0                  | 0                 | 8000                                  | 1             | 8000                               | 43.47826087                                 | 63         |
| PC10                                                          | 0                    | 8000               | 0                 | 8000                                  | 1             | 8000                               | 43.47826087                                 | 63         |
|                                                               | $\Sigma R = 17246$   | $\Sigma Y = 16640$ | $\Sigma B = 8650$ | $\Sigma$ Total Connected Load = 42536 | 1             | $\Sigma$ Demand Load = 42536       | $\Sigma I$ (per phase) = 231.174            |            |
|                                                               | 125A RCCB            | 125A RCCB          | 63A RCCB          |                                       |               | $\Sigma$ Demand Load (KW) = 42.536 | $\Sigma I$ (per phase calculated) = 231.714 |            |
|                                                               | Spare LC 27          | Spare LC 25        | Spare LC 26       |                                       |               | $\Sigma$ Demand Load (KVA) = 53.17 | I (MCCB Rating) = 92.093                    |            |
|                                                               | Spare PC 33,36       | Spare PC 34,37     | Spare PC 32,35    |                                       |               |                                    | 10 Way TPN                                  |            |

| Details of Basement Floor SDB-3(Below parking right side) |                      |                    |                   |                                       |               |                                      |                                             |            |
|-----------------------------------------------------------|----------------------|--------------------|-------------------|---------------------------------------|---------------|--------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                                   | Wattage Per Phase(W) |                    |                   | Total Connected Load                  | Demand Factor | Demand Load                          | Current(I)                                  | MCB Rating |
|                                                           | R                    | Y                  | B                 |                                       |               |                                      |                                             |            |
| LC10                                                      | 576                  | 0                  | 0                 | 576                                   | 1             | 576                                  | 3.130434783                                 | 6          |
| LC11                                                      | 0                    | 710                | 0                 | 710                                   | 1             | 710                                  | 3.858695652                                 | 6          |
| LC12                                                      | 0                    | 0                  | 728               | 728                                   | 1             | 728                                  | 3.956521739                                 | 6          |
| LC13                                                      | 680                  | 0                  | 0                 | 680                                   | 1             | 680                                  | 3.695652174                                 | 6          |
| PC11                                                      | 8000                 | 0                  | 0                 | 8000                                  | 1             | 8000                                 | 43.47826087                                 | 16         |
| PC12                                                      | 0                    | 8000               | 0                 | 8000                                  | 1             | 8000                                 | 43.47826087                                 | 16         |
| PC13                                                      | 0                    | 0                  | 8000              | 8000                                  | 1             | 8000                                 | 43.47826087                                 | 16         |
| PC14                                                      | 8000                 | 0                  | 0                 | 8000                                  | 1             | 8000                                 | 43.47826087                                 | 16         |
| PC15                                                      | 0                    | 3000               | 0                 | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
|                                                           | $\Sigma R = 17256$   | $\Sigma Y = 11710$ | $\Sigma B = 8728$ | $\Sigma$ Total Connected Load = 37694 | 0.95          | $\Sigma$ Demand Load = 36194         | $\Sigma I$ (per phase) = 196.707            |            |
|                                                           | 125A RCCB            | 80A RCCB           | 63A RCCB          |                                       |               | $\Sigma$ Demand Load (KW) = 36.194   | $\Sigma I$ (per phase calculated) = 196.707 |            |
|                                                           | Spare LC 30          | Spare LC 28        | Spare LC 29       |                                       |               | $\Sigma$ Demand Load (KVA) = 45.2425 | I (MCCB Rating) = 78.36                     |            |
|                                                           | Spare PC 39,42       | Spare PC 40,43     | Spare PC 38,41    |                                       |               |                                      | 10 Way TPN                                  |            |

| Details of Basement Floor SDB-4 (Upper parking right side) |                      |                    |                   |                                       |               |                                     |                                             |            |
|------------------------------------------------------------|----------------------|--------------------|-------------------|---------------------------------------|---------------|-------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                                    | Wattage Per Phase(W) |                    |                   | Total Connected Load                  | Demand Factor | Demand Load                         | Current(I)                                  | MCB Rating |
|                                                            | R                    | Y                  | B                 |                                       |               |                                     |                                             |            |
| LC14                                                       | 688                  | 0                  | 0                 | 688                                   | 1             | 688                                 | 3.739130435                                 | 6          |
| LC15                                                       | 0                    | 560                | 0                 | 560                                   | 1             | 612                                 | 3.326086957                                 | 6          |
| LC16                                                       | 0                    | 0                  | 560               | 560                                   | 1             | 748                                 | 4.065217391                                 | 6          |
| LC17                                                       | 560                  | 0                  | 0                 | 560                                   | 1             | 284                                 | 1.543478261                                 | 6          |
| PC16                                                       | 3000                 | 0                  | 0                 | 3000                                  | 0.5           | 264                                 | 1.434782609                                 | 6          |
| PC17                                                       | 0                    | 8000               | 0                 | 8000                                  | 1             | 8000                                | 43.47826087                                 | 63         |
| PC18                                                       | 0                    | 0                  | 8000              | 8000                                  | 1             | 8000                                | 43.47826087                                 | 63         |
| PC19                                                       | 8000                 | 0                  | 0                 | 8000                                  | 1             | 8000                                | 43.47826087                                 | 63         |
| PC20                                                       | 0                    | 8000               | 0                 | 8000                                  | 1             | 8000                                | 43.47826087                                 | 63         |
|                                                            | $\Sigma R = 12248$   | $\Sigma Y = 16560$ | $\Sigma B = 8560$ | $\Sigma$ Total Connected Load = 37368 | 0.95          | $\Sigma$ Demand Load = 34596        | $\Sigma I$ (per phase) = 188.022            |            |
|                                                            | 80A RCCB             | 125A RCCB          | 63A RCCB          |                                       |               | $\Sigma$ Demand Load (KW) = 34.596  | $\Sigma I$ (per phase calculated) = 188.022 |            |
|                                                            | Spare LC 33          | Spare LC 31        | Spare LC 32       |                                       |               | $\Sigma$ Demand Load (KVA) = 43.245 | I (MCCB Rating) = 74.9                      |            |
|                                                            | Spare PC 45,48       | Spare PC 46,49     | Spare PC 44,47    |                                       |               |                                     | 10 Way TPN                                  |            |

| Details of Basement Floor SDB-5(Upper parking left side) |                      |                    |                   |                                       |               |                                   |                                             |            |
|----------------------------------------------------------|----------------------|--------------------|-------------------|---------------------------------------|---------------|-----------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                                  | Wattage Per Phase(W) |                    |                   | Total Connected Load                  | Demand Factor | Demand Load                       | Current(I)                                  | MCB Rating |
|                                                          | R                    | Y                  | B                 |                                       |               |                                   |                                             |            |
| LC18                                                     | 560                  | 0                  | 0                 | 560                                   | 1             | 560                               | 3.043478261                                 | 6          |
| LC19                                                     | 0                    | 560                | 0                 | 560                                   | 1             | 560                               | 3.043478261                                 | 6          |
| LC20                                                     | 0                    | 0                  | 560               | 560                                   | 1             | 560                               | 3.043478261                                 | 6          |
| LC21                                                     | 560                  | 0                  | 0                 | 560                                   | 1             | 560                               | 3.043478261                                 | 6          |
| PC21                                                     | 8000                 | 0                  | 0                 | 8000                                  | 1             | 8000                              | 43.47826087                                 | 63         |
| PC22                                                     | 0                    | 8000               | 0                 | 8000                                  | 1             | 8000                              | 43.47826087                                 | 63         |
| PC23                                                     | 0                    | 0                  | 8000              | 8000                                  | 1             | 8000                              | 43.47826087                                 | 63         |
| PC24                                                     | 8000                 | 0                  | 0                 | 8000                                  | 1             | 8000                              | 43.47826087                                 | 63         |
| PC25                                                     | 0                    | 8000               | 0                 | 8000                                  | 1             | 8000                              | 43.47826087                                 | 63         |
|                                                          | $\Sigma R = 17120$   | $\Sigma Y = 16560$ | $\Sigma B = 8560$ | $\Sigma$ Total Connected Load = 42240 | 1             | $\Sigma$ Demand Load = 42240      | $\Sigma I$ (per phase) = 229.565            |            |
|                                                          | 125A RCCB            | 125A RCCB          | 63A RCCB          |                                       |               | $\Sigma$ Demand Load (KW) = 42.24 | $\Sigma I$ (per phase calculated) = 229.565 |            |
|                                                          | Spare LC 36          | Spare LC 34        | Spare LC 35       |                                       |               | $\Sigma$ Demand Load (KVA) = 52.8 | I (MCCB Rating) = 91.45                     |            |
|                                                          | Spare PC 51,54       | Spare PC 52,55     | Spare PC 50,53    |                                       |               |                                   | 10 ay TPN                                   |            |

b. Ground Floor:

| Details of Ground Floor SDB-1 (lower left area) |                       |                    |                    |                                         |                 |                                      |                                             |            |
|-------------------------------------------------|-----------------------|--------------------|--------------------|-----------------------------------------|-----------------|--------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                         | Wattage Per Phase (W) |                    |                    | Total Connected Load                    | Demand Factor   | Demand Load                          | Current(I)                                  | MCB Rating |
|                                                 | R                     | Y                  | B                  |                                         |                 |                                      |                                             |            |
| LC1                                             | 756                   | 0                  | 0                  | 756                                     | 1               | 756                                  | 4.109                                       | 6          |
| LC2                                             | 0                     | 588                | 0                  | 588                                     | 1               | 588                                  | 3.196                                       | 6          |
| LC7                                             | 0                     | 0                  | 760                | 760                                     | 1               | 760                                  | 4.130                                       | 6          |
| LC8                                             | 298                   | 0                  | 0                  | 298                                     | 1               | 298                                  | 1.620                                       | 6          |
| LC9                                             | 0                     | 504                | 0                  | 504                                     | 1               | 504                                  | 2.739                                       | 6          |
| LC10                                            | 0                     | 0                  | 504                | 504                                     | 1               | 504                                  | 2.739                                       | 6          |
| LC11                                            | 672                   | 0                  | 0                  | 672                                     | 1               | 672                                  | 3.652                                       | 6          |
| PC1                                             | 0                     | 3000               | 0                  | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC2                                             | 0                     | 0                  | 3000               | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC3                                             | 3000                  | 0                  | 0                  | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC4                                             | 0                     | 3000               | 0                  | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC5                                             | 0                     | 0                  | 3000               | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC6                                             | 3000                  | 0                  | 0                  | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC22                                            | 0                     | 3000               | 0                  | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC24                                            | 0                     | 0                  | 3000               | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC25                                            | 3000                  | 0                  | 0                  | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC26                                            | 0                     | 3000               | 0                  | 3000                                    | 0.4             | 1200                                 | 6.522                                       | 16         |
| PC27                                            | 0                     | 0                  | 3000               | 3000                                    | 0.4             | 1200                                 | 6.522                                       | 16         |
| PC28                                            | 2000                  | 0                  | 0                  | 2000                                    | 0.4             | 800                                  | 4.348                                       | 16         |
| PC29                                            | 0                     | 3000               | 0                  | 3000                                    | 0.4             | 1200                                 | 6.522                                       | 16         |
| PC30                                            | 0                     | 0                  | 3000               | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC31                                            | 3000                  | 0                  | 0                  | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC32                                            | 0                     | 3000               | 0                  | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC33                                            | 0                     | 0                  | 3000               | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC34                                            | 3000                  | 0                  | 0                  | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC35                                            | 0                     | 3000               | 0                  | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC40                                            | 0                     | 0                  | 3000               | 3000                                    | 0.5             | 1500                                 | 8.152                                       | 16         |
| 7 LCs, 20 PCs                                   | $\Sigma R = 18726$    | $\Sigma Y = 22092$ | $\Sigma B = 22264$ | $\Sigma$ Total Connected Load = 63082 W | Wt. Avg = 0.515 | $\Sigma$ Demand Load = 32482         | $\Sigma I$ (per phase) = 176.533            |            |
|                                                 | 63A RCCB              | 80A RCCB           | 80A RCCB           |                                         |                 | $\Sigma$ Demand Load (KW) = 32.482   | $\Sigma I$ (per phase calculated) = 176.533 |            |
|                                                 | Spare LC 33           | Spare LC 34        | Spare LC 35        |                                         |                 | $\Sigma$ Demand Load (KVA) = 40.6025 | I (MCCB Rating) = 70.33 A                   |            |
|                                                 | Spare PC 100,101      | Spare PC 102,103   | Spare PC 104,105   |                                         |                 |                                      | 16 Way TPN                                  |            |

| Details of Ground Floor SDB-2 (upper left area) |                       |                    |                    |                                       |                 |                                    |                                             |            |
|-------------------------------------------------|-----------------------|--------------------|--------------------|---------------------------------------|-----------------|------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit Circuit                 | Wattage Per Phase (W) |                    |                    | Total Connected Load                  | Demand Factor   | Demand Load                        | Current(I)                                  | MCB Rating |
|                                                 | R                     | Y                  | B                  |                                       |                 |                                    |                                             |            |
| LC3                                             | 672                   | 0                  | 0                  | 672                                   | 1               | 672                                | 3.652                                       | 6          |
| LC4                                             | 0                     | 504                | 0                  | 504                                   | 1               | 504                                | 2.739                                       | 6          |
| LC5                                             | 0                     | 0                  | 648                | 648                                   | 1               | 648                                | 3.522                                       | 6          |
| LC6                                             | 600                   | 0                  | 0                  | 600                                   | 1               | 600                                | 3.261                                       | 6          |
| LC12                                            | 0                     | 756                | 0                  | 756                                   | 1               | 756                                | 4.109                                       | 6          |
| PC7                                             | 0                     | 0                  | 3000               | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC8                                             | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC9                                             | 0                     | 3000               | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC10                                            | 0                     | 0                  | 3000               | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC11                                            | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC12                                            | 0                     | 3000               | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC13                                            | 0                     | 0                  | 3000               | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC14                                            | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC15                                            | 0                     | 2000               | 0                  | 2000                                  | 0.5             | 1000                               | 5.435                                       | 16         |
| PC16                                            | 0                     | 0                  | 2000               | 2000                                  | 0.4             | 800                                | 4.348                                       | 16         |
| PC17                                            | 3000                  | 0                  | 0                  | 3000                                  | 0.4             | 1200                               | 6.522                                       | 16         |
| PC18                                            | 0                     | 3000               | 0                  | 3000                                  | 0.4             | 1200                               | 6.522                                       | 16         |
| PC19                                            | 0                     | 0                  | 3000               | 3000                                  | 0.4             | 1200                               | 6.522                                       | 16         |
| PC20                                            | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC21                                            | 0                     | 3000               | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC23                                            | 0                     | 0                  | 3000               | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC36                                            | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC37                                            | 0                     | 3000               | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC38                                            | 0                     | 0                  | 3000               | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| PC39                                            | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16         |
| 5 LCs, 20 PCs                                   | $\Sigma R = 22272$    | $\Sigma Y = 18260$ | $\Sigma B = 20648$ | $\Sigma$ Total Connected Load = 61180 | Wt. Avg = 0.508 | $\Sigma$ Demand Load = 31080       | $\Sigma I$ (per phase) = 168.913            |            |
|                                                 | 80A RCCB              | 63A RCCB           | 80A RCCB           |                                       |                 | $\Sigma$ Demand Load (KW) = 31.08  | $\Sigma I$ (per phase calculated) = 168.913 |            |
|                                                 | Spare LC 36           | Spare LC 37        | Spare LC 38        |                                       |                 | $\Sigma$ Demand Load (KVA) = 38.85 | I (MCCB Rating) = 67.29 A                   |            |
|                                                 | Spare PC 106,107      | Spare PC 108,109   | Spare PC 110,111   |                                       |                 |                                    | 14 Way TPN                                  |            |

| Details of Ground Floor SDB-3 (middle area) ` |                       |                       |                       |                                                |                    |                                           |                                                |            |
|-----------------------------------------------|-----------------------|-----------------------|-----------------------|------------------------------------------------|--------------------|-------------------------------------------|------------------------------------------------|------------|
| Light and Power Circuit                       | Wattage Per Phase (W) |                       |                       | Total Connected Load                           | Demand Factor      | Demand Load                               | Current(I)                                     | MCB Rating |
|                                               | R                     | Y                     | B                     |                                                |                    |                                           |                                                |            |
| LC13                                          | 756                   | 0                     | 0                     | 756                                            | 1                  | 756                                       | 4.109                                          | 6          |
| LC14                                          | 0                     | 680                   | 0                     | 680                                            | 1                  | 680                                       | 3.696                                          | 6          |
| LC15                                          | 0                     | 0                     | 720                   | 720                                            | 1                  | 720                                       | 3.913                                          | 6          |
| LC16                                          | 672                   | 0                     | 0                     | 672                                            | 1                  | 672                                       | 3.652                                          | 6          |
| LC17                                          | 0                     | 756                   | 0                     | 756                                            | 1                  | 756                                       | 4.109                                          | 6          |
| LC18                                          | 0                     | 0                     | 756                   | 756                                            | 1                  | 756                                       | 4.109                                          | 6          |
| LC19                                          | 672                   | 0                     | 0                     | 672                                            | 1                  | 672                                       | 3.652                                          | 6          |
| PC41                                          | 0                     | 3000                  | 0                     | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC42                                          | 0                     | 0                     | 3000                  | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC43                                          | 2000                  | 0                     | 0                     | 2000                                           | 0.5                | 1000                                      | 5.435                                          | 16         |
| PC44                                          | 0                     | 3000                  | 0                     | 3000                                           | 0.4                | 1200                                      | 6.522                                          | 16         |
| PC45                                          | 0                     | 0                     | 3000                  | 3000                                           | 0.4                | 1200                                      | 6.522                                          | 16         |
| PC46                                          | 3000                  | 0                     | 0                     | 3000                                           | 0.4                | 1200                                      | 6.522                                          | 16         |
| PC47                                          | 0                     | 3000                  | 0                     | 3000                                           | 0.4                | 1200                                      | 6.522                                          | 16         |
| PC48                                          | 0                     | 0                     | 3000                  | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC49                                          | 3000                  | 0                     | 0                     | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC50                                          | 0                     | 3000                  | 0                     | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC51                                          | 0                     | 0                     | 3000                  | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC52                                          | 3000                  | 0                     | 0                     | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC53                                          | 0                     | 3000                  | 0                     | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC54                                          | 0                     | 0                     | 3000                  | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC55                                          | 3000                  | 0                     | 0                     | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC56                                          | 0                     | 3000                  | 0                     | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC57                                          | 0                     | 0                     | 3000                  | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC58                                          | 3000                  | 0                     | 0                     | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC59                                          | 0                     | 3000                  | 0                     | 3000                                           | 0.5                | 1500                                      | 8.152                                          | 16         |
| PC60                                          | 0                     | 0                     | 2000                  | 2000                                           | 0.5                | 1000                                      | 5.435                                          | 16         |
| PC61                                          | 3000                  | 0                     | 0                     | 3000                                           | 0.4                | 1200                                      | 6.522                                          | 16         |
| PC62                                          | 0                     | 3000                  | 0                     | 3000                                           | 0.4                | 1200                                      | 6.522                                          | 16         |
| PC63                                          | 0                     | 0                     | 3000                  | 3000                                           | 0.4                | 1200                                      | 6.522                                          | 16         |
| PC64                                          | 3000                  | 0                     | 0                     | 3000                                           | 0.4                | 1200                                      | 6.522                                          | 16         |
| 7 LCs,<br>24 PCs                              | $\Sigma R =$<br>25100 | $\Sigma Y =$<br>25436 | $\Sigma B =$<br>24476 | $\Sigma$ Total<br>Connected<br>Load =<br>75012 | Wt. Avg =<br>0.501 | $\Sigma$ Demand<br>Load = 37612           | $\Sigma I$ (per phase) =<br>204.413            |            |
|                                               | 100A<br>RCCB          | 100A<br>RCCB          | 80A<br>RCCB           |                                                |                    | $\Sigma$ Demand<br>Load (KW) =<br>37.612  | $\Sigma I$ (per phase<br>calculated) = 204.413 |            |
|                                               | Spare LC<br>39        | Spare LC<br>40        | Spare LC<br>41        |                                                |                    | $\Sigma$ Demand<br>Load (KVA) =<br>47.015 | I (MCCB Rating) =<br>81.43 A                   |            |
|                                               | Spare PC<br>112,113   | Spare PC<br>114,115   | Spare PC<br>116,117   |                                                |                    |                                           | 16 Way TPN                                     |            |

| Details of Ground Floor SDB-4 (lower right area) |                       |                    |                    |                                      |                 |                                      |                                             |            |
|--------------------------------------------------|-----------------------|--------------------|--------------------|--------------------------------------|-----------------|--------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                          | Wattage Per Phase (W) |                    |                    | Total Connected Load                 | Demand Factor   | Demand Load                          | Current(I)                                  | MCB Rating |
|                                                  | R                     | Y                  | B                  |                                      |                 |                                      |                                             |            |
| LC22                                             | 760                   | 0                  | 0                  | 760                                  | 1               | 760                                  | 4.130                                       | 6          |
| LC23                                             | 0                     | 298                | 0                  | 298                                  | 1               | 298                                  | 1.620                                       | 6          |
| LC24                                             | 0                     | 0                  | 504                | 504                                  | 1               | 504                                  | 2.739                                       | 6          |
| LC25                                             | 504                   | 0                  | 0                  | 504                                  | 1               | 504                                  | 2.739                                       | 6          |
| LC26                                             | 0                     | 672                | 0                  | 672                                  | 1               | 672                                  | 3.652                                       | 6          |
| LC27                                             | 0                     | 0                  | 504                | 504                                  | 1               | 504                                  | 2.739                                       | 6          |
| LC28                                             | 504                   | 0                  | 0                  | 504                                  | 1               | 504                                  | 2.739                                       | 6          |
| LC29                                             | 0                     | 672                | 0                  | 672                                  | 1               | 672                                  | 3.652                                       | 6          |
| PC65                                             | 0                     | 0                  | 3000               | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC66                                             | 3000                  | 0                  | 0                  | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC67                                             | 0                     | 3000               | 0                  | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC68                                             | 0                     | 0                  | 3000               | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC69                                             | 3000                  | 0                  | 0                  | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC70                                             | 0                     | 3000               | 0                  | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC75                                             | 0                     | 0                  | 3000               | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC76                                             | 3000                  | 0                  | 0                  | 3000                                 | 0.4             | 1200                                 | 6.522                                       | 16         |
| PC77                                             | 0                     | 3000               | 0                  | 3000                                 | 0.4             | 1200                                 | 6.522                                       | 16         |
| PC78                                             | 0                     | 0                  | 2000               | 2000                                 | 0.4             | 800                                  | 4.348                                       | 16         |
| PC79                                             | 3000                  | 0                  | 0                  | 3000                                 | 0.4             | 1200                                 | 6.522                                       | 16         |
| PC80                                             | 0                     | 3000               | 0                  | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC93                                             | 0                     | 0                  | 3000               | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC94                                             | 3000                  | 0                  | 0                  | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC95                                             | 0                     | 3000               | 0                  | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC96                                             | 0                     | 0                  | 3000               | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC97                                             | 3000                  | 0                  | 0                  | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC98                                             | 0                     | 3000               | 0                  | 3000                                 | 0.5             | 1500                                 | 8.152                                       | 16         |
| 8 LCs, 18 PCs                                    | $\Sigma R = 19768$    | $\Sigma Y = 19642$ | $\Sigma B = 18008$ | $\Sigma$ Total Connected Load = 7418 | Wt. Avg = 0.519 | $\Sigma$ Demand Load = 29818         | $\Sigma I$ (per phase) = 162.054            |            |
|                                                  | 80A RCCB              | 80A RCCB           | 63A RCCB           |                                      |                 | $\Sigma$ Demand Load (KW) = 29.818   | $\Sigma I$ (per phase calculated) = 162.054 |            |
|                                                  | Spare LC 42           | Spare LC 43        | Spare LC 44        |                                      |                 | $\Sigma$ Demand Load (KVA) = 37.2725 | I (MCCB Rating) = 64.56 A                   |            |
|                                                  | Spare PC 118,119      | Spare PC 120,121   | Spare PC 122,123   |                                      |                 |                                      | 14 Way TPN                                  |            |

| Details of Ground Floor SDB#5 (upper right area) |                       |                    |                    |                                      |                 |                                    |                                             |            |
|--------------------------------------------------|-----------------------|--------------------|--------------------|--------------------------------------|-----------------|------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                          | Wattage Per Phase (W) |                    |                    | Total Connected Load                 | Demand Factor   | Demand Load                        | Current(I)                                  | MCB Rating |
|                                                  | R                     | Y                  | B                  |                                      |                 |                                    |                                             |            |
| LC20                                             | 672                   | 0                  | 0                  | 672                                  | 1               | 672                                | 3.652                                       | 6          |
| LC21                                             | 0                     | 720                | 0                  | 720                                  | 1               | 720                                | 3.913                                       | 6          |
| LC30                                             | 0                     | 0                  | 672                | 672                                  | 1               | 672                                | 3.652                                       | 6          |
| LC31                                             | 504                   | 0                  | 0                  | 504                                  | 1               | 504                                | 2.739                                       | 6          |
| LC32                                             | 0                     | 504                | 0                  | 504                                  | 1               | 504                                | 2.739                                       | 6          |
| PC71                                             | 0                     | 0                  | 3000               | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| PC72                                             | 3000                  | 0                  | 0                  | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| PC73                                             | 0                     | 3000               | 0                  | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| PC74                                             | 0                     | 0                  | 3000               | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| PC81                                             | 3000                  | 0                  | 0                  | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| PC82                                             | 0                     | 3000               | 0                  | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| PC83                                             | 0                     | 0                  | 3000               | 3000                                 | 0.4             | 1200                               | 6.522                                       | 16         |
| PC84                                             | 2000                  | 0                  | 0                  | 2000                                 | 0.4             | 800                                | 4.348                                       | 16         |
| PC85                                             | 0                     | 3000               | 0                  | 3000                                 | 0.4             | 1200                               | 6.522                                       | 16         |
| PC86                                             | 0                     | 0                  | 3000               | 3000                                 | 0.4             | 1200                               | 6.522                                       | 16         |
| PC87                                             | 3000                  | 0                  | 0                  | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| PC88                                             | 0                     | 3000               | 0                  | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| PC89                                             | 0                     | 0                  | 3000               | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| PC90                                             | 3000                  | 0                  | 0                  | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| PC91                                             | 0                     | 3000               | 0                  | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| PC92                                             | 0                     | 0                  | 3000               | 3000                                 | 0.5             | 1500                               | 8.152                                       | 16         |
| 5 LCs, 16 PCs                                    | $\Sigma R = 15176$    | $\Sigma Y = 16224$ | $\Sigma B = 18672$ | $\Sigma$ Total Connected Load = 3072 | Wt. Avg = 0.509 | $\Sigma$ Demand Load = 25472       | $\Sigma I$ (per phase) = 138.435            |            |
|                                                  | 63A RCCB              | 63A RCCB           | 63A RCCB           |                                      |                 | $\Sigma$ Demand Load (KW) = 25.472 | $\Sigma I$ (per phase calculated) = 138.435 |            |
|                                                  | Spare LC 45           | Spare LC 46        | Spare LC 47        |                                      |                 | $\Sigma$ Demand Load (KVA) = 31.84 | I (MCCB Rating) = 55.15 A                   |            |
|                                                  | Spare PC 124,125      | Spare PC 126,127   | Spare PC 128,129   |                                      |                 |                                    | 12 Way TPN                                  |            |

### c. First Floor

| Details of First Floor SDB-1 (lower left area) |                       |      |      |                      |               |             |            |            |
|------------------------------------------------|-----------------------|------|------|----------------------|---------------|-------------|------------|------------|
| Light and Power Circuit                        | Wattage Per Phase (W) |      |      | Total Connected Load | Demand Factor | Demand Load | Current(I) | MCB Rating |
|                                                | R                     | Y    | B    |                      |               |             |            |            |
| LC1                                            | 792                   | 0    | 0    | 792                  | 1             | 792         | 4.304      | 6          |
| LC2                                            | 0                     | 656  | 0    | 656                  | 1             | 656         | 3.565      | 6          |
| LC7                                            | 0                     | 0    | 520  | 520                  | 1             | 520         | 2.826      | 6          |
| LC8                                            | 408                   | 0    | 0    | 408                  | 1             | 408         | 2.217      | 6          |
| LC9                                            | 0                     | 720  | 0    | 720                  | 1             | 720         | 3.913      | 6          |
| LC10                                           | 0                     | 0    | 658  | 658                  | 1             | 658         | 3.576      | 6          |
| LC11                                           | 588                   | 0    | 0    | 588                  | 1             | 588         | 3.196      | 6          |
| LC12                                           | 0                     | 442  | 0    | 442                  | 1             | 442         | 2.402      | 6          |
| PC1                                            | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC2                                            | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC3                                            | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC4                                            | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC5                                            | 3000                  | 0    | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC6                                            | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC7                                            | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC8                                            | 3000                  | 0    | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC9                                            | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC10                                           | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC11                                           | 3000                  | 0    | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC12                                           | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC13                                           | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC14                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC15                                           | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC16                                           | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC17                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC18                                           | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC19                                           | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC20                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC46                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC47                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC48                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC49                                           | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC50                                           | 0                     | 0    | 2000 | 2000                 | 0.4           | 800         | 4.348      | 16         |
| PC51                                           | 3000                  | 0    | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC52                                           | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC53                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC54                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC55                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |

|                      |                         |                         |                         |                                        |                        |                                    |                                            |    |
|----------------------|-------------------------|-------------------------|-------------------------|----------------------------------------|------------------------|------------------------------------|--------------------------------------------|----|
| PC56                 | 0                       | 0                       | 3000                    | 3000                                   | 0.5                    | 1500                               | 8.152                                      | 16 |
| PC57                 | 3000                    | 0                       | 0                       | 3000                                   | 0.5                    | 1500                               | 8.152                                      | 16 |
| PC58                 | 0                       | 3000                    | 0                       | 3000                                   | 0.4                    | 1200                               | 6.522                                      | 16 |
| PC59                 | 0                       | 0                       | 3000                    | 3000                                   | 0.4                    | 1200                               | 6.522                                      | 16 |
| PC60                 | 3000                    | 0                       | 0                       | 3000                                   | 0.4                    | 1200                               | 6.522                                      | 16 |
| PC61                 | 0                       | 3000                    | 0                       | 3000                                   | 0.4                    | 1200                               | 6.522                                      | 16 |
| PC62                 | 0                       | 0                       | 3000                    | 3000                                   | 0.4                    | 1200                               | 6.522                                      | 16 |
| PC63                 | 3000                    | 0                       | 0                       | 3000                                   | 0.5                    | 1500                               | 8.152                                      | 16 |
| PC64                 | 0                       | 3000                    | 0                       | 3000                                   | 0.5                    | 1500                               | 8.152                                      | 16 |
| PC65                 | 0                       | 0                       | 3000                    | 3000                                   | 0.4                    | 1200                               | 6.522                                      | 16 |
| PC66                 | 3000                    | 0                       | 0                       | 3000                                   | 0.4                    | 1200                               | 6.522                                      | 16 |
| PC67                 | 0                       | 3000                    | 0                       | 3000                                   | 0.4                    | 1200                               | 6.522                                      | 16 |
| PC69                 | 0                       | 0                       | 3000                    | 3000                                   | 0.4                    | 1200                               | 6.522                                      | 16 |
| <b>8 LCs, 43 PCs</b> | <b>ΣR = 43788</b>       | <b>ΣY = 43818</b>       | <b>ΣB = 45178</b>       | <b>Σ Total Connected Load = 132784</b> | <b>Wt. Avg = 0.453</b> | <b>Σ Demand Load = 60184</b>       | <b>ΣI (per phase) = 327.087</b>            |    |
|                      | <b>125A RCCB</b>        | <b>125A RCCB</b>        | <b>125A RCCB</b>        |                                        |                        | <b>Σ Demand Load (KW) = 60.184</b> | <b>ΣI (per phase calculated) = 327.087</b> |    |
|                      | <b>Spare LC 43</b>      | <b>Spare LC 44</b>      | <b>Spare LC 45</b>      |                                        |                        | <b>Σ Demand Load (KVA) = 75.23</b> | <b>I (MCCB Rating) = 119.44 A</b>          |    |
|                      | <b>Spare PC 109,110</b> | <b>Spare PC 111,112</b> | <b>Spare PC 113,114</b> |                                        |                        |                                    | <b>24 Way TPN</b>                          |    |

| Details of First Floor SDB-2 (upper left area) |                       |      |      |                      |               |             |            |            |
|------------------------------------------------|-----------------------|------|------|----------------------|---------------|-------------|------------|------------|
| Light and Power Circuit                        | Wattage Per Phase (W) |      |      | Total Connected Load | Demand Factor | Demand Load | Current(I) | MCB Rating |
|                                                | R                     | Y    | B    |                      |               |             |            |            |
| LC3                                            | 544                   | 0    | 0    | 544                  | 1             | 544         | 2.957      | 6          |
| LC4                                            | 0                     | 792  | 0    | 792                  | 1             | 792         | 4.304      | 6          |
| LC5                                            | 0                     | 0    | 380  | 380                  | 1             | 380         | 2.065      | 6          |
| LC6                                            | 540                   | 0    | 0    | 540                  | 1             | 540         | 2.935      | 6          |
| LC13                                           | 0                     | 578  | 0    | 578                  | 1             | 578         | 3.141      | 6          |
| PC21                                           | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC22                                           | 3000                  | 0    | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC23                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC24                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC25                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC26                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC27                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC28                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC29                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC30                                           | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC31                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC32                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC33                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC34                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC35                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC36                                           | 0                     | 0    | 2000 | 2000                 | 0.5           | 1000        | 5.435      | 16         |
| PC37                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC38                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC39                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC40                                           | 3000                  | 0    | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC41                                           | 0                     | 2000 | 0    | 2000                 | 0.4           | 800         | 4.348      | 16         |
| PC42                                           | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC43                                           | 3000                  | 0    | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC44                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC45                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC68                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC70                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC71                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC72                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC73                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC74                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC75                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC76                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC77                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC78                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |

|               |                    |                    |                    |                                        |                 |                                      |                                             |    |
|---------------|--------------------|--------------------|--------------------|----------------------------------------|-----------------|--------------------------------------|---------------------------------------------|----|
| PC79          | 0                  | 3000               | 0                  | 3000                                   | 0.5             | 1500                                 | 8.152                                       | 16 |
| 5 LCs, 36 PCs | $\Sigma R = 37084$ | $\Sigma Y = 36370$ | $\Sigma B = 35380$ | $\Sigma$ Total Connected Load = 108834 | Wt. Avg = 0.495 | $\Sigma$ Demand Load = 53834         | $\Sigma I$ (per phase) = 292.576            |    |
|               | 125A RCCB          | 125A RCCB          | 125A RCCB          |                                        |                 | $\Sigma$ Demand Load (KW) = 53.834   | $\Sigma I$ (per phase calculated) = 292.576 |    |
|               | Spare LC 43        | Spare LC 44        | Spare LC 45        |                                        |                 | $\Sigma$ Demand Load (KVA) = 67.2925 | I (MCCB Rating) = 106.84 A                  |    |
|               | Spare PC 109,110   | Spare PC 111,112   | Spare PC 113,114   |                                        |                 |                                      | 20 Way TPN                                  |    |

| Details of First Floor SDB-3 (middle area) |                       |      |      |                      |               |             |            |            |
|--------------------------------------------|-----------------------|------|------|----------------------|---------------|-------------|------------|------------|
| Light and Power Circuit                    | Wattage Per Phase (W) |      |      | Total Connected Load | Demand Factor | Demand Load | Current(I) | MCB Rating |
|                                            | R                     | Y    | B    |                      |               |             |            |            |
| LC14                                       | 626                   | 0    | 0    | 626                  | 1             | 626         | 3.402      | 6          |
| LC15                                       | 0                     | 680  | 0    | 680                  | 1             | 680         | 3.696      | 6          |
| LC16                                       | 0                     | 0    | 720  | 720                  | 1             | 720         | 3.913      | 6          |
| LC17                                       | 672                   | 0    | 0    | 672                  | 1             | 672         | 3.652      | 6          |
| LC18                                       | 0                     | 756  | 0    | 756                  | 1             | 756         | 4.109      | 6          |
| LC19                                       | 0                     | 0    | 756  | 756                  | 1             | 756         | 4.109      | 6          |
| LC20                                       | 448                   | 0    | 0    | 448                  | 1             | 448         | 2.435      | 6          |
| PC80                                       | 0                     | 2000 | 0    | 2000                 | 0.5           | 1000        | 5.435      | 16         |
| PC81                                       | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC82                                       | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC83                                       | 0                     | 2000 | 0    | 2000                 | 0.5           | 1000        | 5.435      | 16         |
| PC84                                       | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC85                                       | 2000                  | 0    | 0    | 2000                 | 0.5           | 1000        | 5.435      | 16         |
| PC86                                       | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC87                                       | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC88                                       | 3000                  | 0    | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC89                                       | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC90                                       | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC91                                       | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC92                                       | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC93                                       | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC94                                       | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC95                                       | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC96                                       | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC97                                       | 3000                  | 0    | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC98                                       | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC99                                       | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC100                                      | 3000                  | 0    | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC101                                      | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC102                                      | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |

|               |                    |                    |                    |                                       |                 |                                      |                                             |    |
|---------------|--------------------|--------------------|--------------------|---------------------------------------|-----------------|--------------------------------------|---------------------------------------------|----|
| PC103         | 3000               | 0                  | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16 |
| PC104         | 0                  | 3000               | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16 |
| PC105         | 0                  | 0                  | 3000               | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16 |
| 7 LCs, 26 PCs | $\Sigma R = 24746$ | $\Sigma Y = 26436$ | $\Sigma B = 28476$ | $\Sigma$ Total Connected Load = 79658 | Wt. Avg = 0.499 | $\Sigma$ Demand Load = 39758         | $\Sigma I$ (per phase) = 216.076            |    |
|               | 100A RCCB          | 100A RCCB          | 80A RCCB           |                                       |                 | $\Sigma$ Demand Load (KW) = 39.758   | $\Sigma I$ (per phase calculated) = 216.076 |    |
|               | Spare LC 43        | Spare LC 44        | Spare LC 45        |                                       |                 | $\Sigma$ Demand Load (KVA) = 49.6975 | I (MCCB Rating) = 86.08 A                   |    |
|               | Spare PC 109,110   | Spare PC 111,112   | Spare PC 113,114   |                                       |                 |                                      | 18 Way TPN                                  |    |

| Details of First Floor SDB-4 (lower right area) |                       |      |      |                      |               |             |            |            |
|-------------------------------------------------|-----------------------|------|------|----------------------|---------------|-------------|------------|------------|
| Light and Power Circuit                         | Wattage Per Phase (W) |      |      | Total Connected Load | Demand Factor | Demand Load | Current(I) | MCB Rating |
|                                                 | R                     | Y    | B    |                      |               |             |            |            |
| LC23                                            | 760                   | 0    | 0    | 760                  | 1             | 760         | 4.130      | 6          |
| LC24                                            | 0                     | 672  | 0    | 672                  | 1             | 672         | 3.652      | 6          |
| LC25                                            | 0                     | 0    | 504  | 504                  | 1             | 504         | 2.739      | 6          |
| LC26                                            | 504                   | 0    | 0    | 504                  | 1             | 504         | 2.739      | 6          |
| LC27                                            | 0                     | 374  | 0    | 374                  | 1             | 374         | 2.033      | 6          |
| LC28                                            | 0                     | 0    | 588  | 588                  | 1             | 588         | 3.196      | 6          |
| LC29                                            | 510                   | 0    | 0    | 510                  | 1             | 510         | 2.772      | 6          |
| LC30                                            | 0                     | 572  | 0    | 572                  | 1             | 572         | 3.109      | 6          |
| PC125                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC126                                           | 2000                  | 0    | 0    | 2000                 | 0.5           | 1000        | 5.435      | 16         |
| PC127                                           | 0                     | 2000 | 0    | 2000                 | 0.5           | 1000        | 5.435      | 16         |
| PC128                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC129                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC130                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC131                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC132                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC133                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC134                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC135                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC137                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC138                                           | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC139                                           | 2000                  | 0    | 0    | 2000                 | 0.4           | 800         | 4.348      | 16         |
| PC140                                           | 0                     | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC141                                           | 0                     | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.522      | 16         |
| PC142                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC143                                           | 0                     | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC144                                           | 0                     | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152      | 16         |
| PC145                                           | 3000                  | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152      | 16         |

|               |                    |                    |                    |                                       |                 |                                    |                                             |    |
|---------------|--------------------|--------------------|--------------------|---------------------------------------|-----------------|------------------------------------|---------------------------------------------|----|
| PC146         | 0                  | 3000               | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16 |
| PC147         | 0                  | 0                  | 3000               | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16 |
| PC148         | 3000               | 0                  | 0                  | 3000                                  | 0.5             | 1500                               | 8.152                                       | 16 |
| 8 LCs, 23 PCs | $\Sigma R = 23774$ | $\Sigma Y = 21618$ | $\Sigma B = 25092$ | $\Sigma$ Total Connected Load = 70484 | Wt. Avg = 0.516 | $\Sigma$ Demand Load = 36384       | $\Sigma I$ (per phase) = 197.739            |    |
|               | 100A RCCB          | 80A RCCB           | 100A RCCB          |                                       |                 | $\Sigma$ Demand Load (KW) = 36.384 | $\Sigma I$ (per phase calculated) = 197.739 |    |
|               | Spare LC 43        | Spare LC 44        | Spare LC 45        |                                       |                 | $\Sigma$ Demand Load (KVA) = 45.48 | I (MCCB Rating) = 78.77 A                   |    |
|               | Spare PC 109,110   | Spare PC 111,112   | Spare PC 113,114   |                                       |                 |                                    | 16 Way TPN                                  |    |

| Details of First Floor SDB-5 (upper right area) |                       |                    |                    |                                       |                 |                                      |                                             |            |
|-------------------------------------------------|-----------------------|--------------------|--------------------|---------------------------------------|-----------------|--------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                         | Wattage Per Phase (W) |                    |                    | Total Connected Load                  | Demand Factor   | Demand Load                          | Current(I)                                  | MCB Rating |
|                                                 | R                     | Y                  | B                  |                                       |                 |                                      |                                             |            |
| LC21                                            | 672                   | 0                  | 0                  | 672                                   | 1               | 672                                  | 3.652                                       | 6          |
| LC22                                            | 0                     | 788                | 0                  | 788                                   | 1               | 788                                  | 4.283                                       | 6          |
| LC31                                            | 0                     | 0                  | 696                | 696                                   | 1               | 696                                  | 3.783                                       | 6          |
| LC32                                            | 476                   | 0                  | 0                  | 476                                   | 1               | 476                                  | 2.587                                       | 6          |
| LC33                                            | 0                     | 442                | 0                  | 442                                   | 1               | 442                                  | 2.402                                       | 6          |
| PC106                                           | 0                     | 0                  | 3000               | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC107                                           | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC108                                           | 0                     | 3000               | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC109                                           | 0                     | 0                  | 3000               | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC110                                           | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC111                                           | 0                     | 3000               | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC112                                           | 0                     | 0                  | 3000               | 3000                                  | 0.4             | 1200                                 | 6.522                                       | 16         |
| PC113                                           | 2000                  | 0                  | 0                  | 2000                                  | 0.4             | 800                                  | 4.348                                       | 16         |
| PC114                                           | 0                     | 3000               | 0                  | 3000                                  | 0.4             | 1200                                 | 6.522                                       | 16         |
| PC115                                           | 0                     | 0                  | 3000               | 3000                                  | 0.4             | 1200                                 | 6.522                                       | 16         |
| PC116                                           | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC117                                           | 0                     | 2000               | 0                  | 2000                                  | 0.5             | 1000                                 | 5.435                                       | 16         |
| PC118                                           | 0                     | 0                  | 3000               | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC119                                           | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC120                                           | 0                     | 3000               | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC121                                           | 0                     | 0                  | 3000               | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC122                                           | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC123                                           | 0                     | 3000               | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC124                                           | 0                     | 0                  | 3000               | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC136                                           | 3000                  | 0                  | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC149                                           | 0                     | 3000               | 0                  | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| PC150                                           | 0                     | 0                  | 3000               | 3000                                  | 0.5             | 1500                                 | 8.152                                       | 16         |
| 5 LCs, 22 PCs                                   | $\Sigma R = 21148$    | $\Sigma Y = 21230$ | $\Sigma B = 24696$ | $\Sigma$ Total Connected Load = 67074 | Wt. Avg = 0.507 | $\Sigma$ Demand Load = 33974         | $\Sigma I$ (per phase) = 184.641            |            |
|                                                 | 80A RCCB              | 80A RCCB           | 100A RCCB          |                                       |                 | $\Sigma$ Demand Load (KW) = 33.974   | $\Sigma I$ (per phase calculated) = 184.641 |            |
|                                                 | Spare LC 43           | Spare LC 44        | Spare LC 45        |                                       |                 | $\Sigma$ Demand Load (KVA) = 42.4675 | I (MCCB Rating) = 73.56 A                   |            |
|                                                 | Spare PC 109,110      | Spare PC 111,112   | Spare PC 113,114   |                                       |                 |                                      | 16 Way TPN                                  |            |

d. Second Floor:

| Details of Second Floor SDB-1(Left Upper) |                      |                    |                    |                                       |               |                                      |                                             |            |
|-------------------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|--------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                   | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                          | Current(I)                                  | MCB Rating |
|                                           | R                    | Y                  | B                  |                                       |               |                                      |                                             |            |
| LC13                                      | 782                  | 0                  | 0                  | 782                                   | 1             | 782                                  | 4.25                                        | 6          |
| LC14                                      | 0                    | 748                | 0                  | 748                                   | 1             | 748                                  | 4.065217391                                 | 6          |
| LC15                                      | 0                    | 0                  | 374                | 374                                   | 1             | 374                                  | 2.032608696                                 | 6          |
| LC16                                      | 782                  | 0                  | 0                  | 782                                   | 1             | 782                                  | 4.25                                        | 6          |
| LC19                                      | 0                    | 338                | 0                  | 338                                   | 1             | 338                                  | 1.836956522                                 | 6          |
| LC20                                      | 0                    | 0                  | 336                | 336                                   | 1             | 336                                  | 1.826086957                                 | 6          |
| LC21                                      | 782                  | 0                  | 0                  | 782                                   | 1             | 782                                  | 4.25                                        | 6          |
| PC1                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC2                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC3                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC4                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC5                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC6                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC13                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC14                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC15                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC16                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC17                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC18                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC25                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC30                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC31                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC32                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC85                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC86                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC87                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC88                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
|                                           | $\Sigma R = 23346$   | $\Sigma Y = 22086$ | $\Sigma B = 18710$ | $\Sigma$ Total Connected Load = 64142 | 0.62963       | $\Sigma$ Demand Load = 34142         | $\Sigma I$ (per phase) = 185.554            |            |
|                                           | 100A RCCB            | 100A RCCB          | 80A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 34.142   | $\Sigma I$ (per phase calculated) = 185.554 |            |
|                                           | Spare LC 43          | Spare LC 44        | Spare LC 45        |                                       |               | $\Sigma$ Demand Load (KVA) = 42.6775 | I (MCCB Rating) = 73.92                     |            |

|  |                        |                        |                        |  |  |  |            |  |
|--|------------------------|------------------------|------------------------|--|--|--|------------|--|
|  | Spare<br>PC<br>109,110 | Spare<br>PC<br>111,112 | Spare<br>PC<br>113,114 |  |  |  | 16 Way TPN |  |
|--|------------------------|------------------------|------------------------|--|--|--|------------|--|

| Details of Second Floor SDB-2(Left Lower) |                       |                       |                       |                                          |                      |                                         |                                                |               |
|-------------------------------------------|-----------------------|-----------------------|-----------------------|------------------------------------------|----------------------|-----------------------------------------|------------------------------------------------|---------------|
| Light<br>and<br>Power<br>Circuit          | Wattage Per Phase(W)  |                       |                       | Total Connected<br>Load                  | Deman<br>d<br>Factor | Demand<br>Load                          | Current(I)                                     | MCB<br>Rating |
|                                           | R                     | Y                     | B                     |                                          |                      |                                         |                                                |               |
| LC7                                       | 782                   | 0                     | 0                     | 782                                      | 1                    | 782                                     | 4.25                                           | 6             |
| LC8                                       | 0                     | 748                   | 0                     | 748                                      | 1                    | 748                                     | 4.065217391                                    | 6             |
| LC9                                       | 0                     | 0                     | 408                   | 408                                      | 1                    | 408                                     | 2.217391304                                    | 6             |
| LC10                                      | 748                   | 0                     | 0                     | 748                                      | 1                    | 748                                     | 4.065217391                                    | 6             |
| LC11                                      | 0                     | 408                   | 0                     | 408                                      | 1                    | 408                                     | 2.217391304                                    | 6             |
| LC12                                      | 0                     | 0                     | 782                   | 782                                      | 1                    | 782                                     | 4.25                                           | 6             |
| LC17                                      | 408                   | 0                     | 0                     | 408                                      | 1                    | 408                                     | 2.217391304                                    | 6             |
| LC18                                      | 0                     | 696                   | 0                     | 696                                      | 1                    | 696                                     | 3.782608696                                    | 6             |
| PC7                                       | 3000                  | 0                     | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC8                                       | 0                     | 3000                  | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC9                                       | 0                     | 0                     | 3000                  | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC10                                      | 3000                  | 0                     | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC11                                      | 0                     | 3000                  | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC12                                      | 0                     | 0                     | 3000                  | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC19                                      | 3000                  | 0                     | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC20                                      | 0                     | 3000                  | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC21                                      | 0                     | 0                     | 3000                  | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC22                                      | 3000                  | 0                     | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC23                                      | 0                     | 3000                  | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC24                                      | 0                     | 0                     | 2000                  | 2000                                     | 0.5                  | 1000                                    | 5.434782609                                    | 16            |
| PC26                                      | 2000                  | 0                     | 0                     | 2000                                     | 0.5                  | 1000                                    | 5.434782609                                    | 16            |
| PC27                                      | 0                     | 2000                  | 0                     | 2000                                     | 0.5                  | 1000                                    | 5.434782609                                    | 16            |
| PC28                                      | 0                     | 0                     | 3000                  | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC29                                      | 3000                  | 0                     | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC89                                      | 0                     | 3000                  | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC90                                      | 0                     | 0                     | 3000                  | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC91                                      | 3000                  | 0                     | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
| PC92                                      | 0                     | 3000                  | 0                     | 3000                                     | 0.5                  | 1500                                    | 8.152173913                                    | 16            |
|                                           | $\Sigma R =$<br>21938 | $\Sigma Y =$<br>21852 | $\Sigma B =$<br>18190 | $\Sigma$ Total Connected<br>Load = 61980 |                      | $\Sigma$ Demand<br>Load = 33480         | $\Sigma I$ (per phase) =<br>181.957            |               |
|                                           | 100A<br>RCCB          | 100A<br>RCCB          | 100A<br>RCCB          |                                          |                      | $\Sigma$ Demand<br>Load (KW) =<br>33.48 | $\Sigma I$ (per phase<br>calculated) = 179.239 |               |

|  |                        |                        |                        |  |  |                                          |                             |  |
|--|------------------------|------------------------|------------------------|--|--|------------------------------------------|-----------------------------|--|
|  | Spare<br>LC 46         | Spare<br>LC 47         | Spare<br>LC 48         |  |  | $\Sigma$ Demand<br>Load (KVA) =<br>41.85 | I (MCCB Rating) =<br>71.403 |  |
|  | Spare<br>PC<br>115,116 | Spare<br>PC<br>117,118 | Spare<br>PC<br>119,120 |  |  |                                          | 16 Way TPN                  |  |

| Details of Second Floor SDB-3(Middle Upper) |                        |                        |                        |                                                |                  |                                         |                                               |               |
|---------------------------------------------|------------------------|------------------------|------------------------|------------------------------------------------|------------------|-----------------------------------------|-----------------------------------------------|---------------|
| Light<br>and<br>Power<br>Circuit            | Wattage Per Phase(W)   |                        |                        | Total<br>Connected<br>Load                     | Demand<br>Factor | Demand<br>Load                          | Current(I)                                    | MCB<br>Rating |
|                                             | R                      | Y                      | B                      |                                                |                  |                                         |                                               |               |
| LC4                                         | 510                    | 0                      | 0                      | 510                                            | 1                | 510                                     | 2.77173913                                    | 6             |
| LC5                                         | 0                      | 612                    | 0                      | 612                                            | 1                | 612                                     | 3.326086957                                   | 6             |
| LC6                                         | 0                      | 0                      | 510                    | 510                                            | 1                | 510                                     | 2.77173913                                    | 6             |
| LC37                                        | 763                    | 0                      | 0                      | 763                                            | 1                | 763                                     | 4.14673913                                    | 6             |
| LC38                                        | 0                      | 763                    | 0                      | 763                                            | 1                | 763                                     | 4.14673913                                    | 6             |
| LC41                                        | 0                      | 0                      | 150                    | 150                                            | 1                | 150                                     | 0.815217391                                   | 6             |
| LC42                                        | 150                    | 0                      | 0                      | 150                                            | 1                | 150                                     | 0.815217391                                   | 6             |
| PC33                                        | 3000                   | 0                      | 0                      | 3000                                           | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC34                                        | 0                      | 3000                   | 0                      | 3000                                           | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC35                                        | 0                      | 0                      | 3000                   | 3000                                           | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC36                                        | 3000                   | 0                      | 0                      | 3000                                           | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC37                                        | 0                      | 3000                   | 0                      | 3000                                           | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC38                                        | 0                      | 0                      | 3000                   | 3000                                           | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC39                                        | 3000                   | 0                      | 0                      | 3000                                           | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC40                                        | 0                      | 3000                   | 0                      | 3000                                           | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC53                                        | 0                      | 0                      | 2000                   | 2000                                           | 0.5              | 1000                                    | 5.434782609                                   | 16            |
| PC59                                        | 3000                   | 0                      | 0                      | 3000                                           | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC60                                        | 0                      | 3000                   | 0                      | 3000                                           | 0.5              | 1500                                    | 8.152173913                                   | 16            |
|                                             | $\Sigma R =$<br>13423  | $\Sigma Y =$<br>13375  | $\Sigma B =$<br>8660   | $\Sigma$ Total<br>Connected<br>Load =<br>35458 |                  | $\Sigma$ Demand Load =<br>19458         | $\Sigma I$ (per phase) =<br>105.75            |               |
|                                             | 63A<br>RCCB            | 63A<br>RCCB            | 63A<br>RCCB            |                                                |                  | $\Sigma$ Demand Load<br>(KW) = 19.458   | $\Sigma I$ (per phase<br>calculated) = 105.75 |               |
|                                             | Spare<br>LC 49         | Spare<br>LC 50         | Spare<br>LC 51         |                                                |                  | $\Sigma$ Demand Load<br>(KVA) = 24.3225 | I (MCCB Rating) =<br>42.127                   |               |
|                                             | Spare<br>PC<br>121,122 | Spare<br>PC<br>123,124 | Spare<br>PC<br>125,126 |                                                |                  |                                         | 12 Way TPN                                    |               |

| Details of Second Floor SDB-4(Middle Lower) |                       |                       |                       |                                          |               |                                     |                                             |            |
|---------------------------------------------|-----------------------|-----------------------|-----------------------|------------------------------------------|---------------|-------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                     | Wattage Per Phase(W)  |                       |                       | Total Connected Load                     | Demand Factor | Demand Load                         | Current(I)                                  | MCB Rating |
|                                             | R                     | Y                     | B                     |                                          |               |                                     |                                             |            |
| LC1                                         | 612                   | 0                     | 0                     | 612                                      | 1             | 612                                 | 3.326086957                                 | 6          |
| LC2                                         | 0                     | 612                   | 0                     | 612                                      | 1             | 612                                 | 3.326086957                                 | 6          |
| LC3                                         | 0                     | 0                     | 748                   | 748                                      | 1             | 748                                 | 4.065217391                                 | 6          |
| LC39                                        | 284                   | 0                     | 0                     | 284                                      | 1             | 284                                 | 1.543478261                                 | 6          |
| LC40                                        | 0                     | 264                   | 0                     | 264                                      | 1             | 264                                 | 1.434782609                                 | 6          |
| PC41                                        | 3000                  | 0                     | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC42                                        | 0                     | 3000                  | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC43                                        | 0                     | 0                     | 3000                  | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC44                                        | 3000                  | 0                     | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC45                                        | 0                     | 3000                  | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC46                                        | 0                     | 0                     | 3000                  | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC47                                        | 3000                  | 0                     | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC48                                        | 0                     | 3000                  | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC49                                        | 0                     | 0                     | 3000                  | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC50                                        | 3000                  | 0                     | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC51                                        | 0                     | 3000                  | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC52                                        | 0                     | 0                     | 3000                  | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC54                                        | 3000                  | 0                     | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC55                                        | 0                     | 3000                  | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC56                                        | 0                     | 0                     | 3000                  | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC57                                        | 3000                  | 0                     | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC58                                        | 0                     | 3000                  | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC59                                        | 0                     | 0                     | 3000                  | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC60                                        | 3000                  | 0                     | 0                     | 3000                                     | 0.5           | 1500                                | 8.152173913                                 | 16         |
|                                             | $\Sigma R =$<br>21896 | $\Sigma Y =$<br>18876 | $\Sigma B =$<br>18748 | $\Sigma$ Total Connected Load =<br>59520 |               | $\Sigma$ Demand Load =<br>31020     | $\Sigma I$ (per phase) =<br>168.587         |            |
|                                             | 100A RCCB             | 100A RCCB             | 100A RCCB             |                                          |               | $\Sigma$ Demand Load (KW) = 31.02   | $\Sigma I$ (per phase calculated) = 168.587 |            |
|                                             | Spare LC 52           | Spare LC 53           | Spare LC 54           |                                          |               | $\Sigma$ Demand Load (KVA) = 38.775 | I (MCCB Rating) =<br>67.16                  |            |
|                                             | Spare PC 127,128      | Spare PC 129,130      | Spare PC 131,132      |                                          |               |                                     | 14 Way TPN                                  |            |

| Details of Second Floor SDB-5(Right Upper) |                                 |                                 |                                 |                                       |               |                                    |                                             |            |
|--------------------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------------|---------------|------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                    | Wattage Per Phase(W)            |                                 |                                 | Total Connected Load                  | Demand Factor | Demand Load                        | Current(I)                                  | MCB Rating |
|                                            | R                               | Y                               | B                               |                                       |               |                                    |                                             |            |
| LC28                                       | 782                             | 0                               | 0                               | 782                                   | 1             | 782                                | 4.25                                        | 6          |
| LC29                                       | 0                               | 782                             | 0                               | 782                                   | 1             | 782                                | 4.25                                        | 6          |
| LC30                                       | 0                               | 0                               | 748                             | 748                                   | 1             | 748                                | 4.065217391                                 | 6          |
| LC31                                       | 374                             | 0                               | 0                               | 374                                   | 1             | 374                                | 2.032608696                                 | 6          |
| LC32                                       | 0                               | 782                             | 0                               | 782                                   | 1             | 782                                | 4.25                                        | 6          |
| LC34                                       | 0                               | 0                               | 388                             | 388                                   | 1             | 388                                | 2.108695652                                 | 6          |
| LC35                                       | 336                             | 0                               | 0                               | 336                                   | 1             | 336                                | 1.826086957                                 | 6          |
| PC73                                       | 3000                            | 0                               | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC74                                       | 0                               | 3000                            | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC75                                       | 0                               | 0                               | 3000                            | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC76                                       | 3000                            | 0                               | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC77                                       | 0                               | 3000                            | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC78                                       | 0                               | 0                               | 3000                            | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC79                                       | 3000                            | 0                               | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC80                                       | 0                               | 3000                            | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC81                                       | 0                               | 0                               | 3000                            | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC82                                       | 3000                            | 0                               | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC83                                       | 0                               | 3000                            | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC84                                       | 0                               | 0                               | 2000                            | 2000                                  | 0.5           | 1000                               | 5.434782609                                 | 16         |
| PC99                                       | 2000                            | 0                               | 0                               | 2000                                  | 0.5           | 1000                               | 5.434782609                                 | 16         |
| PC101                                      | 0                               | 3000                            | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC102                                      | 0                               | 0                               | 3000                            | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC103                                      | 3000                            | 0                               | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC104                                      | 0                               | 3000                            | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
| PC105                                      | 0                               | 0                               | 2000                            | 2000                                  | 0.5           | 1000                               | 5.434782609                                 | 16         |
| PC106                                      | 3000                            | 0                               | 0                               | 3000                                  | 0.5           | 1500                               | 8.152173913                                 | 16         |
|                                            | $\Sigma R = 21492$<br>100A RCCB | $\Sigma Y = 19564$<br>100A RCCB | $\Sigma B = 17136$<br>100A RCCB | $\Sigma$ Total Connected Load = 58192 |               | $\Sigma$ Demand Load = 31192       | $\Sigma I$ (per phase) = 169.522            |            |
|                                            | Spare LC 55                     | Spare LC 56                     | Spare LC 57                     |                                       |               | $\Sigma$ Demand Load (KW) = 31.192 | $\Sigma I$ (per phase calculated) = 169.522 |            |
|                                            | Spare PC 133,134                | Spare PC 135,136                | Spare PC 137,138                |                                       |               | $\Sigma$ Demand Load (KVA) = 38.99 | I (MCCB Rating) = 67.53                     |            |
|                                            |                                 |                                 |                                 |                                       |               |                                    | 16 Way TPN                                  |            |

| Details of Second Floor SDB-6(Right Lower) |                      |                    |                    |                                       |               |                                     |                                             |            |
|--------------------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|-------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                    | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                         | Current(I)                                  | MCB Rating |
|                                            | R                    | Y                  | B                  |                                       |               |                                     |                                             |            |
| LC22                                       | 782                  | 0                  | 0                  | 782                                   | 1             | 782                                 | 4.25                                        | 6          |
| LC23                                       | 0                    | 748                | 0                  | 748                                   | 1             | 748                                 | 4.065217391                                 | 6          |
| LC24                                       | 0                    | 0                  | 408                | 408                                   | 1             | 408                                 | 2.217391304                                 | 6          |
| LC25                                       | 782                  | 0                  | 0                  | 782                                   | 1             | 782                                 | 4.25                                        | 6          |
| LC26                                       | 0                    | 748                | 0                  | 748                                   | 1             | 748                                 | 4.065217391                                 | 6          |
| LC27                                       | 0                    | 0                  | 408                | 408                                   | 1             | 408                                 | 2.217391304                                 | 6          |
| LC33                                       | 408                  | 0                  | 0                  | 408                                   | 1             | 408                                 | 2.217391304                                 | 6          |
| LC36                                       | 0                    | 696                | 0                  | 696                                   | 1             | 696                                 | 3.782608696                                 | 6          |
| PC61                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC62                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC63                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC64                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC65                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC66                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC67                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC68                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC69                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC70                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC71                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC72                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC93                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC94                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC95                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC96                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC97                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC98                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC100                                      | 2000                 | 0                  | 0                  | 2000                                  | 0.5           | 1000                                | 5.434782609                                 | 16         |
| PC107                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC108                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
|                                            | $\Sigma R = 21972$   | $\Sigma Y = 23192$ | $\Sigma B = 21816$ | $\Sigma$ Total Connected Load = 66980 |               | $\Sigma$ Demand Load = 35980        | $\Sigma I$ (per phase) = 195.543            |            |
|                                            | 100A RCCB            | 100A RCCB          | 100A RCCB          |                                       |               | $\Sigma$ Demand Load (KW) = 35.98   | $\Sigma I$ (per phase calculated) = 195.543 |            |
|                                            | Spare LC 58          | Spare LC 59        | Spare LC 60        |                                       |               | $\Sigma$ Demand Load (KVA) = 44.975 | I (MCCB Rating) = 77.89                     |            |
|                                            | Spare PC 139,140     | Spare PC 141,142   | Spare PC 143,144   |                                       |               |                                     | 16 Way TPN                                  |            |

e. Third Floor:

| Details of Third Floor SDB-1(Left Lower) |                      |                   |                   |                                       |                   |                              |                                 |            |
|------------------------------------------|----------------------|-------------------|-------------------|---------------------------------------|-------------------|------------------------------|---------------------------------|------------|
| Light and Power Circuit                  | Wattage Per Phase(W) |                   |                   | Total Connected Load                  | Demand Factor     | Demand Load                  | Current(I)                      | MCB Rating |
|                                          | R                    | Y                 | B                 |                                       |                   |                              |                                 |            |
| LC7                                      | 792                  | 0                 | 0                 | 792                                   | 1                 | 792                          | 4.304347826                     | 6          |
| LC8                                      | 0                    | 782               | 0                 | 782                                   | 1                 | 782                          | 4.25                            | 6          |
| LC17                                     | 0                    | 0                 | 382               | 382                                   | 1                 | 382                          | 2.076086957                     | 6          |
| LC18                                     | 696                  | 0                 | 0                 | 696                                   | 1                 | 696                          | 3.782608696                     | 6          |
| LC19                                     | 0                    | 180               | 0                 | 180                                   | 1                 | 180                          | 0.97826087                      | 6          |
| LC31                                     | 0                    | 0                 | 750               | 750                                   | 1                 | 750                          | 4.076086957                     | 6          |
| PC1                                      | 3000                 | 0                 | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC2                                      | 0                    | 3000              | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC3                                      | 0                    | 0                 | 3000              | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC4                                      | 2000                 | 0                 | 0                 | 2000                                  | 0.5               | 1000                         | 5.434782609                     | 16         |
| PC5                                      | 0                    | 2000              | 0                 | 2000                                  | 0.5               | 1000                         | 5.434782609                     | 16         |
| PC6                                      | 0                    | 0                 | 3000              | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC7                                      | 3000                 | 0                 | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC8                                      | 0                    | 3000              | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC9                                      | 0                    | 0                 | 3000              | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC10                                     | 3000                 | 0                 | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC11                                     | 0                    | 3000              | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC12                                     | 0                    | 0                 | 3000              | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC13                                     | 3000                 | 0                 | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC14                                     | 0                    | 3000              | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC15                                     | 0                    | 0                 | 2000              | 2000                                  | 0.5               | 1000                         | 5.434782609                     | 16         |
| PC16                                     | 3000                 | 0                 | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC17                                     | 0                    | 3000              | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC18                                     | 0                    | 0                 | 2000              | 2000                                  | 0.5               | 1000                         | 5.434782609                     | 16         |
| PC19                                     | 3000                 | 0                 | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC20                                     | 0                    | 3000              | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC21                                     | 0                    | 0                 | 3000              | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC22                                     | 3000                 | 0                 | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC23                                     | 0                    | 3000              | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC24                                     | 0                    | 0                 | 3000              | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC25                                     | 3000                 | 0                 | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC26                                     | 0                    | 3000              | 0                 | 3000                                  | 0.5               | 1500                         | 8.152173913                     | 16         |
| PC50                                     | 0                    | 0                 | 3000              | 3000                                  | 0.4               | 1200                         | 6.52173913                      | 16         |
| PC51                                     | 3000                 | 0                 | 0                 | 3000                                  | 0.4               | 1200                         | 6.52173913                      | 16         |
| PC52                                     | 0                    | 3000              | 0                 | 3000                                  | 0.4               | 1200                         | 6.52173913                      | 16         |
| PC53                                     | 0                    | 0                 | 3000              | 3000                                  | 0.4               | 1200                         | 6.52173913                      | 16         |
| PC54                                     | 3000                 | 0                 | 0                 | 3000                                  | 0.4               | 1200                         | 6.52173913                      | 16         |
| PC55                                     | 0                    | 3000              | 0                 | 3000                                  | 0.4               | 1200                         | 6.52173913                      | 16         |
| PC56                                     | 0                    | 0                 | 3000              | 3000                                  | 0.4               | 1200                         | 6.52173913                      | 16         |
|                                          | <b>ΣR = 33488</b>    | <b>ΣY = 32962</b> | <b>ΣB = 32132</b> | <b>Σ Total Connected Load = 98582</b> | <b>0.55897436</b> | <b>Σ Demand Load = 48982</b> | <b>ΣI (per phase) = 266.207</b> |            |

|  |                     |                     |                     |  |  |                                               |                                                   |  |
|--|---------------------|---------------------|---------------------|--|--|-----------------------------------------------|---------------------------------------------------|--|
|  | 125A<br>RCCB        | 125A<br>RCCB        | 125A<br>RCCB        |  |  | $\Sigma$ Demand<br>Load<br>(KW) =<br>48.982   | $\Sigma I$ (per phase<br>calculated) =<br>266.207 |  |
|  | Spare LC<br>33      | Spare LC<br>34      | Spare LC<br>35      |  |  | $\Sigma$ Demand<br>Load<br>(KVA) =<br>61.2275 | I (MCCB<br>Rating) =<br>106.049                   |  |
|  | Spare PC<br>143,144 | Spare PC<br>145,146 | Spare PC<br>147,148 |  |  |                                               | 18 Way TPN                                        |  |

| Details of Third Floor SDB-2(Left Upper) |                       |                       |                       |                                          |                  |                                           |                                                   |               |
|------------------------------------------|-----------------------|-----------------------|-----------------------|------------------------------------------|------------------|-------------------------------------------|---------------------------------------------------|---------------|
| Light<br>and<br>Power<br>Circuit         | Wattage Per Phase(W)  |                       |                       | Total Connected<br>Load                  | Demand<br>Factor | Demand<br>Load                            | Current(I)                                        | MCB<br>Rating |
|                                          | R                     | Y                     | B                     |                                          |                  |                                           |                                                   |               |
| LC9                                      | 646                   | 0                     | 0                     | 646                                      | 1                | 646                                       | 3.510869565                                       | 6             |
| LC10                                     | 0                     | 690                   | 0                     | 690                                      | 1                | 690                                       | 3.75                                              | 6             |
| LC20                                     | 0                     | 0                     | 456                   | 456                                      | 1                | 456                                       | 2.47826087                                        | 6             |
| LC21                                     | 328                   | 0                     | 0                     | 328                                      | 1                | 328                                       | 1.782608696                                       | 6             |
| LC29                                     | 0                     | 750                   | 0                     | 750                                      | 1                | 750                                       | 4.076086957                                       | 6             |
| PC27                                     | 0                     | 0                     | 3000                  | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC28                                     | 3000                  | 0                     | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC29                                     | 0                     | 3000                  | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC30                                     | 0                     | 0                     | 3000                  | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC31                                     | 3000                  | 0                     | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC32                                     | 0                     | 3000                  | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC33                                     | 0                     | 0                     | 3000                  | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC34                                     | 3000                  | 0                     | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC35                                     | 0                     | 3000                  | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC36                                     | 0                     | 0                     | 3000                  | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC37                                     | 3000                  | 0                     | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC38                                     | 0                     | 3000                  | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC39                                     | 0                     | 0                     | 3000                  | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC40                                     | 3000                  | 0                     | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC41                                     | 0                     | 3000                  | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC42                                     | 0                     | 0                     | 3000                  | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC43                                     | 3000                  | 0                     | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC44                                     | 0                     | 3000                  | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC45                                     | 0                     | 0                     | 3000                  | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC57                                     | 3000                  | 0                     | 0                     | 3000                                     | 0.5              | 1500                                      | 8.152173913                                       | 16            |
| PC46                                     | 0                     | 3000                  | 0                     | 3000                                     | 0.4              | 1200                                      | 6.52173913                                        | 16            |
| PC47                                     | 0                     | 0                     | 3000                  | 3000                                     | 0.4              | 1200                                      | 6.52173913                                        | 16            |
| PC48                                     | 3000                  | 0                     | 0                     | 3000                                     | 0.4              | 1200                                      | 6.52173913                                        | 16            |
| PC49                                     | 0                     | 3000                  | 0                     | 3000                                     | 0.4              | 1200                                      | 6.52173913                                        | 16            |
|                                          | $\Sigma R =$<br>24974 | $\Sigma Y =$<br>25440 | $\Sigma B =$<br>24456 | $\Sigma$ Total Connected<br>Load = 74870 | 0.57241379       | $\Sigma$ Demand Load =<br>37670           | $\Sigma I$ (per phase)<br>= 204.728               |               |
|                                          | 100A<br>RCCB          | 100A<br>RCCB          | 100A<br>RCCB          |                                          |                  | $\Sigma$ Demand Load<br>(KW) = 37.67      | $\Sigma I$ (per phase<br>calculated) =<br>204.728 |               |
|                                          | Spare<br>LC 36        | Spare<br>LC 37        | Spare<br>LC 38        |                                          |                  | $\Sigma$ Demand<br>Load(KVA) =<br>47.0875 | I (MCCB Rating) =<br>81.558                       |               |

|  |                        |                        |                        |  |  |  |            |  |
|--|------------------------|------------------------|------------------------|--|--|--|------------|--|
|  | Spare<br>PC<br>149,150 | Spare<br>PC<br>151,152 | Spare<br>PC<br>153,154 |  |  |  | 16 Way TPN |  |
|--|------------------------|------------------------|------------------------|--|--|--|------------|--|

| Details of Third Floor SDB-3(Middle Upper) |                        |                        |                        |                                             |                  |                                         |                                               |               |
|--------------------------------------------|------------------------|------------------------|------------------------|---------------------------------------------|------------------|-----------------------------------------|-----------------------------------------------|---------------|
| Light<br>and<br>Power<br>Circuit           | Wattage Per Phase(W)   |                        |                        | Total Connected<br>Load                     | Demand<br>Factor | Demand<br>Load                          | Current(I)                                    | MCB<br>Rating |
|                                            | R                      | Y                      | B                      |                                             |                  |                                         |                                               |               |
| LC4                                        | 510                    | 0                      | 0                      | 510                                         | 1                | 510                                     | 2.77173913                                    | 6             |
| LC5                                        | 0                      | 612                    | 0                      | 612                                         | 1                | 612                                     | 3.326086957                                   | 6             |
| LC6                                        | 0                      | 0                      | 510                    | 510                                         | 1                | 510                                     | 2.77173913                                    | 6             |
| LC15                                       | 315                    | 0                      | 0                      | 315                                         | 1                | 315                                     | 1.711956522                                   | 6             |
| LC16                                       | 0                      | 315                    | 0                      | 315                                         | 1                | 315                                     | 1.711956522                                   | 6             |
| PC58                                       | 3000                   | 0                      | 0                      | 3000                                        | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC59                                       | 0                      | 3000                   | 0                      | 3000                                        | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC60                                       | 0                      | 0                      | 3000                   | 3000                                        | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC61                                       | 3000                   | 0                      | 0                      | 3000                                        | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC62                                       | 0                      | 3000                   | 0                      | 3000                                        | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC63                                       | 0                      | 0                      | 3000                   | 3000                                        | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC70                                       | 3000                   | 0                      | 0                      | 3000                                        | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC71                                       | 0                      | 3000                   | 0                      | 3000                                        | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC72                                       | 0                      | 0                      | 3000                   | 3000                                        | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC73                                       | 3000                   | 0                      | 0                      | 3000                                        | 0.5              | 1500                                    | 8.152173913                                   | 16            |
| PC74                                       | 0                      | 2000                   | 0                      | 2000                                        | 0.4              | 800                                     | 4.347826087                                   | 16            |
|                                            | $\Sigma R =$<br>12825  | $\Sigma Y =$<br>11927  | $\Sigma B =$<br>9510   | $\Sigma$ Total<br>Connected Load<br>= 34262 |                  | $\Sigma$ Demand Load =<br>18062         | $\Sigma I$ (per phase) =<br>98.163            |               |
|                                            | 63A<br>RCCB            | 63A<br>RCCB            | 63A<br>RCCB            |                                             |                  | $\Sigma$ Demand Load<br>(KW) = 18.062   | $\Sigma I$ (per phase<br>calculated) = 98.163 |               |
|                                            | Spare<br>LC 39         | Spare<br>LC 40         | Spare<br>LC 41         |                                             |                  | $\Sigma$ Demand Load<br>(KVA) = 22.5775 | I (MCCB Rating) =<br>39.10                    |               |
|                                            | Spare<br>PC<br>155,156 | Spare<br>PC<br>157,158 | Spare<br>PC<br>159,160 |                                             |                  |                                         | 12 Way TPN                                    |               |

| Details of Third Floor SDB-4(Middle Lower) |                      |                    |                    |                                       |               |                                     |                                                   |            |
|--------------------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|-------------------------------------|---------------------------------------------------|------------|
| Light and Power Circuit                    | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                         | Current(I)                                        | MCB Rating |
|                                            | R                    | Y                  | B                  |                                       |               |                                     |                                                   |            |
| LC1                                        | 612                  | 0                  | 0                  | 612                                   | 1             | 612                                 | 3.326086957                                       | 6          |
| LC2                                        | 0                    | 612                | 0                  | 612                                   | 1             | 612                                 | 3.326086957                                       | 6          |
| LC3                                        | 0                    | 0                  | 748                | 748                                   | 1             | 748                                 | 4.065217391                                       | 6          |
| LC22                                       | 284                  | 0                  | 0                  | 284                                   | 1             | 284                                 | 1.543478261                                       | 6          |
| LC23                                       | 0                    | 264                | 0                  | 264                                   | 1             | 264                                 | 1.434782609                                       | 6          |
| PC64                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC65                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC66                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC67                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC68                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC69                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC75                                       | 0                    | 0                  | 2000               | 2000                                  | 0.4           | 800                                 | 4.347826087                                       | 16         |
| PC76                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                        | 16         |
| PC77                                       | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                        | 16         |
| PC78                                       | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                | 6.52173913                                        | 16         |
| PC79                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                        | 16         |
| PC80                                       | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                        | 16         |
| PC81                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC82                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                        | 16         |
| PC83                                       | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                        | 16         |
| PC84                                       | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                | 6.52173913                                        | 16         |
| PC85                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                        | 16         |
|                                            | $\Sigma R = 18896$   | $\Sigma Y = 15876$ | $\Sigma B = 17748$ | $\Sigma$ Total Connected Load = 52520 |               | $\Sigma$ Demand Load = 24620        | $\Sigma I(\text{per phase}) = 133.804$            |            |
|                                            | 80A RCCB             | 63A RCCB           | 80A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 24.62   | $\Sigma I(\text{per phase calculated}) = 133.804$ |            |
|                                            | Spare LC 42          | Spare LC 43        | Spare LC 44        |                                       |               | $\Sigma$ Demand Load (KVA) = 30.775 | I (MCCB Rating) = 53.3                            |            |
|                                            | Spare PC 161,162     | Spare PC 163,164   | Spare PC 165,166   |                                       |               |                                     | 14 Way TPN                                        |            |

| Details of Third Floor SDB-5(Right Upper) |                      |                    |                    |                                       |               |                                      |                                             |            |
|-------------------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|--------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                   | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                          | Current(I)                                  | MCB Rating |
|                                           | R                    | Y                  | B                  |                                       |               |                                      |                                             |            |
| LC13                                      | 646                  | 0                  | 0                  | 646                                   | 1             | 646                                  | 3.510869565                                 | 6          |
| LC14                                      | 0                    | 690                | 0                  | 690                                   | 1             | 690                                  | 3.75                                        | 6          |
| LC26                                      | 0                    | 0                  | 456                | 456                                   | 1             | 456                                  | 2.47826087                                  | 6          |
| LC27                                      | 328                  | 0                  | 0                  | 328                                   | 1             | 328                                  | 1.782608696                                 | 6          |
| LC30                                      | 0                    | 750                | 0                  | 750                                   | 1             | 750                                  | 4.076086957                                 | 6          |
| PC112                                     | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC113                                     | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC114                                     | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC115                                     | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC116                                     | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC117                                     | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC118                                     | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC119                                     | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC120                                     | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC121                                     | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC122                                     | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC123                                     | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC124                                     | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC125                                     | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC126                                     | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC127                                     | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC128                                     | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC129                                     | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC130                                     | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC131                                     | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                 | 8.152173913                                 | 16         |
| PC136                                     | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                  | 16         |
| PC137                                     | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                 | 6.52173913                                  | 16         |
| PC138                                     | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                  | 16         |
| PC139                                     | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                  | 16         |
|                                           | $\Sigma R = 24974$   | $\Sigma Y = 25440$ | $\Sigma B = 24456$ | $\Sigma$ Total Connected Load = 74870 |               | $\Sigma$ Demand Load = 37670         | $\Sigma I$ (per phase) = 204.728            |            |
|                                           | 100A RCCB            | 100A RCCB          | 100A RCCB          |                                       |               | $\Sigma$ Demand Load (KW) = 37.67    | $\Sigma I$ (per phase calculated) = 204.728 |            |
|                                           | Spare LC 45          | Spare LC 46        | Spare LC 47        |                                       |               | $\Sigma$ Demand Load (KVA) = 47.0875 | I (MCCB Rating) = 81.55                     |            |
|                                           | Spare PC 167,168     | Spare PC 169,170   | Spare PC 171,172   |                                       |               |                                      | 16 Way TPN                                  |            |

| Details of Third Floor SDB-6(Right Lower) |                      |      |      |                      |               |             |             |            |
|-------------------------------------------|----------------------|------|------|----------------------|---------------|-------------|-------------|------------|
| Light and Power Circuit                   | Wattage Per Phase(W) |      |      | Total Connected Load | Demand Factor | Demand Load | Current(I)  | MCB Rating |
|                                           | R                    | Y    | B    |                      |               |             |             |            |
| LC11                                      | 782                  | 0    | 0    | 782                  | 1             | 782         | 4.25        | 6          |
| LC12                                      | 0                    | 792  | 0    | 792                  | 1             | 792         | 4.304347826 | 6          |
| LC24                                      | 0                    | 0    | 382  | 382                  | 1             | 382         | 2.076086957 | 6          |
| LC25                                      | 696                  | 0    | 0    | 696                  | 1             | 696         | 3.782608696 | 6          |
| LC28                                      | 0                    | 180  | 0    | 180                  | 1             | 180         | 0.97826087  | 6          |
| LC32                                      | 0                    | 0    | 750  | 750                  | 1             | 750         | 4.076086957 | 6          |
| PC86                                      | 3000                 | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC87                                      | 0                    | 2000 | 0    | 2000                 | 0.5           | 1000        | 5.434782609 | 16         |
| PC88                                      | 0                    | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC89                                      | 2000                 | 0    | 0    | 2000                 | 0.5           | 1000        | 5.434782609 | 16         |
| PC90                                      | 0                    | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC91                                      | 0                    | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC92                                      | 3000                 | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC93                                      | 0                    | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC94                                      | 0                    | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC95                                      | 3000                 | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC96                                      | 0                    | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC97                                      | 0                    | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC98                                      | 3000                 | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC99                                      | 0                    | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC100                                     | 0                    | 0    | 2000 | 2000                 | 0.5           | 1000        | 5.434782609 | 16         |
| PC101                                     | 2000                 | 0    | 0    | 2000                 | 0.5           | 1000        | 5.434782609 | 16         |
| PC102                                     | 0                    | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC103                                     | 0                    | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC104                                     | 3000                 | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC105                                     | 0                    | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC106                                     | 0                    | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC107                                     | 3000                 | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC108                                     | 0                    | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC109                                     | 0                    | 0    | 3000 | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC110                                     | 3000                 | 0    | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC111                                     | 0                    | 3000 | 0    | 3000                 | 0.5           | 1500        | 8.152173913 | 16         |
| PC132                                     | 0                    | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.52173913  | 16         |
| PC133                                     | 3000                 | 0    | 0    | 3000                 | 0.4           | 1200        | 6.52173913  | 16         |
| PC134                                     | 0                    | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.52173913  | 16         |
| PC135                                     | 0                    | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.52173913  | 16         |
| PC140                                     | 3000                 | 0    | 0    | 3000                 | 0.4           | 1200        | 6.52173913  | 16         |
| PC141                                     | 0                    | 3000 | 0    | 3000                 | 0.4           | 1200        | 6.52173913  | 16         |
| PC142                                     | 0                    | 0    | 3000 | 3000                 | 0.4           | 1200        | 6.52173913  | 16         |

|                       |                       |                       |                                             |  |                                            |                                                         |  |
|-----------------------|-----------------------|-----------------------|---------------------------------------------|--|--------------------------------------------|---------------------------------------------------------|--|
| $\Sigma R =$<br>32478 | $\Sigma Y =$<br>32972 | $\Sigma B =$<br>33132 | $\Sigma$ Total<br>Connected<br>Load = 98582 |  | $\Sigma$ Demand Load<br>= 48982            | $\Sigma I(\text{per phase}) =$<br>266.207               |  |
| 125A<br>RCCB          | 125A<br>RCCB          | 125A<br>RCCB          |                                             |  | $\Sigma$ Demand Load<br>(KW) = 48.982      | $\Sigma I(\text{per phase}$<br>calculated) =<br>266.207 |  |
| Spare LC<br>48        | Spare LC<br>49        | Spare LC<br>50        |                                             |  | $\Sigma$ Demand Load<br>(KVA) =<br>61.2275 | I (MCCB<br>Rating) = 106.05                             |  |
| Spare PC<br>173,174   | Spare PC<br>175,176   | Spare PC<br>177,178   |                                             |  |                                            | 16 ay<br>TPN                                            |  |

f. Fourth Floor:

| Details of Fourth Floor SDB-1(Left Lower) |                      |      |      |                            |                  |                |             |               |
|-------------------------------------------|----------------------|------|------|----------------------------|------------------|----------------|-------------|---------------|
| Light<br>and<br>Power<br>Circuit          | Wattage Per Phase(W) |      |      | Total<br>Connected<br>Load | Demand<br>Factor | Demand<br>Load | Current(I)  | MCB<br>Rating |
|                                           | R                    | Y    | B    |                            |                  |                |             |               |
| LC1                                       | 692                  | 0    | 0    | 692                        | 1                | 692            | 3.760869565 | 6             |
| LC2                                       | 0                    | 748  | 0    | 748                        | 1                | 748            | 4.065217391 | 6             |
| LC3                                       | 0                    | 0    | 786  | 786                        | 1                | 786            | 4.27173913  | 6             |
| LC4                                       | 700                  | 0    | 0    | 700                        | 1                | 700            | 3.804347826 | 6             |
| LC5                                       | 0                    | 568  | 0    | 568                        | 1                | 568            | 3.086956522 | 6             |
| LC6                                       | 0                    | 0    | 748  | 748                        | 1                | 748            | 4.065217391 | 6             |
| PC1                                       | 3000                 | 0    | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC2                                       | 0                    | 3000 | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC3                                       | 0                    | 0    | 3000 | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC4                                       | 3000                 | 0    | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC5                                       | 0                    | 3000 | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC6                                       | 0                    | 0    | 3000 | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC7                                       | 3000                 | 0    | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC8                                       | 0                    | 3000 | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC9                                       | 0                    | 0    | 3000 | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC10                                      | 2500                 | 0    | 0    | 2500                       | 0.5              | 1250           | 6.793478261 | 16            |
| PC11                                      | 0                    | 3000 | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC12                                      | 0                    | 0    | 3000 | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC13                                      | 3000                 | 0    | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC14                                      | 0                    | 3000 | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC15                                      | 0                    | 0    | 3000 | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC16                                      | 3000                 | 0    | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC17                                      | 0                    | 3000 | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC18                                      | 0                    | 0    | 3000 | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |
| PC19                                      | 2500                 | 0    | 0    | 2500                       | 0.5              | 1250           | 6.793478261 | 16            |
| PC20                                      | 0                    | 3000 | 0    | 3000                       | 0.5              | 1500           | 8.152173913 | 16            |

|      |                             |                             |                             |                                               |                   |                                             |                                                    |    |
|------|-----------------------------|-----------------------------|-----------------------------|-----------------------------------------------|-------------------|---------------------------------------------|----------------------------------------------------|----|
| PC21 | 0                           | 0                           | 2500                        | 2500                                          | 0.5               | 1250                                        | 6.793478261                                        | 16 |
| PC22 | 3000                        | 0                           | 0                           | 3000                                          | 0.5               | 1500                                        | 8.152173913                                        | 16 |
| PC23 | 0                           | 3000                        | 0                           | 3000                                          | 0.5               | 1500                                        | 8.152173913                                        | 16 |
| PC24 | 0                           | 0                           | 3000                        | 3000                                          | 0.5               | 1500                                        | 8.152173913                                        | 16 |
| PC25 | 3000                        | 0                           | 0                           | 3000                                          | 0.5               | 1500                                        | 8.152173913                                        | 16 |
| PC26 | 0                           | 3000                        | 0                           | 3000                                          | 0.5               | 1500                                        | 8.152173913                                        | 16 |
| PC27 | 0                           | 0                           | 3000                        | 3000                                          | 0.5               | 1500                                        | 8.152173913                                        | 16 |
| PC28 | 3000                        | 0                           | 0                           | 3000                                          | 0.5               | 1500                                        | 8.152173913                                        | 16 |
| PC29 | 0                           | 3000                        | 0                           | 3000                                          | 0.5               | 1500                                        | 8.152173913                                        | 16 |
| PC30 | 0                           | 0                           | 3000                        | 3000                                          | 0.5               | 1500                                        | 8.152173913                                        | 16 |
|      | <b>ΣR =<br/>30392</b>       | <b>ΣY =<br/>31316</b>       | <b>ΣB =<br/>31034</b>       | <b>Σ Total<br/>Connected<br/>Load = 92742</b> | <b>0.58333333</b> | <b>Σ Demand<br/>Load = 48492</b>            | <b>ΣI(per phase) =<br/>263.543</b>                 |    |
|      | <b>125A<br/>RCCB</b>        | <b>125A<br/>RCCB</b>        | <b>125A<br/>RCCB</b>        |                                               |                   | <b>Σ Demand<br/>Load (KW) =<br/>48.492</b>  | <b>ΣI (per phase<br/>calculated) =<br/>263.543</b> |    |
|      | <b>Spare LC<br/>32</b>      | <b>Spare LC<br/>33</b>      | <b>Spare LC<br/>34</b>      |                                               |                   | <b>Σ Demand<br/>Load (KVA) =<br/>60.615</b> | <b>I (MCCB<br/>Rating) =<br/>104.988 A</b>         |    |
|      | <b>Spare PC<br/>143,146</b> | <b>Spare PC<br/>144,147</b> | <b>Spare PC<br/>145,148</b> |                                               |                   |                                             | <b>18 Way TPN</b>                                  |    |

| Details of Fourth Floor SDB-2(Left Upper) |                                     |                                     |                                     |                                       |               |                                     |                                               |            |
|-------------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|---------------------------------------|---------------|-------------------------------------|-----------------------------------------------|------------|
| Light and Power Circuit                   | Wattage Per Phase(W)                |                                     |                                     | Total Connected Load                  | Demand Factor | Demand Load                         | Current(I)                                    | MCB Rating |
|                                           | R                                   | Y                                   | B                                   |                                       |               |                                     |                                               |            |
| LC7                                       | 494                                 | 0                                   | 0                                   | 494                                   | 1             | 494                                 | 2.684782609                                   | 6          |
| LC8                                       | 0                                   | 525                                 | 0                                   | 525                                   | 1             | 525                                 | 2.85326087                                    | 6          |
| LC9                                       | 0                                   | 0                                   | 525                                 | 525                                   | 1             | 525                                 | 2.85326087                                    | 6          |
| LC10                                      | 525                                 | 0                                   | 0                                   | 525                                   | 1             | 525                                 | 2.85326087                                    | 6          |
| LC11                                      | 0                                   | 525                                 | 0                                   | 525                                   | 1             | 525                                 | 2.85326087                                    | 6          |
| PC31                                      | 0                                   | 0                                   | 2500                                | 2500                                  | 0.5           | 1250                                | 6.793478261                                   | 16         |
| PC32                                      | 3000                                | 0                                   | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC33                                      | 0                                   | 3000                                | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC34                                      | 0                                   | 0                                   | 3000                                | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC35                                      | 3000                                | 0                                   | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC36                                      | 0                                   | 3000                                | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC37                                      | 0                                   | 0                                   | 3000                                | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC38                                      | 3000                                | 0                                   | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC39                                      | 0                                   | 3000                                | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC40                                      | 0                                   | 0                                   | 3000                                | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC41                                      | 3000                                | 0                                   | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC42                                      | 0                                   | 3000                                | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC43                                      | 0                                   | 0                                   | 3000                                | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC44                                      | 3000                                | 0                                   | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC45                                      | 0                                   | 3000                                | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC46                                      | 0                                   | 0                                   | 3000                                | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC47                                      | 2500                                | 0                                   | 0                                   | 2500                                  | 0.5           | 1250                                | 6.793478261                                   | 16         |
| PC48                                      | 0                                   | 3000                                | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC49                                      | 0                                   | 0                                   | 3000                                | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC50                                      | 2500                                | 0                                   | 0                                   | 2500                                  | 0.5           | 1250                                | 6.793478261                                   | 16         |
| PC51                                      | 0                                   | 3000                                | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC52                                      | 0                                   | 0                                   | 3000                                | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC53                                      | 3000                                | 0                                   | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC54                                      | 0                                   | 3000                                | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC55                                      | 0                                   | 0                                   | 3000                                | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC56                                      | 3000                                | 0                                   | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC57                                      | 0                                   | 3000                                | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC58                                      | 0                                   | 0                                   | 3000                                | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
| PC59                                      | 3000                                | 0                                   | 0                                   | 3000                                  | 0.5           | 1500                                | 8.152173913                                   | 16         |
|                                           | $\Sigma R = 30019$<br>125 A<br>RCCB | $\Sigma Y = 28050$<br>125 A<br>RCCB | $\Sigma B = 30025$<br>125 A<br>RCCB | $\Sigma$ Total Connected Load = 88094 | 0.5862069     | $\Sigma$ Demand Load = 37844        | $\Sigma I$ (per phase) = 205.674              |            |
|                                           |                                     |                                     |                                     |                                       |               | $\Sigma$ Demand Load (KW) = 37.844  | $\Sigma I$ (per phase calculated) = 205.674 A |            |
|                                           | Spare LC 36                         | Spare LC 37                         | Spare LC 35                         |                                       |               | $\Sigma$ Demand Load (KVA) = 47.305 | I (MCCB Rating) = 81.934 A                    |            |
|                                           | Spare PC 150,153                    | Spare PC 151,154                    | Spare PC 149,152                    |                                       |               |                                     | 14 Way TPN                                    |            |

| Details of Fourth Floor SDB-3(Middle Upper) |                      |                    |                    |                                       |               |                                    |                                              |            |
|---------------------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|------------------------------------|----------------------------------------------|------------|
| Light and Power Circuit                     | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                        | Current(I)                                   | MCB Rating |
|                                             | R                    | Y                  | B                  |                                       |               |                                    |                                              |            |
| LC12                                        | 676                  | 0                  | 0                  | 676                                   | 1             | 676                                | 3.673913043                                  | 6          |
| LC13                                        | 0                    | 730                | 0                  | 730                                   | 1             | 730                                | 3.967391304                                  | 6          |
| LC14                                        | 0                    | 0                  | 730                | 730                                   | 1             | 730                                | 3.967391304                                  | 6          |
| LC15                                        | 700                  | 0                  | 0                  | 700                                   | 1             | 700                                | 3.804347826                                  | 6          |
| LC16                                        | 0                    | 612                | 0                  | 612                                   | 1             | 612                                | 3.326086957                                  | 6          |
| PC60                                        | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
| PC61                                        | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
| PC62                                        | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
| PC63                                        | 1000                 | 0                  | 0                  | 1000                                  | 0.5           | 500                                | 2.717391304                                  | 16         |
| PC64                                        | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
| PC65                                        | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
| PC66                                        | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
| PC67                                        | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
| PC68                                        | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
| PC69                                        | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
| PC70                                        | 0                    | 1000               | 0                  | 1000                                  | 0.5           | 500                                | 2.717391304                                  | 16         |
| PC71                                        | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
| PC72                                        | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
| PC73                                        | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                               | 8.152173913                                  | 16         |
|                                             | $\Sigma R = 14376$   | $\Sigma Y = 14342$ | $\Sigma B = 12730$ | $\Sigma$ Total Connected Load = 41448 | 0.63157895    | $\Sigma$ Demand Load = 17948       | $\Sigma I$ (per phase) = 97.543              |            |
|                                             | 63A RCCB             | 63A RCCB           | 63A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 17.948 | $\Sigma I$ (per phase calculated) = 97.543 A |            |
|                                             | Spare LC 39          | Spare LC 40        | Spare LC 38        |                                       |               | $\Sigma$ Demand Load(KVA) = 22.435 | I (MCCB Rating) = 38.8585 A                  |            |
|                                             | Spare PC 156,159     | Spare PC 157,160   | Spare PC 155,158   |                                       |               |                                    | 12 Way TPN                                   |            |

| Details of Fourth Floor SDB-4(Middle Lower) |                         |                         |                         |                                       |                   |                                      |                                              |            |
|---------------------------------------------|-------------------------|-------------------------|-------------------------|---------------------------------------|-------------------|--------------------------------------|----------------------------------------------|------------|
| Light and Power Circuit                     | Wattage Per Phase(W)    |                         |                         | Total Connected Load                  | Demand Factor     | Demand Load                          | Current(I)                                   | MCB Rating |
|                                             | R                       | Y                       | B                       |                                       |                   |                                      |                                              |            |
| LC17                                        | 612                     | 0                       | 0                       | 612                                   | 1                 | 612                                  | 3.326086957                                  | 6          |
| LC18                                        | 0                       | 544                     | 0                       | 544                                   | 1                 | 544                                  | 2.956521739                                  | 6          |
| LC19                                        | 0                       | 0                       | 318                     | 318                                   | 1                 | 318                                  | 1.72826087                                   | 6          |
| LC20                                        | 580                     | 0                       | 0                       | 580                                   | 1                 | 580                                  | 3.152173913                                  | 6          |
| PC74                                        | 0                       | 3000                    | 0                       | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
| PC75                                        | 0                       | 0                       | 3000                    | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
| PC76                                        | 3000                    | 0                       | 0                       | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
| PC77                                        | 0                       | 3000                    | 0                       | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
| PC78                                        | 0                       | 0                       | 3000                    | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
| PC79                                        | 2000                    | 0                       | 0                       | 2000                                  | 0.5               | 1000                                 | 5.434782609                                  | 16         |
| PC80                                        | 0                       | 3000                    | 0                       | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
| PC81                                        | 0                       | 0                       | 2000                    | 2000                                  | 0.5               | 1000                                 | 5.434782609                                  | 16         |
| PC82                                        | 3000                    | 0                       | 0                       | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
| PC83                                        | 0                       | 3000                    | 0                       | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
| PC84                                        | 0                       | 0                       | 3000                    | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
| PC85                                        | 3000                    | 0                       | 0                       | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
| PC86                                        | 0                       | 3000                    | 0                       | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
| PC87                                        | 0                       | 0                       | 3000                    | 3000                                  | 0.5               | 1500                                 | 8.152173913                                  | 16         |
|                                             | <b>ΣR = 12192</b>       | <b>ΣY = 15544</b>       | <b>ΣB = 14318</b>       | <b>Σ Total Connected Load = 42054</b> | <b>0.61111111</b> | <b>Σ Demand Load = 22054</b>         | <b>ΣI(per phase) = 119.859</b>               |            |
|                                             | <b>63 A RCCB</b>        | <b>63A RCCB</b>         | <b>63 A RCCB</b>        |                                       |                   | <b>Σ Demand Load (KW) = 22.054</b>   | <b>ΣI (per phase calculated) = 119.859 A</b> |            |
|                                             | <b>Spare LC 43</b>      | <b>Spare LC 41</b>      | <b>Spare LC 42</b>      |                                       |                   | <b>Σ Demand Load (KVA) = 27.5675</b> | <b>I (MCCB Rating) = 47.748 A</b>            |            |
|                                             | <b>Spare PC 162,165</b> | <b>Spare PC 163,166</b> | <b>Spare PC 161,164</b> |                                       |                   |                                      | <b>12 Way TPN</b>                            |            |

| Details of Fourth Floor SDB-5(Right lower) |                      |                    |                    |                                       |               |                                     |                                             |            |
|--------------------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|-------------------------------------|---------------------------------------------|------------|
| Light and Power Circuit                    | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                         | Current(I)                                  | MCB Rating |
|                                            | R                    | Y                  | B                  |                                       |               |                                     |                                             |            |
| LC21                                       | 692                  | 0                  | 0                  | 692                                   | 1             | 692                                 | 3.760869565                                 | 6          |
| LC22                                       | 0                    | 748                | 0                  | 748                                   | 1             | 748                                 | 4.065217391                                 | 6          |
| LC23                                       | 0                    | 0                  | 786                | 786                                   | 1             | 786                                 | 4.27173913                                  | 6          |
| LC24                                       | 700                  | 0                  | 0                  | 700                                   | 1             | 700                                 | 3.804347826                                 | 6          |
| LC25                                       | 0                    | 568                | 0                  | 568                                   | 1             | 568                                 | 3.086956522                                 | 6          |
| LC26                                       | 0                    | 0                  | 748                | 748                                   | 1             | 748                                 | 4.065217391                                 | 6          |
| PC88                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC89                                       | 0                    | 2500               | 0                  | 2500                                  | 0.5           | 1250                                | 6.793478261                                 | 16         |
| PC90                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC91                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC92                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC93                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC94                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC95                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC96                                       | 0                    | 0                  | 2500               | 2500                                  | 0.5           | 1250                                | 6.793478261                                 | 16         |
| PC97                                       | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC98                                       | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC99                                       | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC100                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC101                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC102                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC103                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC104                                      | 0                    | 2500               | 0                  | 2500                                  | 0.5           | 1250                                | 6.793478261                                 | 16         |
| PC105                                      | 0                    | 0                  | 2000               | 2000                                  | 0.5           | 1000                                | 5.434782609                                 | 16         |
| PC106                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC107                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC108                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC109                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC110                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC111                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC112                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC113                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC114                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
| PC115                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                 | 16         |
|                                            | $\Sigma R = 31392$   | $\Sigma Y = 27316$ | $\Sigma B = 27034$ | $\Sigma$ Total Connected Load = 85742 | 0.58823529    | $\Sigma$ Demand Load = 37492        | $\Sigma I$ (per phase) = 203.761            |            |
|                                            | 125A RCCB            | 125A RCCB          | 125 A RCCB         |                                       |               | $\Sigma$ Demand Load (KW) = 37.492  | $\Sigma I$ (per phase calculated) = 203.761 |            |
|                                            | Spare LC 44          | Spare LC 45        | Spare LC 46        |                                       |               | $\Sigma$ Demand Load (KVA) = 46.865 | I (MCCB Rating) = 81.17 A                   |            |
|                                            | Spare PC 169,172     | Spare PC 167,170   | Spare PC 168,171   |                                       |               |                                     | 18 Way TPN                                  |            |

| Details of Fourth Floor SDB-6(Right Upper) |                      |                    |                    |                                               |               |                                             |                                                   |            |
|--------------------------------------------|----------------------|--------------------|--------------------|-----------------------------------------------|---------------|---------------------------------------------|---------------------------------------------------|------------|
| Light and Power Circuit                    | Wattage Per Phase(W) |                    |                    | Total Connected Load                          | Demand Factor | Demand Load                                 | Current(I)                                        | MCB Rating |
|                                            | R                    | Y                  | B                  |                                               |               |                                             |                                                   |            |
| LC27                                       | 494                  | 0                  | 0                  | 494                                           | 1             | 494                                         | 2.684782609                                       | 6          |
| LC28                                       | 0                    | 525                | 0                  | 525                                           | 1             | 525                                         | 2.85326087                                        | 6          |
| LC29                                       | 0                    | 0                  | 525                | 525                                           | 1             | 525                                         | 2.85326087                                        | 6          |
| LC30                                       | 525                  | 0                  | 0                  | 525                                           | 1             | 525                                         | 2.85326087                                        | 6          |
| LC31                                       | 0                    | 525                | 0                  | 525                                           | 1             | 525                                         | 2.85326087                                        | 6          |
| PC116                                      | 0                    | 0                  | 3000               | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC117                                      | 3000                 | 0                  | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC118                                      | 0                    | 3000               | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC119                                      | 0                    | 0                  | 2500               | 2500                                          | 0.5           | 1250                                        | 6.793478261                                       | 16         |
| PC120                                      | 3000                 | 0                  | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC121                                      | 0                    | 3000               | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC122                                      | 0                    | 0                  | 3000               | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC123                                      | 3000                 | 0                  | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC124                                      | 0                    | 3000               | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC125                                      | 0                    | 0                  | 3000               | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC126                                      | 3000                 | 0                  | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC127                                      | 0                    | 3000               | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC128                                      | 0                    | 0                  | 3000               | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC129                                      | 3000                 | 0                  | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC130                                      | 0                    | 3000               | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC131                                      | 0                    | 0                  | 3000               | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC132                                      | 3000                 | 0                  | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC133                                      | 0                    | 3000               | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC134                                      | 0                    | 0                  | 3000               | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC135                                      | 3000                 | 0                  | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC136                                      | 0                    | 3000               | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC137                                      | 0                    | 0                  | 3000               | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC138                                      | 3000                 | 0                  | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC139                                      | 0                    | 3000               | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC140                                      | 0                    | 0                  | 3000               | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC141                                      | 3000                 | 0                  | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
| PC142                                      | 0                    | 3000               | 0                  | 3000                                          | 0.5           | 1500                                        | 8.152173913                                       | 16         |
|                                            | $\Sigma R = 28019$   | $\Sigma Y = 28050$ | $\Sigma B = 27025$ | $\Sigma \text{ Total Connected Load} = 83094$ | 0.578125      | $\Sigma \text{ Demand Load} = 42844$        | $\Sigma I(\text{per phase}) = 232.848$            |            |
|                                            | 125A RCCB            | 125A RCCB          | 125 A RCCB         |                                               |               | $\Sigma \text{ Demand Load (KW)} = 42.844$  | $\Sigma I(\text{per phase calculated}) = 232.848$ |            |
|                                            | Spare LC 48          | Spare LC 49        | Spare LC 47        |                                               |               | $\Sigma \text{ Demand Load (KVA)} = 53.555$ | I (MCCB Rating) = 92.76 A                         |            |
|                                            | Spare PC 173,176     | Spare PC 174,177   | Spare PC 175,178   |                                               |               |                                             | 16 way TPN                                        |            |

g. Fifth Floor:

| Details of Fifth Floor SDB-1 |                      |                    |                    |                                       |               |                                    |                                                   |            |
|------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|------------------------------------|---------------------------------------------------|------------|
| Light and Power Circuit      | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                        | Current(I)                                        | MCB Rating |
|                              | R                    | Y                  | B                  |                                       |               |                                    |                                                   |            |
| LC15                         | 578                  | 0                  | 0                  | 578                                   | 1             | 578                                | 3.141304348                                       | 6          |
| LC16                         | 0                    | 726                | 0                  | 726                                   | 1             | 726                                | 3.945652174                                       | 6          |
| LC10                         | 736                  | 0                  | 0                  | 736                                   | 1             | 736                                | 4                                                 | 6          |
| PC1                          | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC2                          | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC3                          | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC4                          | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC5                          | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC6                          | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC7                          | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC8                          | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC9                          | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC10                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC11                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC12                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC13                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC14                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC15                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC16                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC17                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC18                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC19                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC20                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC47                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC48                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC49                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC50                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC51                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC52                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC53                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC54                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC55                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
| PC56                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                               | 6.52173913                                        | 16         |
|                              | $\Sigma R = 28314$   | $\Sigma Y = 27726$ | $\Sigma B = 27000$ | $\Sigma$ Total Connected Load = 83040 | 0.46          | $\Sigma$ Demand Load = 38040       | $\Sigma I(\text{per phase}) = 206.739$            |            |
|                              | 80A RCCB             | 100A RCCB          | 80A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 38.04  | $\Sigma I(\text{per phase calculated}) = 206.739$ |            |
|                              | Spare LC 26          | Spare LC 27        | Spare LC 28        |                                       |               | $\Sigma$ Demand Load (KVA) = 47.55 | I (MCCB Rating) = 74.564                          |            |
|                              | Spare PC 129,130     | Spare PC 131,132   | Spare PC 133,134   |                                       |               |                                    | 16 Way TPN                                        |            |

| Details of Fifth Floor SDB-2 |                      |                    |                    |                                       |               |                                       |                                                   |            |
|------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|---------------------------------------|---------------------------------------------------|------------|
| Light and Power Circuit      | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                           | Current(I)                                        | MCB Rating |
|                              | R                    | Y                  | B                  |                                       |               |                                       |                                                   |            |
| LC11                         | 0                    | 692                | 0                  | 692                                   | 1             | 692                                   | 3.760869565                                       | 6          |
| LC12                         | 0                    | 0                  | 551                | 551                                   | 1             | 551                                   | 2.994565217                                       | 6          |
| LC13                         | 388                  | 0                  | 0                  | 388                                   | 1             | 388                                   | 2.108695652                                       | 6          |
| LC14                         | 0                    | 784                | 0                  | 784                                   | 1             | 784                                   | 4.260869565                                       | 6          |
| PC23                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC24                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC25                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC26                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC27                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC28                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC30                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC31                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC32                         | 0                    | 0                  | 2000               | 2000                                  | 0.4           | 800                                   | 4.347826087                                       | 16         |
| PC33                         | 2000                 | 0                  | 0                  | 2000                                  | 0.4           | 800                                   | 4.347826087                                       | 16         |
| PC34                         | 0                    | 2000               | 0                  | 2000                                  | 0.4           | 800                                   | 4.347826087                                       | 16         |
| PC35                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC36                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC37                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC38                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC39                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC40                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC41                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC42                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC43                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC44                         |                      |                    | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC45                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC46                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
|                              | $\Sigma R = 23388$   | $\Sigma Y = 24476$ | $\Sigma B = 20551$ | $\Sigma$ Total Connected Load = 68415 | 0.488889      | $\Sigma$ Demand Load = 28815          | $\Sigma I(\text{per phase}) = 156.603$            |            |
|                              | 80A RCCB             | 80A RCCB           | 80A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 28.815    | $\Sigma I(\text{per phase calculated}) = 156.603$ |            |
|                              | Spare LC 29          | Spare LC 30        | Spare LC 31        |                                       |               | $\Sigma$ Demand Load (KVA) = 36.01875 | I (MCCB Rating) = 46.8                            |            |
|                              | Spare PC 135,136     | Spare PC 137,138   | Spare PC 139,140   |                                       |               |                                       | 16 Way TPN                                        |            |

| Details of Fifth Floor SDB-3 |                      |                    |                    |                                       |               |                                      |                                                   |            |
|------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|--------------------------------------|---------------------------------------------------|------------|
| Light and Power Circuit      | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                          | Current(I)                                        | MCB Rating |
|                              | R                    | Y                  | B                  |                                       |               |                                      |                                                   |            |
| LC7                          | 603                  | 0                  | 0                  | 603                                   | 1             | 603                                  | 3.277173913                                       | 6          |
| LC8                          | 0                    | 652                | 0                  | 652                                   | 1             | 652                                  | 3.543478261                                       | 6          |
| LC9                          | 0                    | 0                  | 656                | 656                                   | 1             | 656                                  | 3.565217391                                       | 6          |
| LC14                         | 763                  | 0                  | 0                  | 763                                   | 1             | 763                                  | 4.14673913                                        | 6          |
| PC21                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC22                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC29                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC57                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC58                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC59                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC60                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC61                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC62                         | 0                    | 0                  | 2000               | 2000                                  | 0.4           | 800                                  | 4.347826087                                       | 16         |
| PC63                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC64                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC65                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC66                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
| PC67                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                        | 16         |
|                              | $\Sigma R = 16366$   | $\Sigma Y = 15652$ | $\Sigma B = 11656$ | $\Sigma$ Total Connected Load = 43674 | 0.533333      | $\Sigma$ Demand Load = 19074         | $\Sigma I(\text{per phase}) = 103.663$            |            |
|                              | 63A RCCB             | 63A RCCB           | 63A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 19.074   | $\Sigma I(\text{per phase calculated}) = 102.147$ |            |
|                              | Spare LC 32          | Spare LC 33        | Spare LC 34        |                                       |               | $\Sigma$ Demand Load (KVA) = 23.8425 | I (MCCB Rating) = 40.69                           |            |
|                              | Spare PC 141,142     | Spare PC 143,144   | Spare PC 145,146   |                                       |               |                                      | 12 Way TPN                                        |            |

| Details of Fifth Floor SDB-4 |                      |                    |                    |                                       |               |                                       |                                                   |            |
|------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|---------------------------------------|---------------------------------------------------|------------|
| Light and Power Circuit      | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                           | Current(I)                                        | MCB Rating |
|                              | R                    | Y                  | B                  |                                       |               |                                       |                                                   |            |
| LC23                         | 635                  | 0                  | 0                  | 635                                   | 1             | 635                                   | 3.451086957                                       | 6          |
| LC24                         | 0                    | 476                | 0                  | 476                                   | 1             | 476                                   | 2.586956522                                       | 6          |
| LC19                         | 0                    | 0                  | 736                | 736                                   | 1             | 736                                   | 4                                                 | 6          |
| LC25                         | 576                  | 0                  | 0                  | 576                                   | 1             | 576                                   | 3.130434783                                       | 6          |
| PC56                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC68                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC69                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC70                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC71                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC72                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC73                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC74                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC75                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC76                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC77                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC78                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC79                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC80                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC81                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC82                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
| PC83                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                        | 16         |
|                              | $\Sigma R = 19211$   | $\Sigma Y = 18476$ | $\Sigma B = 15736$ | $\Sigma$ Total Connected Load = 53423 | 0.514286      | $\Sigma$ Demand Load = 22823          | $\Sigma I(\text{per phase}) = 124.038$            |            |
|                              | 63A RCCB             | 63A RCCB           | 63A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 22.823    | $\Sigma I(\text{per phase calculated}) = 124.038$ |            |
|                              | Spare LC 35          | Spare LC 36        | Spare LC 37        |                                       |               | $\Sigma$ Demand Load (KVA) = 28.52875 | I (MCCB Rating) = 49.41                           |            |
|                              | Spare PC 147,148     | Spare PC 149,150   | Spare PC 151,152   |                                       |               |                                       | 14 Way TPN                                        |            |

| Details of Fifth Floor SDB-5 |                      |                    |                    |                                       |               |                                       |                                                  |            |
|------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|---------------------------------------|--------------------------------------------------|------------|
| Light and Power Circuit      | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                           | Current(I)                                       | MCB Rating |
|                              | R                    | Y                  | B                  |                                       |               |                                       |                                                  |            |
| LC20                         | 692                  | 0                  | 0                  | 692                                   | 1             | 692                                   | 3.760869565                                      | 6          |
| LC21                         | 0                    | 551                | 0                  | 551                                   | 1             | 551                                   | 2.994565217                                      | 6          |
| LC22                         | 0                    | 0                  | 388                | 388                                   | 1             | 388                                   | 2.108695652                                      | 6          |
| PC84                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC85                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC86                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC87                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC88                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC89                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC90                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC91                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC92                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC93                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC94                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC95                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC96                         | 2000                 | 0                  | 0                  | 2000                                  | 0.4           | 800                                   | 4.347826087                                      | 16         |
| PC97                         | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC98                         | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC99                         | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC100                        | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC101                        | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC102                        | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC103                        | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC104                        | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC105                        | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
| PC106                        | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                  | 6.52173913                                       | 16         |
|                              | $\Sigma R = 23692$   | $\Sigma Y = 24551$ | $\Sigma B = 21388$ | $\Sigma$ Total Connected Load = 69631 | 0.469231      | $\Sigma$ Demand Load = 28831          | $\Sigma I(\text{per phase}) = 156.69$            |            |
|                              | 80A RCCB             | 80A RCCB           | 80A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 28.831    | $\Sigma I(\text{per phase calculated}) = 156.69$ |            |
|                              | Spare LC 38          | Spare LC 39        | Spare LC 40        |                                       |               | $\Sigma$ Demand Load (KVA) = 30.03875 | I (MCCB Rating) = 62.42                          |            |
|                              | Spare PC 153,154     | Spare PC 155,156   | Spare PC 157,158   |                                       |               |                                       | 14 Way TPN                                       |            |

| Details of Fifth Floor DB-6 |                      |                    |                    |                                       |               |                                      |                                            |            |
|-----------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|--------------------------------------|--------------------------------------------|------------|
| Light and Power Circuit     | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                          | Current(I)                                 | MCB Rating |
|                             | R                    | Y                  | B                  |                                       |               |                                      |                                            |            |
| LC1                         | 612                  | 0                  | 0                  | 612                                   | 1             | 612                                  | 3.326086957                                | 6          |
| LC2                         | 0                    | 612                | 0                  | 612                                   | 1             | 612                                  | 3.326086957                                | 6          |
| LC3                         | 0                    | 0                  | 748                | 748                                   | 1             | 748                                  | 4.065217391                                | 6          |
| LC4                         | 510                  | 0                  | 0                  | 510                                   | 1             | 510                                  | 2.77173913                                 | 6          |
| LC5                         | 0                    | 612                | 0                  | 612                                   | 1             | 612                                  | 3.326086957                                | 6          |
| LC6                         | 0                    | 0                  | 510                | 510                                   | 1             | 510                                  | 2.77173913                                 | 6          |
| LC17                        | 726                  | 0                  | 0                  | 726                                   | 1             | 726                                  | 3.945652174                                | 6          |
| LC18                        | 528                  | 0                  | 0                  | 528                                   | 1             | 528                                  | 2.869565217                                | 6          |
| PC107                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC108                       | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC109                       | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC110                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC111                       | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC112                       | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC113                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC114                       | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC115                       | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC116                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC117                       | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC118                       | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC119                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC120                       | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC121                       | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC122                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC123                       | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC124                       | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC125                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC126                       | 0                    | 3000               | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC127                       | 0                    | 0                  | 3000               | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC128                       | 3000                 | 0                  | 0                  | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
|                             | $\Sigma R = 26376$   | $\Sigma Y = 22224$ | $\Sigma B = 22258$ | $\Sigma$ Total Connected Load = 70858 | 0.56          | $\Sigma$ Demand Load = 31258         | $\Sigma I$ (per phase) = 169.88            |            |
|                             | 100A RCCB            | 100A RCCB          | 63A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 31.258   | $\Sigma I$ (per phase calculated) = 169.88 |            |
|                             | Spare LC 41          | Spare LC 42        | Spare LC 43        |                                       |               | $\Sigma$ Demand Load (KVA) = 39.0725 | I (MCCB Rating) = 67.67                    |            |
|                             | Spare PC 159,160     | Spare PC 161,162   | Spare PC 163,164   |                                       |               |                                      | 16 way TPN                                 |            |

## h. Sixth Floor:

| Details of Sixth Floor SDB-1(Left Upper) |                      |                   |                   |                                       |               |                                      |                                            |            |
|------------------------------------------|----------------------|-------------------|-------------------|---------------------------------------|---------------|--------------------------------------|--------------------------------------------|------------|
| Light and Power Circuit                  | Wattage Per Phase(W) |                   |                   | Total Connected Load                  | Demand Factor | Demand Load                          | Current(I)                                 | MCB Rating |
|                                          | R                    | Y                 | B                 |                                       |               |                                      |                                            |            |
| LC7                                      | 356                  | 0                 | 0                 | 356                                   | 1             | 356                                  | 1.934782609                                | 6          |
| LC8                                      | 0                    | 558               | 0                 | 558                                   | 1             | 558                                  | 3.032608696                                | 6          |
| PC3                                      | 0                    | 0                 | 3000              | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC4                                      | 3000                 | 0                 | 0                 | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC5                                      |                      | 3000              | 0                 | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC6                                      | 0                    | 0                 | 3000              | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC7                                      | 3000                 | 0                 | 0                 | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
| PC8                                      | 0                    | 3000              | 0                 | 3000                                  | 0.4           | 1200                                 | 6.52173913                                 | 16         |
|                                          | $\Sigma R = 6356$    | $\Sigma Y = 6558$ | $\Sigma B = 6000$ | $\Sigma$ Total Connected Load = 18914 | 0.55          | $\Sigma$ Demand Load = 8114          | $\Sigma I$ (per phase) = 44.098            |            |
|                                          | 25A RCCB             | 25A RCCB          | 25A RCCB          |                                       |               | $\Sigma$ Demand Load (KW) = 8.114    | $\Sigma I$ (per phase calculated) = 44.098 |            |
|                                          | Spare LC 19          | Spare LC 20       | Spare LC 21       |                                       |               | $\Sigma$ Demand Load (KVA) = 10.1425 | I (MCB Rating) = 17.56                     |            |
|                                          | Spare PC 60,61       | Spare PC 62,63    | Spare PC 64, 65   |                                       |               |                                      | 8 Way TPN                                  |            |

| Details of Sixth Floor SDB-2(Left Lower) |                      |                    |                    |                                       |               |                                       |                                            |            |
|------------------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|---------------------------------------|--------------------------------------------|------------|
| Light and Power Circuit                  | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                           | Current(I)                                 | MCB Rating |
|                                          | R                    | Y                  | B                  |                                       |               |                                       |                                            |            |
| LC9                                      | 796                  | 0                  | 0                  | 796                                   | 1             | 796                                   | 4.326086957                                | 6          |
| LC10                                     | 0                    | 773                | 0                  | 773                                   | 1             | 773                                   | 4.201086957                                | 6          |
| PC1                                      |                      |                    | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC2                                      | 3000                 |                    |                    | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC9                                      |                      | 3000               |                    | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC10                                     |                      |                    | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC11                                     | 3000                 |                    |                    | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC12                                     |                      | 3000               |                    | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC13                                     |                      |                    | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC14                                     | 3000                 |                    |                    | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC15                                     |                      | 3000               |                    | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC16                                     |                      |                    | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC17                                     | 3000                 |                    |                    | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC18                                     |                      | 3000               |                    | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC58                                     |                      |                    | 3000               | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC59                                     | 3000                 |                    |                    | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
|                                          | $\Sigma R = 15796$   | $\Sigma Y = 12773$ | $\Sigma B = 15000$ | $\Sigma$ Total Connected Load = 37569 | 0.485714      | $\Sigma$ Demand Load = 15969          | $\Sigma I$ (per phase) = 86.788            |            |
|                                          | 63A RCCB             | 40A RCCB           | 63A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 15.969    | $\Sigma I$ (per phase calculated) = 86.788 |            |
|                                          | Spare LC 22          | Spare LC 23        | Spare LC 24        |                                       |               | $\Sigma$ Demand Load (KVA) = 19.96125 | I (MCB Rating) = 34.57                     |            |
|                                          | Spare PC 66,67       | Spare PC 68,69     | Spare PC 70,71     |                                       |               |                                       | 10 Way TPN                                 |            |

| Details of Sixth Floor SDB-3(Middle Upper) |                       |                       |                       |                                       |                 |                                      |                                           |            |
|--------------------------------------------|-----------------------|-----------------------|-----------------------|---------------------------------------|-----------------|--------------------------------------|-------------------------------------------|------------|
| Light and Power Circuit                    | Wattage Per Phase(W)  |                       |                       | Total Connected Load                  | Demand Factor   | Demand Load                          | Current(I)                                | MCB Rating |
|                                            | R                     | Y                     | B                     |                                       |                 |                                      |                                           |            |
| LC4                                        | 510                   | 0                     | 0                     | 510                                   | 1               | 510                                  | 2.77173913                                | 6          |
| LC5                                        | 0                     | 612                   | 0                     | 612                                   | 1               | 612                                  | 3.326086957                               | 6          |
| LC6                                        | 0                     | 0                     | 510                   | 510                                   | 1               | 510                                  | 2.77173913                                | 6          |
| LC11                                       | 763                   | 0                     | 0                     | 763                                   | 1               | 763                                  | 4.14673913                                | 6          |
| LC15                                       | 0                     | 763                   | 0                     | 763                                   | 1               | 763                                  | 4.14673913                                | 6          |
| PC19                                       | 3000                  | 0                     | 0                     | 3000                                  | 0.4             | 1200                                 | 6.52173913                                | 16         |
| PC20                                       | 0                     | 3000                  | 0                     | 3000                                  | 0.4             | 1200                                 | 6.52173913                                | 16         |
| PC21                                       | 0                     | 0                     | 3000                  | 3000                                  | 0.4             | 1200                                 | 6.52173913                                | 16         |
| PC22                                       | 3000                  | 0                     | 0                     | 3000                                  | 0.4             | 1200                                 | 6.52173913                                | 16         |
| PC23                                       | 0                     | 3000                  | 0                     | 3000                                  | 0.4             | 1200                                 | 6.52173913                                | 16         |
| PC24                                       | 0                     | 0                     | 3000                  | 3000                                  | 0.4             | 1200                                 | 6.52173913                                | 16         |
| PC25                                       | 3000                  | 0                     | 0                     | 3000                                  | 0.4             | 1200                                 | 6.52173913                                | 16         |
| PC26                                       | 0                     | 3000                  | 0                     | 3000                                  | 0.4             | 1200                                 | 6.52173913                                | 16         |
|                                            | <b>ΣR = 10273</b>     | <b>ΣY = 10375</b>     | <b>ΣB = 6510</b>      | <b>Σ Total Connected Load = 27158</b> | <b>0.630769</b> | <b>Σ Demand Load = 12758</b>         | <b>ΣI (per phase) = 69.337</b>            |            |
|                                            | <b>63A RCCB</b>       | <b>63A RCCB</b>       | <b>40A RCCB</b>       |                                       |                 | <b>Σ Demand Load (KW) = 12.758</b>   | <b>ΣI (per phase calculated) = 69.337</b> |            |
|                                            | <b>Spare LC 25</b>    | <b>Spare LC 26</b>    | <b>Spare LC 27</b>    |                                       |                 | <b>Σ Demand Load (KVA) = 15.9475</b> | <b>I (MCCB Rating) = 27.621</b>           |            |
|                                            | <b>Spare PC 72,73</b> | <b>Spare PC 74,75</b> | <b>Spare PC 76,77</b> |                                       |                 |                                      | <b>10 Way TPN</b>                         |            |

| Details of Sixth Floor SDB-4(Middle Lower) |                      |                   |                    |                                       |               |                                     |                                           |            |
|--------------------------------------------|----------------------|-------------------|--------------------|---------------------------------------|---------------|-------------------------------------|-------------------------------------------|------------|
| Light and Power Circuit                    | Wattage Per Phase(W) |                   |                    | Total Connected Load                  | Demand Factor | Demand Load                         | Current(I)                                | MCB Rating |
|                                            | R                    | Y                 | B                  |                                       |               |                                     |                                           |            |
| LC1                                        | 612                  | 0                 | 0                  | 612                                   | 1             | 612                                 | 3.326086957                               | 6          |
| LC2                                        | 0                    | 612               | 0                  | 612                                   | 1             | 612                                 | 3.326086957                               | 6          |
| LC3                                        | 0                    | 0                 | 748                | 748                                   | 1             | 748                                 | 4.065217391                               | 6          |
| LC15                                       | 440                  | 0                 | 0                  | 440                                   | 1             | 440                                 | 2.391304348                               | 6          |
| LC16                                       | 0                    | 312               | 0                  | 312                                   | 1             | 312                                 | 1.695652174                               | 6          |
| LC17                                       | 0                    | 0                 | 760                | 760                                   | 1             | 760                                 | 4.130434783                               | 6          |
| PC27                                       | 0                    | 3000              | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                | 16         |
| PC28                                       | 0                    | 0                 | 3000               | 3000                                  | 0.4           | 1200                                | 6.52173913                                | 16         |
| PC29                                       | 3000                 | 0                 | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                | 16         |
| PC30                                       | 0                    | 3000              | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                | 16         |
| PC31                                       | 0                    | 0                 | 3000               | 3000                                  | 0.4           | 1200                                | 6.52173913                                | 16         |
| PC32                                       | 3000                 | 0                 | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                | 16         |
| PC33                                       | 0                    | 3000              | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                | 16         |
| PC34                                       | 0                    | 0                 | 3000               | 3000                                  | 0.4           | 1200                                | 6.52173913                                | 16         |
| PC35                                       | 3000                 | 0                 | 0                  | 3000                                  | 0.4           | 1200                                | 6.52173913                                | 16         |
|                                            | $\Sigma R = 10052$   | $\Sigma Y = 9924$ | $\Sigma B = 10508$ | $\Sigma$ Total Connected Load = 30484 | 0.64          | $\Sigma$ Demand Load = 14284        | $\Sigma I$ (per phase) = 77.63            |            |
|                                            | 63A RCCB             | 63A RCCB          | 63A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 14.284  | $\Sigma I$ (per phase calculated) = 77.63 |            |
|                                            | Spare LC 28          | Spare LC 29       | Spare LC 30        |                                       |               | $\Sigma$ Demand Load (KVA) = 17.855 | I (MCCB Rating) = 30.92                   |            |
|                                            | Spare PC 78,79       | Spare PC 80,81    | Spare PC 82,83     |                                       |               |                                     | 10 Way TPN                                |            |

| Details of Sixth Floor SDB-5(Right Upper) |                      |                   |                   |                                       |               |                                       |                                            |            |
|-------------------------------------------|----------------------|-------------------|-------------------|---------------------------------------|---------------|---------------------------------------|--------------------------------------------|------------|
| Light and Power Circuit                   | Wattage Per Phase(W) |                   |                   | Total Connected Load                  | Demand Factor | Demand Load                           | Current(I)                                 | MCB Rating |
|                                           | R                    | Y                 | B                 |                                       |               |                                       |                                            |            |
| LC12                                      | 558                  | 0                 | 0                 | 558                                   | 1             | 558                                   | 3.032608696                                | 6          |
| LC13                                      | 0                    | 356               | 0                 | 356                                   | 1             | 356                                   | 1.934782609                                | 6          |
| LC14                                      | 0                    | 0                 | 773               | 773                                   | 1             | 773                                   | 4.201086957                                | 6          |
| PC36                                      | 3000                 | 0                 | 0                 | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC37                                      | 0                    | 3000              | 0                 | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC38                                      | 0                    | 0                 | 3000              | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC39                                      | 3000                 | 0                 | 0                 | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC40                                      | 0                    | 3000              | 0                 | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC41                                      | 0                    | 0                 | 3000              | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC42                                      | 3000                 | 0                 | 0                 | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
| PC43                                      | 0                    | 3000              | 0                 | 3000                                  | 0.4           | 1200                                  | 6.52173913                                 | 16         |
|                                           | $\Sigma R = 9558$    | $\Sigma Y = 9356$ | $\Sigma B = 6773$ | $\Sigma$ Total Connected Load = 25687 | 0.563636      | $\Sigma$ Demand Load = 11287          | $\Sigma I$ (per phase) = 61.342            |            |
|                                           | 40A RCCB             | 40A RCCB          | 25A RCCB          |                                       |               | $\Sigma$ Demand Load (KW) = 11.287    | $\Sigma I$ (per phase calculated) = 61.342 |            |
|                                           | Spare LC 31          | Spare LC 32       | Spare LC 33       |                                       |               | $\Sigma$ Demand Load (KVA) = 14.10875 | I (MCCB Rating) = 24.43                    |            |
|                                           | Spare PC 84,85       | Spare PC 86,87    | Spare PC 88,89    |                                       |               |                                       | 10 Way TPN                                 |            |

| Details of Sixth Floor SDB-6(Right Lower) |                      |                    |                    |                                       |               |                                     |                                                   |            |
|-------------------------------------------|----------------------|--------------------|--------------------|---------------------------------------|---------------|-------------------------------------|---------------------------------------------------|------------|
| Light and Power Circuit                   | Wattage Per Phase(W) |                    |                    | Total Connected Load                  | Demand Factor | Demand Load                         | Current(I)                                        | MCB Rating |
|                                           | R                    | Y                  | B                  |                                       |               |                                     |                                                   |            |
| LC18                                      | 796                  | 0                  | 0                  | 796                                   | 1             | 796                                 | 4.326086957                                       | 6          |
| PC44                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC45                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC46                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC47                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC48                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC49                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC50                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC51                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC52                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC53                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC54                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC55                                      | 0                    | 0                  | 3000               | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC56                                      | 3000                 | 0                  | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
| PC57                                      | 0                    | 3000               | 0                  | 3000                                  | 0.5           | 1500                                | 8.152173913                                       | 16         |
|                                           | $\Sigma R = 15796$   | $\Sigma Y = 15000$ | $\Sigma B = 12000$ | $\Sigma$ Total Connected Load = 42796 | 0.533333      | $\Sigma$ Demand Load = 21796        | $\Sigma I(\text{per phase}) = 118.457$            |            |
|                                           | 63A RCCB             | 63A RCCB           | 63A RCCB           |                                       |               | $\Sigma$ Demand Load (KW) = 21.796  | $\Sigma I(\text{per phase calculated}) = 118.457$ |            |
|                                           | Spare LC 34          | Spare LC 35        | Spare LC 36        |                                       |               | $\Sigma$ Demand Load (KVA) = 27.245 | I (MCCB Rating) = 47.19                           |            |
|                                           | Spare PC 90,91       | Spare PC 92,93     | Spare PC 94,95     |                                       |               |                                     | 10 ay TPN                                         |            |

## 6. Cable Calculations (FDB-SDB):

The cables used will be 4-cored PVC insulated armored Aluminum cables. The formula to calculate the minimum area of the conductor is,  $A \geq 1.5KI l / \Delta V$

where,

$$K = \text{resistivity of aluminum} = 0.03 \Omega \text{mm}^2/\text{m}$$

I = current in Amperes

l = length of the conductor from MDB to SDB in meters

$\Delta V$  = permissible Voltage Drop = 2% of 230V = 4.6V

### 6.1. BASEMENT FLOOR

#### 6.1.1. FDB to SDB-1

$$I = 79.574 \text{ A}$$

$$L = 58.048 \text{ m} + 15\% = 66.7552 \text{ m}$$

$$A \geq \frac{1.5KI l}{\Delta V} \text{ mm}^2 = 51.965 \text{ mm}^2$$

$\therefore$  The nearest cable size is 70 mm<sup>2</sup>/115A (pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.1.2. FDB to SDB-2

$$I = 92.093 \text{ A}$$

$$L = 14.80 \text{ m} + 15\% = 17.02 \text{ m}$$

$$A \geq \frac{1.5KI l}{\Delta V} \text{ mm}^2 = 15.335 \text{ mm}^2$$

$\therefore$  The nearest cable size is 50 mm<sup>2</sup>/95A (pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.1.3. FDB to SDB-3

$$I = 78.36 \text{ A}$$

$$L = 70.993 \text{ m} + 15\% = 81.642 \text{ m}$$

$$A \geq \frac{1.5KI l}{\Delta V} \text{ mm}^2 = 62.533 \text{ mm}^2$$

$\therefore$  The nearest cable size is 70 mm<sup>2</sup>/115A (pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.1.4. FDB to SDB-4

$$I = 74.9 \text{ A}$$

$$L = 103.983 \text{ m} + 15\% = 119.58 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 87.6185 \text{ mm}^2$$

∴ The nearest cable size is 95mm<sup>2</sup>/140A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.1.5. FDB to SDB-5

$$I = 91.45 \text{ A}$$

$$L = 86.2 \text{ m} + 15\% = 99.13 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 88.685 \text{ mm}^2$$

∴ The nearest cable size is 95 mm<sup>2</sup>/140A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.2. GROUND FLOOR

#### 6.2.1. FDB to SDB-1

$$I = 70.33 \text{ A}$$

$$L = 47.053 \text{ m} + 15\% = 54.111 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 37.22 \text{ mm}^2$$

∴ The nearest cable size is 50 mm<sup>2</sup>/95A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.2.2. FDB to SDB-2

$$I = 67.29 \text{ A}$$

$$L = 66.744 \text{ m} + 15\% = 76.7556 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 50.526 \text{ mm}^2$$

∴ The nearest cable size is 70 mm<sup>2</sup>/115A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.2.3. FDB to SDB-3

$$I = 81.43 \text{ A}$$

$$L = 45.502 \text{ m} + 15\% = 52.3273 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 41.68 \text{ mm}^2$$

∴ The nearest cable size is 50 mm<sup>2</sup>/95A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.2.4. FDB to SDB-4

$$I = 64.56A$$

$$L = 16.759m + 15\% = 19.273 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 12.17 \text{ mm}^2$$

∴ The nearest cable size is 35mm<sup>2</sup>/77A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.2.5. FDB to SDB-5

$$I = 55.15A$$

$$L = 93.47m + 15\% = 107.49m$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 57.992 \text{ mm}^2$$

∴ The nearest cable size is 70 mm<sup>2</sup>/115A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.3. FIRST FLOOR

#### 6.3.1. FDB to SDB-1

$$I = 119.44A$$

$$L = 47.053 \text{ m} + 15\% = 54.111 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 63.22 \text{ mm}^2$$

∴ The nearest cable size is 95 mm<sup>2</sup>/140A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.3.2. FDB to SDB-2

$$I = 106.84A$$

$$L = 66.744 \text{ m} + 15\% = 76.7556 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 80.22 \text{ mm}^2$$

∴ The nearest cable size is 95 mm<sup>2</sup>/140A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.3.3. FDB to SDB-3

$$I = 86.08A$$

$$L = 45.502 \text{ m} + 15\% = 52.3273 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 44.06 \text{ mm}^2$$

∴ The nearest cable size is 50 mm<sup>2</sup>/95A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.3.4. FDB to SDB-4

$$I = 78.77\text{A}$$

$$L = 16.759\text{m} + 15\% = 19.273\text{ m}$$

$$A \geq \frac{1.5\text{kIL}}{\Delta v} \text{ mm}^2 = 14.85 \text{ mm}^2$$

∴ The nearest cable size is 50mm<sup>2</sup>/95A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.3.5. FDB to SDB-5

$$I = 73.56\text{A}$$

$$L = 93.47\text{m} + 15\% = 107.49\text{m}$$

$$A \geq \frac{1.5\text{kIL}}{\Delta v} \text{ mm}^2 = 77.35 \text{ mm}^2$$

∴ The nearest cable size is 95 mm<sup>2</sup>/140A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.4. SECOND FLOOR

#### 6.4.1. FDB to SDB-1

$$I = 73.92\text{ A}$$

$$L = 67.181\text{ m} + 15\% = 77.25\text{ m}$$

$$A \geq \frac{1.5\text{kIL}}{\Delta v} \text{ mm}^2 = 37.24 \text{ mm}^2$$

∴ The nearest cable size is 50 mm<sup>2</sup>/95A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.4.2. FDB to SDB-2

$$I = 71.403\text{ A}$$

$$L = 43.302\text{ m} + 15\% = 54.4\text{ m}$$

$$A \geq \frac{1.5\text{kIL}}{\Delta v} \text{ mm}^2 = 37.99 \text{ mm}^2$$

∴ The nearest cable size is 50 mm<sup>2</sup>/95A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.4.3. FDB to SDB-3

$$I = 42.127\text{ A}$$

$$L = 45.502\text{ m} + 15\% = 52.327\text{ m}$$

$$A \geq \frac{1.5\text{kIL}}{\Delta v} \text{ mm}^2 = 21.56 \text{ mm}^2$$

∴ The nearest cable size is 35 mm<sup>2</sup>/77A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.4.4. FDB to SDB-4

$$I = 67.106 \text{ A}$$

$$L = 16.75 \text{ m} + 15\% = 19.272 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 12.66 \text{ mm}^2$$

∴ The nearest cable size is 35 mm<sup>2</sup>/77A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.4.5. FDB to SDB-5

$$I = 67.53 \text{ A}$$

$$L = 94.05 \text{ m} + 15\% = 108.15 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 71.44 \text{ mm}^2$$

∴ The nearest cable size is 95 mm<sup>2</sup>/140A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.4.6. FDB to SDB-6

$$I = 77.89 \text{ A}$$

$$L = 74.478 \text{ m} + 15\% = 85.65 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 65.26 \text{ mm}^2$$

∴ The nearest cable size is 70 mm<sup>2</sup>/115A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.5. THIRD FLOOR

#### 6.5.1. FDB to SDB-1

$$I = 106.049 \text{ A}$$

$$L = 46.053 \text{ m} + 15\% = 54.111 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 56.1366 \text{ mm}^2$$

∴ The nearest cable size is 70 mm<sup>2</sup>/115A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.5.2. FDB to SDB-2

$$I = 81.558 \text{ A}$$

$$L = 66.744 \text{ m} + 15\% = 76.7556 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 61.24 \text{ mm}^2$$

∴ The nearest cable size is 70mm<sup>2</sup>/115A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.5.3. FDB to SDB-3

$$I = 42.127A$$

$$L = 45.502 \text{ m} + 15\% = 52.3273 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 21.5647 \text{ mm}^2$$

∴ The nearest cable size is 25mm<sup>2</sup>/63A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.5.4. FDB to SDB-4

$$I = 67.16A$$

$$L = 16.759\text{m} + 15\% = 19.273 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 12.66 \text{ mm}^2$$

∴ The nearest cable size is 35mm<sup>2</sup>/77A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.5.5. FDB to SDB-5

$$I = 81.55A$$

$$L = 93.47\text{m} + 15\% = 107.49\text{m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 85.75 \text{ mm}^2$$

∴ The nearest cable size is 95 mm<sup>2</sup>/ 140A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.5.6. FDB to SDB-6

$$I = 106.05A$$

$$L = 74.554\text{m} + 15\% = 85.737 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 88.947 \text{ mm}^2$$

∴ The nearest cable size is 95 mm<sup>2</sup>/140A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

## 6.6. FOURTH FLOOR

### 6.6.1. FDB to SDB-1

$$I = 104.988 \text{ A}$$

$$L = 43.35 \text{ m} + 15\% = 49.853 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 46.598 \text{ mm}^2$$

∴ The nearest cable size is 70 mm<sup>2</sup>/115A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.6.2. FDB to SDB-2

$$I = 81.934 \text{ A}$$

$$L = 59.17 \text{ m} + 15\% = 68.046 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 54.54 \text{ mm}^2$$

∴ The nearest cable size is 70mm<sup>2</sup>/115A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.6.3. FDB to SDB-3

$$I = 38.858 \text{ A}$$

$$L = 5.3475 \text{ m} + 15\% = 6.15 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 2.337 \text{ mm}^2$$

∴ The nearest cable size is 10 mm<sup>2</sup>/39A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.6.4. FDB to SDB-4

$$I = 47.748 \text{ A}$$

$$L = 13.716 \text{ m} + 15\% = 15.7734 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 5.796 \text{ mm}^2$$

∴ The nearest cable size is 16mm<sup>2</sup>/50A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.6.5. FDB to SDB-5

$$I = 81.17 \text{ A}$$

$$L = 57.1323 \text{ m} + 15\% = 65.7 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 52.171 \text{ mm}^2$$

∴ The nearest cable size is 70 mm<sup>2</sup>/ 115A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.6.6. FDB to SDB-6

$$I = 92.76 \text{ A}$$

$$L = 74.02 \text{ m} + 15\% = 85.128 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 77.2455 \text{ mm}^2$$

∴ The nearest cable size is 95 mm<sup>2</sup>/140A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

## 6.7. FIFTH FLOOR

### 6.7.1. FDB to SDB-1

$$I = 74.564 \text{ A}$$

$$L = 61.2357 \text{ m} + 15\% = 70.421 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 51.367 \text{ mm}^2$$

∴ The nearest cable size is 70 mm<sup>2</sup>/115A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.7.2. FDB to SDB-2

$$I = 51.932 \text{ A}$$

$$L = 51.93174 \text{ m} + 15\% = 59.72 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 27.34 \text{ mm}^2$$

∴ The nearest cable size is 35 mm<sup>2</sup>/77A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.7.3. FDB to SDB-3

$$I = 40.69 \text{ A}$$

$$L = 1.40313 \text{ m} + 15\% = 1.6136 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 0.64 \text{ mm}^2$$

∴ The nearest cable size is 16 mm<sup>2</sup>/50A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.7.4. FDB to SDB-4

$$I = 49.41 \text{ A}$$

$$L = 55.62874 \text{ m} + 15\% = 63.973 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 30.92 \text{ mm}^2$$

∴ The nearest cable size is 35mm<sup>2</sup>/77A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.7.5. FDB to SDB-5

$$I = 62.42 \text{ A}$$

$$L = 86.84 \text{ m} + 15\% = 99.866 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 60.981 \text{ mm}^2$$

∴ The nearest cable size is 70 mm<sup>2</sup>/115A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.7.6. FDB to SDB-6

$$I = 67.67 \text{ A}$$

$$L = 5.39598 \text{ m} + 15\% = 6.2 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 4.1078 \text{ mm}^2$$

∴ The nearest cable size is 35 mm<sup>2</sup>/77A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 6.8. SIXTH FLOOR

#### 6.8.1. FDB to SDB-1

$$I = 17.56 \text{ A}$$

$$L = 42.96 \text{ m} + 15\% = 49.4 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 8.437 \text{ mm}^2$$

∴ The nearest cable size is 10 mm<sup>2</sup>/39A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.8.2. FDB to SDB-2

$$I = 34.57 \text{ A}$$

$$L = 52.96 \text{ m} + 15\% = 60.9 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 20.597 \text{ mm}^2$$

∴ The nearest cable size is 25 mm<sup>2</sup>/63A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.8.3. FDB to SDB-3

$$I = 27.62 \text{ A}$$

$$L = 7.5 \text{ m} + 15\% = 8.618 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 2.328 \text{ mm}^2$$

∴ The nearest cable size is 6 mm<sup>2</sup>/30A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.8.4. FDB to SDB-4

$$I = 30.92 \text{ A}$$

$$L = 16.66 \text{ m} + 15\% = 19.162 \text{ m}$$

$$A \geq \frac{1.5 \text{ kIL}}{\Delta v} \text{ mm}^2 = 5.796 \text{ mm}^2$$

∴ The nearest cable size is 10mm<sup>2</sup>/40A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.8.5. FDB to SDB-5

$$I = 24.43\text{A}$$

$$L = 66.869\text{m} + 15\% = 76.9\text{ m}$$

$$A \geq \frac{1.5\text{kIL}}{\Delta v} \text{ mm}^2 = 18.377 \text{ mm}^2$$

∴ The nearest cable size is 25 mm<sup>2</sup>/63A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

#### 6.8.6. FDB to SDB-6

$$I = 47.19\text{A}$$

$$L = 77.376 \text{ m} + 15\% = 88.98 \text{ m}$$

$$A \geq \frac{1.5\text{kIL}}{\Delta v} \text{ mm}^2 = 41.078 \text{ mm}^2$$

∴ The nearest cable size is 50 mm<sup>2</sup>/95A(pg.21) rated and it can carry the required current when used in duct. So, we choose it.

## 7. Cable Calculations (FDB-MDB):

The cables that are going to be used will be 3.5-core XLPE insulated armored Aluminum cables. The formula to calculate the minimum area of the conductor is,  $A \geq 1.5KI/\Delta V$

where,

$K$ = resistivity of aluminum =  $0.03 \Omega\text{mm}^2/\text{m}$

$I$ = current in Amperes

$L$ = length of the conductor from MDB to SDB in meters

$\Delta V$ = permissible Voltage Drop = 2% of 230V = 4.6V

### 7.1. BASEMENT FLOOR

#### 7.1.1. MDB to FDB

$I = 320 \text{ A}$

$L = 3.5 \text{ m} + 15\% = 4.025 \text{ m}$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 12.6 \text{ mm}^2$$

$\therefore$  The nearest cable size is  $400 \text{ mm}^2/350\text{A}$ (pg.32) rated and it can carry the required current when used in duct. So, we choose it.

### 7.2. GROUND FLOOR

#### 7.2.1. MDB to FDB

$I = 250 \text{ A}$

$L = 1.5 \text{ m} + 15\% = 1.725 \text{ m}$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 4.218 \text{ mm}^2$$

$\therefore$  The nearest cable size is  $240 \text{ mm}^2/270\text{A}$ (pg.21) rated and it can carry the required current when used in duct. So, we choose it.

### 7.3. FIRST FLOOR

#### 7.3.1. MDB to FDB

$I = 350 \text{ A}$

$L = 3.5 \text{ m} + 15\% = 4.025 \text{ m}$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 13.78 \text{ mm}^2$$

$\therefore$  The nearest cable size is  $400 \text{ mm}^2/350\text{A}$ (pg.32) rated and it can carry the required current when used in duct. So, we choose it.

## **7.4. SECOND FLOOR**

### **7.4.1. MDB to FDB**

$$I = 320 \text{ A}$$

$$L = 7 \text{ m} + 15\% = 8.05 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 25.2 \text{ mm}^2$$

∴ The nearest cable size is 400 mm<sup>2</sup>/350A(pg.32) rated and it can carry the required current when used in duct. So, we choose it.

## **7.5. THIRD FLOOR**

### **7.5.1. MDB to FDB**

$$I = 400 \text{ A}$$

$$L = 10.5 \text{ m} + 15\% = 12.075 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 47.25 \text{ mm}^2$$

∴ The nearest cable size is 500 mm<sup>2</sup>/405A(pg.32) rated and it can carry the required current when used in duct. So, we choose it.

## **7.6. FOURTH FLOOR**

### **7.6.1. MDB to FDB**

$$I = 320 \text{ A}$$

$$L = 14 \text{ m} + 15\% = 16.1 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 50.4 \text{ mm}^2$$

∴ The nearest cable size is 400 mm<sup>2</sup>/350A(pg.32) rated and it can carry the required current when used in duct. So, we choose it.

## **7.7. FIFTH FLOOR**

### **7.7.1. MDB to FDB**

$$I = 320 \text{ A}$$

$$L = 17.5 \text{ m} + 15\% = 20.125 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 63 \text{ mm}^2$$

∴ The nearest cable size is 400 mm<sup>2</sup>/350A(pg.32) rated and it can carry the required current when used in duct. So, we choose it.

## 7.8. SIXTH FLOOR

### 7.8.1. MDB to FDB

$$I = 160 \text{ A}$$

$$L = 21 \text{ m} + 15\% = 24.15 \text{ m}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 37.8 \text{ mm}^2$$

∴ The nearest cable size is 95 mm<sup>2</sup>/165A(pg.32) rated and it can carry the required current when used in duct.  
So, we choose it.

## 8. Cable Calculations (MDB-Transformer):

The cables that are going to be used will be 3.5-core XLPE insulated armored Aluminum cables. The formula to calculate the minimum area of the conductor is,  $A \geq 1.5KIL/\Delta V$   
where,

$$K = \text{resistivity of aluminum} = 0.03 \text{ } \Omega\text{mm}^2/\text{m}$$

I= current in Amperes

l= length of the conductor from MDB to SDB in meters

ΔV= permissible Voltage Drop = 2% of 230V = 4.6V

## 8.1. ACB CALCULATION

| Floors       | Incomer Current Rating (FDB) (A) |
|--------------|----------------------------------|
| Basement     | 320                              |
| Ground Floor | 250                              |
| First Floor  | 350                              |
| Second Floor | 320                              |
| Third Floor  | 400                              |
| Fourth Floor | 320                              |
| Fifth Floor  | 320                              |
| Sixth Floor  | 160                              |
| <b>Total</b> | <b>2440 @ 80% = 1952A</b>        |

Therefore, we choose ACB of rating 2000A.

## 8.2. CABLE CALCULATION

$$I = 2000 \text{ A}$$

$$L = (20-35) \text{ m} = 30 \text{ m (Assume)}$$

$$A \geq \frac{1.5kIL}{\Delta v} \text{ mm}^2 = 586.95 \text{ mm}^2$$

∴ The nearest cable size is 630 mm<sup>2</sup>/723A(pg.32) rated. However, it cannot fulfill current requirement of 2000A. So, we must choose multiple runs of cable to be used in air.

$$\text{No. of run (r)} = 2000/723 \approx 2.77 \approx 3$$

$$\text{Conductor AC resistance (R)} = 0.0648 \text{ ohm/km}$$

$$\text{Reactance of Cable at 50 Hz (X)} = 0.069 \text{ ohm/km}$$

$$\begin{aligned} \text{Voltage drop in cable ( } \Delta V \text{ )} &= \frac{\sqrt{3} \times I \times (R \cos \theta + X \sin \theta) \times L \times 100}{400 \times r \times 1000} \\ &= \frac{\sqrt{3} \times 2000 \times (0.0648 \times 0.8 + 0.069 \times 0.6) \times 30 \times 100}{400 \times 3 \times 1000} \\ &= 0.807\% < 2.5\% \text{ Permissible voltage drop} \end{aligned}$$

So, we choose 3 parallel run of 630 mm<sup>2</sup>, 723A aluminum 3.5 core insulated armored XLPE cable.

### 8.3. TRANSFORMER RATING CALCULATION

| Floors                         | Total Load (KW)  |
|--------------------------------|------------------|
| Basement                       | 192.32           |
| Ground Floor                   | 156.464          |
| First Floor                    | 224.134          |
| Second Floor                   | 185.267          |
| Third Floor                    | 253.656          |
| Fourth Floor                   | 206.674          |
| Fifth Floor                    | 170.041          |
| Sixth Floor                    | 84.208           |
| <b>Total</b>                   | <b>1472.764</b>  |
| <b>Total+20% Safety Factor</b> | <b>1767.3168</b> |
| <b>In KVA</b>                  | <b>2209.146</b>  |

Therefore, closest transformer rating for 2209.146 KVA is 2500 KVA.

## 9. Conclusion:

In this way, using the Light fixture calculation tables and the DB calculation tables, a power distribution system was designed for a large building. While, doing our calculations, we have assumed values for the factor of simultaneity ( $K_s$ ) and the factor for maximum utilization ( $K_u$ ). These assumptions are based on the general everyday notions of government buildings and offices in general. A detailed load study must be carried out to obtain accurate values for  $K_u$  and  $K_s$ . If the accurate values are obtained, then some changes must be made in the main MCCB selection and in the transformer selection.

## 10. Appendix:

Suspended/ Surface mounted architectural luminaire provides uniquely clean and simple lighting system with soft light and glare free illumination. Available in continuous mounting option, Consta is the ideal solution for every space.

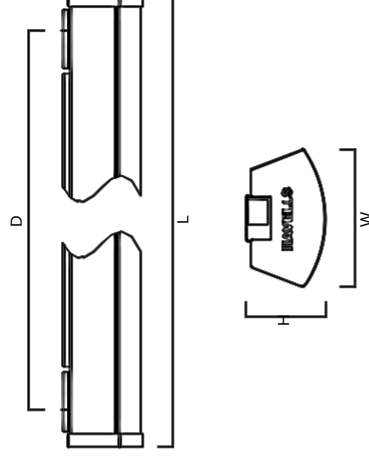
## CONSTA LED SURFACE / SUSPENDED

### TECHNICAL DATA

| Description                  | System Power | Voltage (V) | CRI | Lumen (lm) | Colour Temp. (CCT) | System Luminous Efficacy lm/ W |
|------------------------------|--------------|-------------|-----|------------|--------------------|--------------------------------|
| CONSTA544LP20WLED830SPMMAWH  | 20 W         | 240         | ≥80 | 1800       | 3000 K             | ≥90                            |
| CONSTA544LP20WLED840SPMMAWH  | 20 W         | 240         | ≥80 | 1900       | 4000 K             | ≥95                            |
| CONSTA544LP20WLED857SPMMAWH  | 20 W         | 240         | ≥80 | 2000       | 5700 K             | ≥100                           |
| CONSTA1144LP40WLED830SPMMAWH | 40 W         | 240         | ≥80 | 3600       | 3000 K             | ≥90                            |
| CONSTA1144LP40WLED840SPMMAWH | 40 W         | 240         | ≥80 | 3800       | 4000 K             | ≥95                            |
| CONSTA1144LP40WLED857SPMMAWH | 40 W         | 240         | ≥80 | 4000       | 5700 K             | ≥100                           |

Driver code : LSSCRX1763 for 20 W,  
LSSCRX1763 - 2 nos. for 40 W.

### TECHNICAL DRAWING



| System Power | Length l (mm) | Width w (mm) | Height h (mm) | Suspension Wire Distance (d) |
|--------------|---------------|--------------|---------------|------------------------------|
| 20 W         | 600           | 106          | 60            | 544                          |
| 40 W         | 1200          | 106          | 60            | 1144                         |

All dimensions are in mm  
Tolerance: ± 5 mm

Ultra modern recess mounting luminaire with high performance LEDs, suitable for mounting with Armstrong/Grid ceiling.



## VENUS NEO HE 2X2 18-42 W MOD

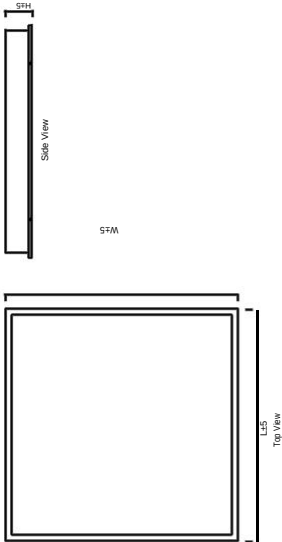
### TECHNICAL DATA

| Description                     | System Power | Voltage (V) | CRI | Lumen (lm) | Colour Temp. (CCT) | System Luminous Efficacy lm/W |
|---------------------------------|--------------|-------------|-----|------------|--------------------|-------------------------------|
| VENUSNEO1-10V2X2PLR34WLED840MOD | 18-42 W      | 36          | ≥80 | 1800-3600  | 3000 K             | ≥100                          |
| VENUSNEO1-10V2X2PLR34WLED857MOD | 18-42 W      | 36          | ≥80 | 1880-3780  | 4000 K             | ≥100                          |
| VENUSNEOHE2X2PLR18-42WLED840MOD | 18-42 W      | 36          | ≥80 | 2000-3960  | 5700 K             | ≥110                          |
| VENUSNEOHE2X2PLR18-42WLED857MOD | 18-42 W      | 36          | ≥80 | 2020-3980  | 6500 K             | ≥110                          |

Driver code:

LEDDRIVER36V18 WS for 18 W,  
LEDDRIVER36V24 WS for 24 W,  
LEDDRIVER36V28 WS for 28 W,  
LEDDRIVER36V34 WS for 34 W /  
36 W,  
LSSLOW0221 for 42 W.

### TECHNICAL DRAWING



| System Power | Length l (mm) | Width w (mm) | Height h (mm) | Cutout Size (in mm) |
|--------------|---------------|--------------|---------------|---------------------|
| 18W/24W/28W  | 595           | 595          | 65            | 580 x 580           |
| 34W/36W/42W  | 595           | 595          | 65            | 580 x 580           |

All dimensions are in mm  
Toleran

Ultra modern recess mounting luminaire with high performance LEDs, suitable for mounting with Armstrong/ Grid ceiling.

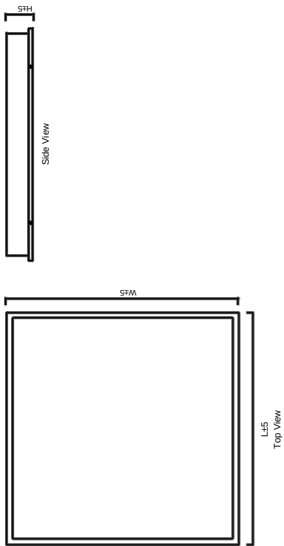


## VENUS 42 W TUNABLE WHITE

### TECHINCAL DATA

| Description                 | System Power | Voltage (V) | CRI | Lumen (lm) | Colour Temp. (CCT) | System Luminous Efficacy lm/ W |
|-----------------------------|--------------|-------------|-----|------------|--------------------|--------------------------------|
| VENUSNEOTW2X2PLR42WLEDTWMOD | 42 W         | 30          | ≥80 | 4000-4300  | 3000 K - 5700 K    | up to ≥100                     |

### TECHNICAL DRAWING



| System Power | Length l (mm) | Width w (mm) | Height h (mm) | Cutout Size (in mm) |
|--------------|---------------|--------------|---------------|---------------------|
| 42 W         | 595           | 595          | 65            | 580 x 580           |

All dimensions are in mm  
Tolerance: ± 5 mm

Environmental friendly, new generation, aesthetically designed, ultra modern recess mounting luminaire with high brightness SMD LEDs as light source, suitable for Armstrong/Grid ceiling.

## VENUS NEO HE 4X1

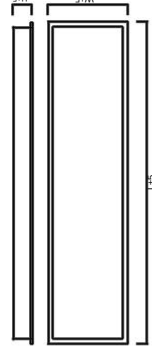
### TECHNICAL DATA

| Description                     | System Power | Voltage (V) | CRI | Lumen (lm) | Colour Temp. (CCT) | System Luminous Efficacy lm/ W |
|---------------------------------|--------------|-------------|-----|------------|--------------------|--------------------------------|
| VENUSNEOHE4X1PLR28-36WLED830MOD | 28-36 W      | 36          | ≥80 | 2700-3560  | 3000 K             | ≥100                           |
| VENUSNEOHE4X1PLR28-36WLED840MOD | 28-36 W      | 36          | ≥80 | 2850-3750  | 4000 K             | ≥105                           |
| VENUSNEOHE4X1PLR28-36WLED857MOD | 28-36 W      | 36          | ≥80 | 3000-3700  | 5700 K             | ≥105                           |

Driver codes: -

LEDDRIVER36V28WS for 28 W,  
LEDDRIVER36V34WS for 34 W,  
LEDDRIVER36V36WS for 36 W

### TECHNICAL DRAWING



| System Power | Length l (mm) | Width w (mm) | Height h (mm) | Cutout Size (in mm) |
|--------------|---------------|--------------|---------------|---------------------|
| 28 W         | 1195          | 295          | 70            | 1180 x 280          |
| 36 W         | 1195          | 295          | 70            | 1180 x 280          |

All dimensions are in mm  
Tolerance: ± 5 mm

High efficacy 2x1 recess flat panel with high brightness LED for glare-free homogeneous illumination, suitable for recessed mounting, Armstrong/POP/GB ceiling.

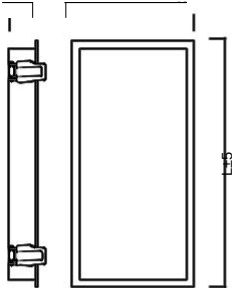
# PLUTO 2X1



## TECHNICAL DATA

| Description           | System Power | Voltage (V) | CRI | Lumen (lm) | Colour Temp. (CCT) | System Luminous Efficacy lm/ W |
|-----------------------|--------------|-------------|-----|------------|--------------------|--------------------------------|
| PLUTO2X1PLR20WLED840S | 20 W         | 240         | ≥80 | 1920       | 4000 K             | ≥95                            |
| PLUTO2X1PLR20WLED857S | 20 W         | 240         | ≥80 | 2000       | 5700 K             | ≥100                           |
| PLUTO2X1PLR28WLED840S | 28 W         | 240         | ≥80 | 2840       | 4000 K             | ≥100                           |
| PLUTO2X1PLR28WLED857S | 28 W         | 240         | ≥80 | 2960       | 5700 K             | ≥105                           |

## TECHNICAL DRAWING



| System Power | Length l (mm) | Width w (mm) | Height h (mm) | Cutout Size (in mm) |
|--------------|---------------|--------------|---------------|---------------------|
| 20 W         | 595           | 295          | 70            | 580 x 280           |
| 28 W         | 595           | 295          | 70            | 580 x 280           |

All dimensions are in mm  
Tolerance: ± 5 mm

Aesthetically designed 1x1 LED flat panels for surface mounting which provides soft-light and glare-free symmetrical illumination.

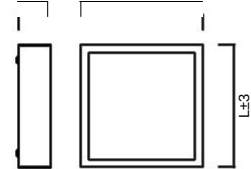


# PLUTO SURFACE 1X1

## TECHNICAL DATA

| Description                   | System Power | Voltage (V) | CRI | Lumen (lm) | Colour Temp. (CCT) | System Luminous Efficacy lm/ W |
|-------------------------------|--------------|-------------|-----|------------|--------------------|--------------------------------|
| PLUTOSURFACE1X1PLS15 WLED840S | 15 W         | 240         | ≥80 | 1125       | 4000 K             | ≥75                            |
| PLUTOSURFACE1X1PLS15 WLED857S | 15 W         | 240         | ≥80 | 1180       | 5700 K             | ≥78                            |

## TECHNICAL DRAWING



| System Power | Length l (mm) | Width w (mm) | Height h (mm) |
|--------------|---------------|--------------|---------------|
| 15 W         | 295           | 295          | 70            |
| 15 W         | 295           | 295          | 70            |

All dimensions are in mm  
Tolerance: ± 3 mm

Recess mounted downlighter with pressure die-cast aluminium heat sink & PC diffuser in white powder coated finish with integrated driver.

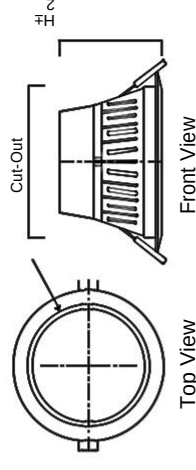


## INTEGRA NEO

### TECHNICAL DATA

| Description                | System Power | Voltage (V) | CRI | Lumen (lm) | Colour Temp. (CCT) | System Luminous Efficacy lm/ W |
|----------------------------|--------------|-------------|-----|------------|--------------------|--------------------------------|
| INTEGRANEODLR10WLED830S    | 10 W         | 240         | ≥80 | 810        | 3000 K             | ≥81                            |
| INTEGRANEODLR10WLED840S    | 10 W         | 240         | ≥80 | 850        | 4000 K             | ≥85                            |
| INTEGRANEODLR10WLED857S    | 10 W         | 240         | ≥80 | 1000       | 5700 K             | ≥100                           |
| INTEGRANEODLR12WLED840S    | 12 W         | 240         | ≥80 | 900        | 4000 K             | ≥75                            |
| INTEGRANEODLR12WLED857S    | 12 W         | 240         | ≥80 | 1000       | 5700 K             | ≥83                            |
| INTEGRANEODLR15WLED830S    | 15 W         | 240         | ≥80 | 1125       | 3000 K             | ≥75                            |
| INTEGRANEODLR15WLED840S    | 15 W         | 240         | ≥80 | 1215       | 4000 K             | ≥81                            |
| INTEGRANEODLR15WLED857S    | 15 W         | 240         | ≥80 | 1500       | 5700 K             | ≥100                           |
| INTEGRANEODLR18WLED830S    | 18 W         | 240         | ≥80 | 1530       | 3000 K             | ≥85                            |
| INTEGRANEODLR18WLED840S    | 18 W         | 240         | ≥80 | 1620       | 4000 K             | ≥90                            |
| INTEGRANEODLR18WLED857S    | 18 W         | 240         | ≥80 | 1800       | 5700 K             | ≥100                           |
| INTEGRANEODLR24WLED830S    | 24 W         | 240         | ≥80 | 1920       | 3000 K             | ≥80                            |
| INTEGRANEODLR24WLED840S    | 24 W         | 240         | ≥80 | 2040       | 4000 K             | ≥85                            |
| INTEGRANEODLR24WLED857S    | 24 W         | 240         | ≥80 | 2400       | 5700 K             | ≥100                           |
| INTEGRANEODLR15WLED840MOD* | 15 W DALI    | 75-85       | ≥80 | 1350       | 4000 K             | ≥90                            |
| INTEGRANEODLR15WLED857MOD* | 15 W DALI    | 75-85       | ≥80 | 1425       | 5700 K             | ≥95                            |

### TECHNICAL DRAWING



| System Power | Dia Ø (mm) | Height h (mm) | Cutout Size (in mm) |
|--------------|------------|---------------|---------------------|
| 10 W         | 111        | 90            | 90                  |
| 12 W         | 158        | 90            | 138                 |
| 15 W         | 158        | 90            | 138                 |
| 24 W         | 197        | 90            | 175                 |

All dimensions are in mm  
Tolerance: ± 5 mm



Recess mounted downlighter with pressure die-cast aluminium heat sink & PC diffuser in white powder coated finish with integrated driver.

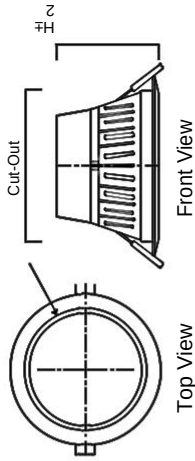


# INTEGRA GRAND

## TECHNICAL DATA

| Description                   | System Power | Voltage (V) | CRI | Lumen (lm) | Colour Temp. (CCT) | System Luminous Efficacy lm/ W |
|-------------------------------|--------------|-------------|-----|------------|--------------------|--------------------------------|
| INTEGRAGRANDDDLR10WLED830S    | 10 W         | 240         | ≥80 | 900        | 3000 K             | ≥90                            |
| INTEGRAGRANDDDLR10WLED840S    | 10 W         | 240         | ≥80 | 940        | 4000 K             | ≥94                            |
| INTEGRAGRANDDDLR10WLED857S    | 10 W         | 240         | ≥80 | 1000       | 5700 K             | ≥100                           |
| INTEGRAGRANDDDLR15WLED830S    | 15 W         | 240         | ≥80 | 1280       | 3000 K             | ≥85                            |
| INTEGRAGRANDDDLR15WLED840S    | 15 W         | 240         | ≥80 | 1350       | 4000 K             | ≥90                            |
| INTEGRAGRANDDDLR15WLED857S    | 15 W         | 240         | ≥80 | 1500       | 5700 K             | ≥100                           |
| INTEGRAGRANDDDLR15WLED840MOD* | 15 W DALI    | 75-85       | ≥80 | 1350       | 4000 K             | ≥90                            |
| INTEGRAGRANDDDLR15WLED857MOD* | 15 W DALI    | 75-85       | ≥80 | 1425       | 5700 K             | ≥95                            |

## TECHNICAL DRAWING



| System Power | Dia Ø (mm) | Height h (mm) | Cutout Size (in mm) |
|--------------|------------|---------------|---------------------|
| 10W/15W      | 197        | 90            | 175                 |

All dimensions are in mm  
Tolerance: ± 5 mm

Stout LED is a complete range of LED battens to replace traditional T8 battens (600mm & 1200mm). The range is perfect for lighting large areas such as corridors, warehouses & general purpose spaces.

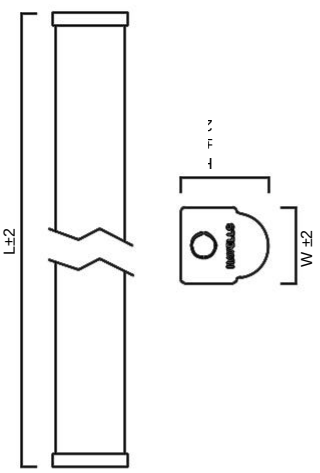
# STOUT PLUS



## TECHNICAL DATA

| Description                   | System Power | Voltage (V) | CRI | Lumen (lm) | Colour Temp. (CCT) | System Luminous Efficacy lm/ W |
|-------------------------------|--------------|-------------|-----|------------|--------------------|--------------------------------|
| STOUTPLUS600BS20WLED840SPCWH  | 20 W         | 240         | ≥80 | 2000       | 4000 K             | ≥100                           |
| STOUTPLUS600BS20WLED857SPCWH  | 20 W         | 240         | ≥80 | 2100       | 5700 K             | ≥105                           |
| STOUTPLUS1200BS40WLED840SPCWH | 40 W         | 240         | ≥80 | 3800       | 4000 K             | ≥90                            |
| STOUTPLUS1200BS40WLED857SPCWH | 40 W         | 240         | ≥80 | 4000       | 5700 K             | ≥95                            |

## TECHNICAL DRAWING



| System Power | Length l (mm) | Width w (mm) | Height h (mm) | Suspension Wire Distance (d) |
|--------------|---------------|--------------|---------------|------------------------------|
| 20 W         | 605           | 65.5         | 76            | At Centre                    |
| 40 W         | 1205          | 65.5         | 76            | At Centre                    |

All dimensions are in mm  
Tolerance: ± 5 mm



# Hexa Half Cut Wall Pack Waterproof SEMI Cut Off

| SAP Description                                      | Switchable Wattage (W) | Power Factor | THD (%) | CRI (Ra) | Maximum wattage Lumens (lm)<br>5000 K / 4000 K / 3000 K | System Efficacy (lm/W) |
|------------------------------------------------------|------------------------|--------------|---------|----------|---------------------------------------------------------|------------------------|
| SAP Code                                             |                        |              |         |          |                                                         |                        |
| HEXAHCWPA75WLED850ASYMTGSWSCIP65<br>LOEOBCP8P46J075  | 75 W / 60 W / 45 W     | 0.90         | ≤20     | ≥80      | 9000 lm / 9000 lm / 9000 lm                             | ≥120 lm/W              |
|                                                      |                        |              |         |          |                                                         |                        |
| HEXAHCWPA120WLED850ASYMTGSWSCIP65<br>LHEOBCP8P46J120 | 120 W / 96 W / 72 W    | 0.90         | ≤20     | ≥80      | 14400 lm / 14400 lm / 14400 lm                          | ≥120 lm/W              |
|                                                      |                        |              |         |          |                                                         |                        |

Nominal Voltage (V) : 120 V-277 V

Weight : 2.7 kg

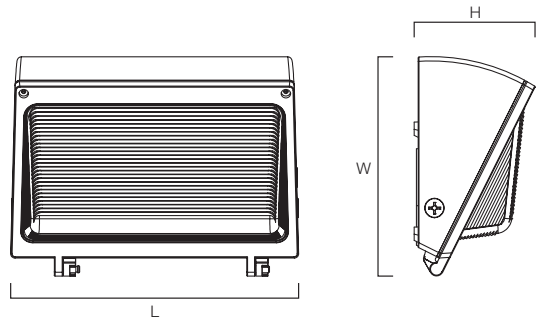
Colour of Fixture : Grey

Accessories : Hanging Chain

Optics: Polycarbonate

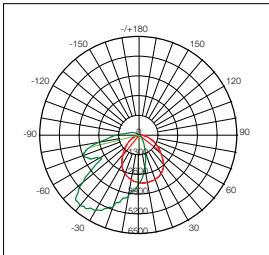
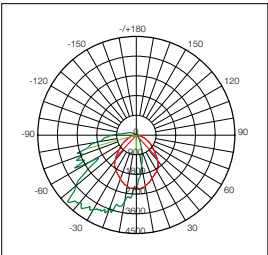


## TECHNICAL DRAWING



## DIMENSION

| System Power | Length (l) | Width (w) | Height (h) |
|--------------|------------|-----------|------------|
| 120W SWSC    | 329 mm     | 251 mm    | 138 mm     |
| 70W SWSC     | 329 mm     | 251 mm    | 138 mm     |



# Product datasheet

Specifications



## Miniature circuit breaker (MCB), Acti9 xC60, 1P, 6A, B curve, 10kA (IS/IEC60947-2)

A9N1P06BIN

### Main

|                           |                                         |
|---------------------------|-----------------------------------------|
| Device application        | Distribution                            |
| Range of product          | Acti9                                   |
| Range                     | Acti9                                   |
| product name              | XC60                                    |
| Product or component type | Miniature circuit-breaker               |
| Device short name         | xC60                                    |
| Poles description         | 1P                                      |
| Number of protected poles | 1                                       |
| [In] rated current        | 6 A                                     |
| Network type              | AC<br>DC                                |
| Trip unit technology      | Thermal-magnetic                        |
| Curve code                | B                                       |
| Utilisation category      | Category A conforming to IS/IEC 60947-2 |
| Suitability for isolation | Yes conforming to IS/IEC 60947-2        |

### Complementary

|                                        |                                                                                                                                                                                 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network frequency                      | 50/60 Hz                                                                                                                                                                        |
| [Ue] rated operational voltage         | 220...240 V AC 50/60 Hz<br>12...48 V DC<br>60 V DC                                                                                                                              |
| Magnetic tripping limit                | 4 x In +/- 20 %                                                                                                                                                                 |
| Breaking capacity                      | 10 kA Icu at 220...240 V AC 50/60 Hz conforming to IS/IEC 60947-2<br>15 kA Icu at 12...48 V DC conforming to IS/IEC 60947-2<br>6 kA Icu at 60 V DC conforming to IS/IEC 60947-2 |
| [Ics] rated service breaking capacity  | 15 kA conforming to IS/IEC 60947-2 - 12...48 V DC<br>6 kA conforming to IS/IEC 60947-2 - 60 V DC<br>7.5 kA conforming to IS/IEC 60947-2 - 220...240 V AC 50/60 Hz               |
| Limitation class                       | 3 conforming to IS/IEC 60947-2                                                                                                                                                  |
| [Ui] rated insulation voltage          | 500 V AC 50/60 Hz conforming to IS/IEC 60947-2                                                                                                                                  |
| [Uimp] rated impulse withstand voltage | 4 kV conforming to IS/IEC 60947-2                                                                                                                                               |
| Contact position indicator             | Yes                                                                                                                                                                             |
| control type                           | Toggle                                                                                                                                                                          |
| Local signalling                       | ON/OFF indication                                                                                                                                                               |

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# Product datasheet

Specifications



## Miniature circuit breaker (MCB), Acti9 xC60, 1P, 16A, B curve, 10kA (IS/IEC60947-2)

A9N1P16BIN

### Main

|                           |                                         |
|---------------------------|-----------------------------------------|
| Device application        | Distribution                            |
| Range of product          | Acti9                                   |
| Range                     | Acti9                                   |
| product name              | XC60                                    |
| Product or component type | Miniature circuit-breaker               |
| Device short name         | xC60                                    |
| Poles description         | 1P                                      |
| Number of protected poles | 1                                       |
| [In] rated current        | 16 A                                    |
| Network type              | DC<br>AC                                |
| Trip unit technology      | Thermal-magnetic                        |
| Curve code                | B                                       |
| Utilisation category      | Category A conforming to IS/IEC 60947-2 |
| Suitability for isolation | Yes conforming to IS/IEC 60947-2        |

### Complementary

|                                        |                                                                                                                                                                                 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network frequency                      | 50/60 Hz                                                                                                                                                                        |
| [Ue] rated operational voltage         | 220...240 V AC 50/60 Hz<br>12...48 V DC<br>60 V DC                                                                                                                              |
| Magnetic tripping limit                | 4 x In +/- 20 %                                                                                                                                                                 |
| Breaking capacity                      | 10 kA Icu at 220...240 V AC 50/60 Hz conforming to IS/IEC 60947-2<br>15 kA Icu at 12...48 V DC conforming to IS/IEC 60947-2<br>6 kA Icu at 60 V DC conforming to IS/IEC 60947-2 |
| [Ics] rated service breaking capacity  | 15 kA conforming to IS/IEC 60947-2 - 12...48 V DC<br>6 kA conforming to IS/IEC 60947-2 - 60 V DC<br>7.5 kA conforming to IS/IEC 60947-2 - 220...240 V AC 50/60 Hz               |
| Limitation class                       | 3 conforming to IS/IEC 60947-2                                                                                                                                                  |
| [Ui] rated insulation voltage          | 500 V AC 50/60 Hz conforming to IS/IEC 60947-2                                                                                                                                  |
| [Uimp] rated impulse withstand voltage | 4 kV conforming to IS/IEC 60947-2                                                                                                                                               |
| Contact position indicator             | Yes                                                                                                                                                                             |
| control type                           | Toggle                                                                                                                                                                          |
| Local signalling                       | ON/OFF indication                                                                                                                                                               |

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# Product datasheet

Specifications



## Miniature circuit breaker (MCB), Acti9 xC60, 1P, 63A, B curve, 10kA (IS/IEC60947-2)

A9N1P63BIN

### Main

|                           |                                         |
|---------------------------|-----------------------------------------|
| Device application        | Distribution                            |
| Range of product          | Acti9                                   |
| Range                     | Acti9                                   |
| product name              | XC60                                    |
| Product or component type | Miniature circuit-breaker               |
| Device short name         | xC60                                    |
| Poles description         | 1P                                      |
| Number of protected poles | 1                                       |
| [In] rated current        | 63 A                                    |
| Network type              | DC<br>AC                                |
| Trip unit technology      | Thermal-magnetic                        |
| Curve code                | B                                       |
| Utilisation category      | Category A conforming to IS/IEC 60947-2 |
| Suitability for isolation | Yes conforming to IS/IEC 60947-2        |

### Complementary

|                                        |                                                                                                                                                                                 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network frequency                      | 50/60 Hz                                                                                                                                                                        |
| [Ue] rated operational voltage         | 220...240 V AC 50/60 Hz<br>12...48 V DC<br>60 V DC                                                                                                                              |
| Magnetic tripping limit                | 4 x In +/- 20 %                                                                                                                                                                 |
| Breaking capacity                      | 10 kA Icu at 220...240 V AC 50/60 Hz conforming to IS/IEC 60947-2<br>15 kA Icu at 12...48 V DC conforming to IS/IEC 60947-2<br>6 kA Icu at 60 V DC conforming to IS/IEC 60947-2 |
| [Ics] rated service breaking capacity  | 15 kA conforming to IS/IEC 60947-2 - 12...48 V DC<br>6 kA conforming to IS/IEC 60947-2 - 60 V DC<br>7.5 kA conforming to IS/IEC 60947-2 - 220...240 V AC 50/60 Hz               |
| Limitation class                       | 3 conforming to IS/IEC 60947-2                                                                                                                                                  |
| [Ui] rated insulation voltage          | 500 V AC 50/60 Hz conforming to IS/IEC 60947-2                                                                                                                                  |
| [Uimp] rated impulse withstand voltage | 4 kV conforming to IS/IEC 60947-2                                                                                                                                               |
| Contact position indicator             | Yes                                                                                                                                                                             |
| control type                           | Toggle                                                                                                                                                                          |
| Local signalling                       | ON/OFF indication                                                                                                                                                               |

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# Product datasheet

Specifications



## Residual current circuit breaker (RCCB), Acti9 iID, 2P, 25A, AC type, 10mA

A9R10225

### Main

|                                     |                                         |
|-------------------------------------|-----------------------------------------|
| Range                               | Acti9                                   |
| product name                        | Acti9 iID                               |
| Product or component type           | Residual current circuit breaker (RCCB) |
| Device short name                   | iID                                     |
| Poles description                   | 2P                                      |
| Neutral position                    | Left                                    |
| [In] rated current                  | 25 A                                    |
| Network type                        | AC                                      |
| Earth-leakage sensitivity           | 10 mA                                   |
| Earth-leakage protection time delay | Instantaneous                           |
| Earth-leakage protection class      | Type AC                                 |

### Complementary

|                                         |                         |
|-----------------------------------------|-------------------------|
| Device location in system               | Outgoer                 |
| Network frequency                       | 50/60 Hz                |
| [Ue] rated operational voltage          | 220...240 V AC 50/60 Hz |
| Residual current tripping technology    | Voltage independent     |
| Rated breaking and making capacity      | Idm 1500 A<br>Im 1500 A |
| Rated conditional short-circuit current | 10 kA                   |
| [Ui] rated insulation voltage           | 500 V AC 50/60 Hz       |
| [Uimp] rated impulse withstand voltage  | 6 kV                    |
| Contact position indicator              | Yes                     |
| control type                            | Toggle                  |
| Mounting mode                           | Clip-on                 |
| Mounting support                        | DIN rail                |
| 9 mm pitches                            | 4                       |
| Height                                  | 91 mm                   |
| Width                                   | 36 mm                   |
| Depth                                   | 73.5 mm                 |
| Net weight                              | 0.21 kg                 |

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# Product datasheet

Specifications



## Residual current circuit breaker (RCCB), Acti9 iID, 2P, 40A, AC type, 100mA

A9R12240

### Main

|                                     |                                         |
|-------------------------------------|-----------------------------------------|
| Range                               | Acti9                                   |
| product name                        | Acti9 iID                               |
| Product or component type           | Residual current circuit breaker (RCCB) |
| Device short name                   | iID                                     |
| Poles description                   | 2P                                      |
| Neutral position                    | Left                                    |
| [In] rated current                  | 40 A                                    |
| Network type                        | AC                                      |
| Earth-leakage sensitivity           | 100 mA                                  |
| Earth-leakage protection time delay | Instantaneous                           |
| Earth-leakage protection class      | Type AC                                 |

### Complementary

|                                         |                         |
|-----------------------------------------|-------------------------|
| Device location in system               | Outgoer                 |
| Network frequency                       | 50/60 Hz                |
| [Ue] rated operational voltage          | 220...240 V AC 50/60 Hz |
| Residual current tripping technology    | Voltage independent     |
| Rated breaking and making capacity      | Idm 1500 A<br>Im 1500 A |
| Rated conditional short-circuit current | 10 kA                   |
| [Ui] rated insulation voltage           | 500 V AC 50/60 Hz       |
| [Uimp] rated impulse withstand voltage  | 6 kV                    |
| Contact position indicator              | Yes                     |
| control type                            | Toggle                  |
| Mounting mode                           | Clip-on                 |
| Mounting support                        | DIN rail                |
| 9 mm pitches                            | 4                       |
| Height                                  | 91 mm                   |
| Width                                   | 36 mm                   |
| Depth                                   | 73.5 mm                 |
| Net weight                              | 0.21 kg                 |

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# Product datasheet

Specifications



## Residual current circuit breaker (RCCB), Acti9 iID, 2P, 63A, AC type, 100mA

A9R12263

### Main

|                                     |                                         |
|-------------------------------------|-----------------------------------------|
| Range                               | Acti9                                   |
| product name                        | Acti9 iID                               |
| Product or component type           | Residual current circuit breaker (RCCB) |
| Device short name                   | iID                                     |
| Poles description                   | 2P                                      |
| Neutral position                    | Left                                    |
| [In] rated current                  | 63 A                                    |
| network type                        | AC                                      |
| Earth-leakage sensitivity           | 100 mA                                  |
| Earth-leakage protection time delay | Instantaneous                           |
| Earth-leakage protection class      | Type AC                                 |

### Complementary

|                                         |                         |
|-----------------------------------------|-------------------------|
| Device location in system               | Outgoer                 |
| Network frequency                       | 50/60 Hz                |
| [Ue] rated operational voltage          | 220...240 V AC 50/60 Hz |
| Residual current tripping technology    | Voltage independent     |
| Rated breaking and making capacity      | Idm 1500 A<br>Im 1500 A |
| Rated conditional short-circuit current | 10 kA                   |
| [Ui] rated insulation voltage           | 500 V AC 50/60 Hz       |
| [Uimp] rated impulse withstand voltage  | 6 kV                    |
| Contact position indicator              | Yes                     |
| Control type                            | Toggle                  |
| Mounting mode                           | Clip-on                 |
| Mounting support                        | DIN rail                |
| 9 mm pitches                            | 4                       |
| Height                                  | 91 mm                   |
| Width                                   | 36 mm                   |
| Depth                                   | 73.5 mm                 |
| Net weight                              | 0.21 kg                 |

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# Product datasheet

Specifications



## Residual current circuit breaker (RCCB), Acti9 iID, 2P, 80A, AC type, 30mA

A9R11280

### Main

|                                     |                                         |
|-------------------------------------|-----------------------------------------|
| Range                               | Acti9                                   |
| product name                        | Acti9 iID                               |
| Product or component type           | Residual current circuit breaker (RCCB) |
| Device short name                   | iID                                     |
| Poles description                   | 2P                                      |
| Neutral position                    | Left                                    |
| [In] rated current                  | 80 A                                    |
| network type                        | AC                                      |
| Earth-leakage sensitivity           | 30 mA                                   |
| Earth-leakage protection time delay | Instantaneous                           |
| Earth-leakage protection class      | Type AC                                 |

### Complementary

|                                         |                         |
|-----------------------------------------|-------------------------|
| Device location in system               | Outgoer                 |
| Network frequency                       | 50/60 Hz                |
| [Ue] rated operational voltage          | 220...240 V AC 50/60 Hz |
| Residual current tripping technology    | Voltage independent     |
| Rated breaking and making capacity      | Idm 1500 A<br>Im 1500 A |
| Rated conditional short-circuit current | 10 kA                   |
| [Ui] rated insulation voltage           | 500 V AC 50/60 Hz       |
| [Uimp] rated impulse withstand voltage  | 6 kV                    |
| Contact position indicator              | Yes                     |
| Control type                            | Toggle                  |
| Mounting mode                           | Clip-on                 |
| Mounting support                        | DIN rail                |
| 9 mm pitches                            | 4                       |
| Height                                  | 91 mm                   |
| Width                                   | 36 mm                   |
| Depth                                   | 73.5 mm                 |
| Net weight                              | 0.21 kg                 |

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# Product datasheet

Specifications



## Residual current circuit breaker (RCCB), Acti9 iID, 2P, 100A, AC type, 30mA

A9R11291

### Main

|                                     |                                         |
|-------------------------------------|-----------------------------------------|
| Range                               | Acti9                                   |
| product name                        | Acti9 iID                               |
| Product or component type           | Residual current circuit breaker (RCCB) |
| Device short name                   | iID                                     |
| Poles description                   | 2P                                      |
| Neutral position                    | Left                                    |
| [In] rated current                  | 100 A                                   |
| network type                        | AC                                      |
| Earth-leakage sensitivity           | 30 mA                                   |
| Earth-leakage protection time delay | Instantaneous                           |
| Earth-leakage protection class      | Type AC                                 |

### Complementary

|                                         |                         |
|-----------------------------------------|-------------------------|
| Device location in system               | Outgoer                 |
| Network frequency                       | 50/60 Hz                |
| [Ue] rated operational voltage          | 220...240 V AC 50/60 Hz |
| Residual current tripping technology    | Voltage independent     |
| Rated breaking and making capacity      | Idm 1500 A<br>Im 1500 A |
| Rated conditional short-circuit current | 10 kA                   |
| [Ui] rated insulation voltage           | 500 V AC 50/60 Hz       |
| [Uimp] rated impulse withstand voltage  | 6 kV                    |
| Contact position indicator              | Yes                     |
| Control type                            | Toggle                  |
| Mounting mode                           | Clip-on                 |
| Mounting support                        | DIN rail                |
| 9 mm pitches                            | 4                       |
| Height                                  | 91 mm                   |
| Width                                   | 36 mm                   |
| Depth                                   | 73.5 mm                 |
| Net weight                              | 0.21 kg                 |

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# Product datasheet

Specifications



## residual current circuit breaker ID Fi - 2 poles - 125 A - 30mA - class AC

16966

⚠ Discontinued on: 31-Dec-2024

⚠ To be discontinued

### Main

|                                         |                                                       |
|-----------------------------------------|-------------------------------------------------------|
| Range                                   | Acti9                                                 |
| product name                            | Acti9 RCCB-ID                                         |
| Product or component type               | Residual current circuit breaker (RCCB)               |
| Device short name                       | RCCB-ID                                               |
| Poles description                       | 2P                                                    |
| Neutral position                        | Left                                                  |
| [In] rated current                      | 125 A                                                 |
| network type                            | AC                                                    |
| Earth-leakage sensitivity               | 30 mA                                                 |
| Earth-leakage protection time delay     | Instantaneous                                         |
| Earth-leakage protection class          | Type AC                                               |
| Rated breaking and making capacity      | I <sub>m</sub> = 1250 A 230 V conforming to IEC 61008 |
| Rated conditional short-circuit current | I <sub>nc</sub> 10 kA 125 A                           |

### Complementary

|                                        |                                          |
|----------------------------------------|------------------------------------------|
| Device location in system              | Outgoer                                  |
| Network frequency                      | 50/60 Hz                                 |
| [Ue] rated operational voltage         | 230 V AC 50 Hz conforming to IEC 61008   |
| Residual current tripping technology   | Electromechanical                        |
| [Ui] rated insulation voltage          | 440 V AC 50 Hz conforming to IEC 61008-1 |
| [Uimp] rated impulse withstand voltage | 4 kV conforming to IEC 61008-1           |
| Control type                           | Toggle                                   |
| Mounting mode                          | Fixed                                    |
| Mounting support                       | 35 mm symmetrical DIN rail               |
| Connection pitch                       | 18 mm between phases                     |
| 9 mm pitches                           | 4                                        |
| Height                                 | 86 mm                                    |
| Width                                  | 36 mm                                    |
| Depth                                  | 76 mm                                    |

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

# Product datasheet

Specifications



## circuit breaker EasyPact EZC100B - TMD - 25 A - 3 poles 3d

EZC100B3025

### Main

|                                        |                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                                                                                                                                                                 |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                                                                                                                                                          |
| Device short name                      | EasyPact EZC100B                                                                                                                                                                                                                                                                                                                                                                         |
| Circuit breaker name                   | EasyPact EZC100B                                                                                                                                                                                                                                                                                                                                                                         |
| Device application                     | Distribution                                                                                                                                                                                                                                                                                                                                                                             |
| Poles description                      | 3P                                                                                                                                                                                                                                                                                                                                                                                       |
| Protected poles description            | 3t                                                                                                                                                                                                                                                                                                                                                                                       |
| Network type                           | DC<br>AC                                                                                                                                                                                                                                                                                                                                                                                 |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                                                                                                                                                                 |
| [In] rated current                     | 25 A at 40 °C                                                                                                                                                                                                                                                                                                                                                                            |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                              |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                           |
| [Ue] rated operational voltage         | 550 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                              |
| Breaking capacity code                 | B                                                                                                                                                                                                                                                                                                                                                                                        |
| Breaking capacity                      | 10 kA Icu at 110...130 V AC 50/60 Hz conforming to IEC 60947-2<br>10 kA Icu at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>5 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>2.5 kA Icu at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>7.5 kA Icu at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>7.5 kA Icu at 400...415 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 0.625 kA at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>1.25 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>1.875 kA at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>1.875 kA at 400/415 V AC 50/60 Hz conforming to IEC 60947-2<br>2.5 kA at 110/130 V AC 50/60 Hz conforming to IEC 60947-2<br>2.5 kA at 220/230/240 V AC 50/60 Hz conforming to IEC 60947-2                |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                            |
| Utilisation category                   | Category A                                                                                                                                                                                                                                                                                                                                                                               |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                                                                                                                                                     |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                                                                                                                                                         |
| Trip unit rating                       | 25 A at 50 °C                                                                                                                                                                                                                                                                                                                                                                            |
| Protection type                        | Overload protection<br>Short-circuit protection                                                                                                                                                                                                                                                                                                                                          |
| Pollution degree                       | 3 conforming to IEC 60664-1<br>3 conforming to IEC 947-1                                                                                                                                                                                                                                                                                                                                 |

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

# Product datasheet

Specifications



## circuit breaker Easypact EZC100B - TMD - 40 A - 3 poles 3d

EZC100B3040

### Main

|                                        |                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                                                                                                                                                                 |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                                                                                                                                                          |
| Device short name                      | Easypact EZC100B                                                                                                                                                                                                                                                                                                                                                                         |
| Circuit breaker name                   | Easypact EZC100B                                                                                                                                                                                                                                                                                                                                                                         |
| Device application                     | Distribution                                                                                                                                                                                                                                                                                                                                                                             |
| Poles description                      | 3P                                                                                                                                                                                                                                                                                                                                                                                       |
| Protected poles description            | 3t                                                                                                                                                                                                                                                                                                                                                                                       |
| Network type                           | DC<br>AC                                                                                                                                                                                                                                                                                                                                                                                 |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                                                                                                                                                                 |
| [In] rated current                     | 40 A at 40 °C                                                                                                                                                                                                                                                                                                                                                                            |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                              |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                           |
| [Ue] rated operational voltage         | 550 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                              |
| Breaking capacity code                 | B                                                                                                                                                                                                                                                                                                                                                                                        |
| Breaking capacity                      | 10 kA Icu at 110...130 V AC 50/60 Hz conforming to IEC 60947-2<br>10 kA Icu at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>5 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>2.5 kA Icu at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>7.5 kA Icu at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>7.5 kA Icu at 400...415 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 0.625 kA at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>1.25 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>1.875 kA at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>1.875 kA at 400/415 V AC 50/60 Hz conforming to IEC 60947-2<br>2.5 kA at 110/130 V AC 50/60 Hz conforming to IEC 60947-2<br>2.5 kA at 220/230/240 V AC 50/60 Hz conforming to IEC 60947-2                |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                            |
| Utilisation category                   | Category A                                                                                                                                                                                                                                                                                                                                                                               |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                                                                                                                                                     |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                                                                                                                                                         |
| Trip unit rating                       | 40 A at 50 °C                                                                                                                                                                                                                                                                                                                                                                            |
| Protection type                        | Overload protection<br>Short-circuit protection                                                                                                                                                                                                                                                                                                                                          |
| Pollution degree                       | 3 conforming to IEC 60664-1<br>3 conforming to IEC 947-1                                                                                                                                                                                                                                                                                                                                 |

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# Product datasheet

Specifications



## circuit breaker EasyPact EZC100F - TMD - 100 A - 3 poles 3d

EZC100F3100

### Main

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Device short name                      | EasyPact EZC100F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Circuit breaker name                   | EasyPact EZC100F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Device application                     | Distribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Poles description                      | 3P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Protected poles description            | 3t                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Network type                           | DC<br>AC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| [In] rated current                     | 100 A at 40 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| [Ue] rated operational voltage         | 550 V AC 50/60 Hz conforming to IEC 60947-2<br>250 V DC conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                |
| Breaking capacity code                 | F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Breaking capacity                      | 25 kA Icu at 110...130 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA Icu at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>5 kA Icu at 125 V DC 1P conforming to IEC 60947-2<br>5 kA Icu at 250 V DC 2P conforming to IEC 60947-2<br>5 kA Icu at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>10 kA Icu at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>10 kA Icu at 400...415 V AC 50/60 Hz conforming to IEC 60947-2<br>7.5 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 12.5 kA at 110/130 V AC 50/60 Hz conforming to IEC 60947-2<br>12.5 kA at 220/230/240 V AC 50/60 Hz conforming to IEC 60947-2<br>2.5 kA at 125 V DC conforming to IEC 60947-2<br>2.5 kA at 250 V DC conforming to IEC 60947-2<br>2.5 kA at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>5 kA at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>5 kA at 400/415 V AC 50/60 Hz conforming to IEC 60947-2<br>3.75 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2                                |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Utilisation category                   | Category A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Trip unit rating                       | 100 A at 50 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Protection type                        | Short-circuit protection<br>Overload protection                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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# Product datasheet

Specifications



## circuit breaker Easycompact EZC100N - TMD - 63 A - 3 poles 3d

EZC100N3063

### Main

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Device short name                      | Easycompact EZC100N                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Circuit breaker name                   | Easycompact EZC100N                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Device application                     | Distribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Poles description                      | 3P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Protected poles description            | 3t                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Network type                           | DC<br>AC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| [In] rated current                     | 63 A at 40 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| [Ue] rated operational voltage         | 550 V AC 50/60 Hz conforming to IEC 60947-2<br>250 V DC conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                               |
| Breaking capacity code                 | N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Breaking capacity                      | 10 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA Icu at 110...130 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA Icu at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>5 kA Icu at 125 V DC 1P conforming to IEC 60947-2<br>5 kA Icu at 250 V DC 2P conforming to IEC 60947-2<br>5 kA Icu at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>15 kA Icu at 415 V AC 50/60 Hz conforming to IEC 60947-2<br>18 kA Icu at 380...400 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 5 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>2.5 kA at 125 V DC conforming to IEC 60947-2<br>2.5 kA at 250 V DC conforming to IEC 60947-2<br>2.5 kA at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>12.5 kA at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>12.5 kA at 110...130 V AC 50/60 Hz conforming to IEC 60947-2<br>9 kA at 380...400 V AC 50/60 Hz conforming to IEC 60947-2<br>7.5 kA at 415 V AC 50/60 Hz conforming to IEC 60947-2                              |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Utilisation category                   | Category A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Trip unit rating                       | 63 A at 50 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Protection type                        | Overload protection<br>Short-circuit protection                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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# Product datasheet

Specifications



## circuit breaker Easycompact EZC100N - TMD - 80 A - 3 poles 3d

EZC100N3080

### Main

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Device short name                      | Easycompact EZC100N                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Circuit breaker name                   | Easycompact EZC100N                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Device application                     | Distribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Poles description                      | 3P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Protected poles description            | 3t                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Network type                           | AC<br>DC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| [In] rated current                     | 80 A at 40 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| [Ue] rated operational voltage         | 550 V AC 50/60 Hz conforming to IEC 60947-2<br>250 V DC conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                               |
| Breaking capacity code                 | N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Breaking capacity                      | 10 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>15 kA Icu at 400...415 V AC 50/60 Hz conforming to IEC 60947-2<br>18 kA Icu at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA Icu at 110...130 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA Icu at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>5 kA Icu at 125 V DC 1P conforming to IEC 60947-2<br>5 kA Icu at 250 V DC 2P conforming to IEC 60947-2<br>5 kA Icu at 550 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 5 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>9 kA at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>12.5 kA at 110/130 V AC 50/60 Hz conforming to IEC 60947-2<br>12.5 kA at 220/230/240 V AC 50/60 Hz conforming to IEC 60947-2<br>2.5 kA at 125 V DC conforming to IEC 60947-2<br>2.5 kA at 250 V DC conforming to IEC 60947-2<br>2.5 kA at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>7.5 kA at 400/415 V AC 50/60 Hz conforming to IEC 60947-2                                |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Utilisation category                   | Category A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Trip unit rating                       | 80 A at 50 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Protection type                        | Overload protection<br>Short-circuit protection                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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# Product datasheet

Specifications



## circuit breaker Easycompact EZC250F - TMD - 125 A - 3 poles 3d

EZC250F3125

### Main

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Device short name                      | Easycompact EZC250F                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Circuit breaker name                   | Easycompact EZC250F                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Device application                     | Distribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Poles description                      | 3P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Protected poles description            | 3t                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Network type                           | DC<br>AC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| [In] rated current                     | 125 A at 40 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| [Ue] rated operational voltage         | 550 V AC 50/60 Hz conforming to IEC 60947-2<br>250 V DC conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                               |
| Breaking capacity code                 | F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Breaking capacity                      | 10 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>15 kA Icu at 400...415 V AC 50/60 Hz conforming to IEC 60947-2<br>18 kA Icu at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA Icu at 110...130 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA Icu at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>5 kA Icu at 125 V DC 1P conforming to IEC 60947-2<br>5 kA Icu at 250 V DC 2P conforming to IEC 60947-2<br>5 kA Icu at 550 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 9 kA at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>12.5 kA at 110/130 V AC 50/60 Hz conforming to IEC 60947-2<br>12.5 kA at 220/230/240 V AC 50/60 Hz conforming to IEC 60947-2<br>2.5 kA at 125 V DC conforming to IEC 60947-2<br>2.5 kA at 250 V DC conforming to IEC 60947-2<br>2.5 kA at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>9 kA at 400/415 V AC 50/60 Hz conforming to IEC 60947-2<br>7.5 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2                                |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Utilisation category                   | Category A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Trip unit rating                       | 125 A at 50 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Protection type                        | Overload protection<br>Short-circuit protection                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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# Product datasheet

Specifications



## circuit breaker Easycompact EZC250F - TMD - 150 A - 3 poles 3d

EZC250F3150

### Main

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Device short name                      | Easycompact EZC250F                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Circuit breaker name                   | Easycompact EZC250F                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Device application                     | Distribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Poles description                      | 3P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Protected poles description            | 3t                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Network type                           | AC<br>DC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| [In] rated current                     | 150 A at 40 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| [Ue] rated operational voltage         | 550 V AC 50/60 Hz conforming to IEC 60947-2<br>250 V DC conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                               |
| Breaking capacity code                 | F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Breaking capacity                      | 10 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>15 kA Icu at 400...415 V AC 50/60 Hz conforming to IEC 60947-2<br>18 kA Icu at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA Icu at 110...130 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA Icu at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>5 kA Icu at 125 V DC 1P conforming to IEC 60947-2<br>5 kA Icu at 250 V DC 2P conforming to IEC 60947-2<br>5 kA Icu at 550 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 9 kA at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>12.5 kA at 110/130 V AC 50/60 Hz conforming to IEC 60947-2<br>12.5 kA at 220/230/240 V AC 50/60 Hz conforming to IEC 60947-2<br>2.5 kA at 125 V DC conforming to IEC 60947-2<br>2.5 kA at 250 V DC conforming to IEC 60947-2<br>2.5 kA at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>9 kA at 400/415 V AC 50/60 Hz conforming to IEC 60947-2<br>7.5 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2                                |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Utilisation category                   | Category A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Trip unit rating                       | 150 A at 50 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Protection type                        | Overload protection<br>Short-circuit protection                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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# Product datasheet

Specifications



## circuit breaker EasyPact Ezc250N - TMD - 160 A - 4 poles 3d

Ezc250N4160

### Main

|                                        |                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                                                                                                                                            |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                                                                                                                                     |
| Device short name                      | EasyPact Ezc250N                                                                                                                                                                                                                                                                                                                                                    |
| Circuit breaker name                   | EasyPact Ezc250N                                                                                                                                                                                                                                                                                                                                                    |
| Device application                     | Distribution                                                                                                                                                                                                                                                                                                                                                        |
| Poles description                      | 4P                                                                                                                                                                                                                                                                                                                                                                  |
| Protected poles description            | 3t                                                                                                                                                                                                                                                                                                                                                                  |
| Network type                           | AC<br>DC                                                                                                                                                                                                                                                                                                                                                            |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                                                                                                                                            |
| [In] rated current                     | 160 A at 40 °C                                                                                                                                                                                                                                                                                                                                                      |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                         |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                      |
| [Ue] rated operational voltage         | 550 V AC 50/60 Hz conforming to IEC 60947-2<br>250 V DC conforming to IEC 60947-2                                                                                                                                                                                                                                                                                   |
| Breaking capacity code                 | N                                                                                                                                                                                                                                                                                                                                                                   |
| Breaking capacity                      | 20 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>50 kA Icu at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>20 kA Icu at 125 V DC 1P conforming to IEC 60947-2<br>20 kA Icu at 250 V DC 2P conforming to IEC 60947-2<br>8 kA Icu at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA Icu at 380...415 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 10 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>10 kA at 125 V DC conforming to IEC 60947-2<br>10 kA at 250 V DC conforming to IEC 60947-2<br>4 kA at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>12.5 kA at 380...415 V AC 50/60 Hz conforming to IEC 60947-2                             |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                       |
| Utilisation category                   | Category A                                                                                                                                                                                                                                                                                                                                                          |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                                                                                                                                |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                                                                                                                                    |
| Trip unit rating                       | 160 A at 50 °C                                                                                                                                                                                                                                                                                                                                                      |
| Protection type                        | Short-circuit protection<br>Overload protection                                                                                                                                                                                                                                                                                                                     |
| Pollution degree                       | 3 conforming to IEC 60947                                                                                                                                                                                                                                                                                                                                           |

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# Product datasheet

Specifications



## circuit breaker EasyPact EZC250N - TMD - 250 A - 4 poles 3d

EZC250N4250

### Main

|                                        |                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                                                                                                                                            |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                                                                                                                                     |
| Device short name                      | EasyPact EZC250N                                                                                                                                                                                                                                                                                                                                                    |
| Circuit breaker name                   | EasyPact EZC250N                                                                                                                                                                                                                                                                                                                                                    |
| Device application                     | Distribution                                                                                                                                                                                                                                                                                                                                                        |
| Poles description                      | 4P                                                                                                                                                                                                                                                                                                                                                                  |
| Protected poles description            | 3t                                                                                                                                                                                                                                                                                                                                                                  |
| Network type                           | AC<br>DC                                                                                                                                                                                                                                                                                                                                                            |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                                                                                                                                            |
| [In] rated current                     | 250 A at 40 °C                                                                                                                                                                                                                                                                                                                                                      |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                         |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                      |
| [Ue] rated operational voltage         | 550 V AC 50/60 Hz conforming to IEC 60947-2<br>250 V DC conforming to IEC 60947-2                                                                                                                                                                                                                                                                                   |
| Breaking capacity code                 | N                                                                                                                                                                                                                                                                                                                                                                   |
| Breaking capacity                      | 20 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>50 kA Icu at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>20 kA Icu at 125 V DC 1P conforming to IEC 60947-2<br>20 kA Icu at 250 V DC 2P conforming to IEC 60947-2<br>8 kA Icu at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA Icu at 380...415 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 10 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA at 220...240 V AC 50/60 Hz conforming to IEC 60947-2<br>10 kA at 125 V DC conforming to IEC 60947-2<br>10 kA at 250 V DC conforming to IEC 60947-2<br>4 kA at 550 V AC 50/60 Hz conforming to IEC 60947-2<br>12.5 kA at 380...415 V AC 50/60 Hz conforming to IEC 60947-2                             |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                                                                                                                                       |
| Utilisation category                   | Category A                                                                                                                                                                                                                                                                                                                                                          |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                                                                                                                                |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                                                                                                                                    |
| Trip unit rating                       | 250 A at 50 °C                                                                                                                                                                                                                                                                                                                                                      |
| Protection type                        | Short-circuit protection<br>Overload protection                                                                                                                                                                                                                                                                                                                     |
| Pollution degree                       | 3 conforming to IEC 60947                                                                                                                                                                                                                                                                                                                                           |

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# Product datasheet

Specifications



## circuit breaker EasyPact EZC400H - TMD - 320 A - 4 poles 3d

EZC400H4320N

### Main

|                                        |                                                                                                                                                                                                                                                          |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                                 |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                          |
| Device short name                      | EasyPact EZC400H                                                                                                                                                                                                                                         |
| Circuit breaker name                   | EasyPact EZC400H                                                                                                                                                                                                                                         |
| Device application                     | Distribution                                                                                                                                                                                                                                             |
| Poles description                      | 4P                                                                                                                                                                                                                                                       |
| Protected poles description            | 3t                                                                                                                                                                                                                                                       |
| Network type                           | AC                                                                                                                                                                                                                                                       |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                                 |
| [In] rated current                     | 400 A at 40 °C                                                                                                                                                                                                                                           |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                              |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                           |
| [Ue] rated operational voltage         | 440 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                              |
| Breaking capacity code                 | H                                                                                                                                                                                                                                                        |
| Breaking capacity                      | 50 kA Icu at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>50 kA Icu at 400...415 V AC 50/60 Hz conforming to IEC 60947-2<br>50 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>70 kA Icu at 220...240 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 25 kA at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA at 400/415 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>35 kA at 220/230 V AC 50/60 Hz conforming to IEC 60947-2                     |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                            |
| Utilisation category                   | Category A                                                                                                                                                                                                                                               |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                     |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                         |
| Trip unit rating                       | 320 A at 50 °C                                                                                                                                                                                                                                           |
| Protection type                        | Short-circuit protection<br>Overload protection                                                                                                                                                                                                          |
| Pollution degree                       | 3 conforming to IEC 60947                                                                                                                                                                                                                                |

### Complementary

|               |        |
|---------------|--------|
| control type  | Toggle |
| Mounting mode | Fixed  |

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# Product datasheet

Specifications



## circuit breaker EasyPact EZC400H - TMD - 350 A - 4 poles 3d

EZC400H4350N

### Main

|                                        |                                                                                                                                                                                                                                                          |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                                 |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                          |
| Device short name                      | EasyPact EZC400H                                                                                                                                                                                                                                         |
| Circuit breaker name                   | EasyPact EZC400H                                                                                                                                                                                                                                         |
| Device application                     | Distribution                                                                                                                                                                                                                                             |
| Poles description                      | 4P                                                                                                                                                                                                                                                       |
| Protected poles description            | 3t                                                                                                                                                                                                                                                       |
| Network type                           | AC                                                                                                                                                                                                                                                       |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                                 |
| [In] rated current                     | 400 A at 40 °C                                                                                                                                                                                                                                           |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                              |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                           |
| [Ue] rated operational voltage         | 440 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                              |
| Breaking capacity code                 | H                                                                                                                                                                                                                                                        |
| Breaking capacity                      | 50 kA Icu at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>50 kA Icu at 400...415 V AC 50/60 Hz conforming to IEC 60947-2<br>50 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>70 kA Icu at 220...240 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 25 kA at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA at 400/415 V AC 50/60 Hz conforming to IEC 60947-2<br>25 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>35 kA at 220/230 V AC 50/60 Hz conforming to IEC 60947-2                     |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                            |
| Utilisation category                   | Category A                                                                                                                                                                                                                                               |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                     |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                         |
| Trip unit rating                       | 350 A at 50 °C                                                                                                                                                                                                                                           |
| Protection type                        | Short-circuit protection<br>Overload protection                                                                                                                                                                                                          |
| Pollution degree                       | 3 conforming to IEC 60947                                                                                                                                                                                                                                |

### Complementary

|               |        |
|---------------|--------|
| control type  | Toggle |
| Mounting mode | Fixed  |

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# Product datasheet

Specifications



## circuit breaker EasyPact EZC400N - TMD - 400 A - 4 poles 3d

EZC400N4400N

### Main

|                                        |                                                                                                                                                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                             |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                      |
| Device short name                      | EasyPact EZC400N                                                                                                                                                                                                                                     |
| Circuit breaker name                   | EasyPact EZC400N                                                                                                                                                                                                                                     |
| Device application                     | Distribution                                                                                                                                                                                                                                         |
| Poles description                      | 4P                                                                                                                                                                                                                                                   |
| Protected poles description            | 3t                                                                                                                                                                                                                                                   |
| Network type                           | AC                                                                                                                                                                                                                                                   |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                             |
| [In] rated current                     | 400 A at 40 °C                                                                                                                                                                                                                                       |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                          |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                       |
| [Ue] rated operational voltage         | 440 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                          |
| Breaking capacity code                 | N                                                                                                                                                                                                                                                    |
| Breaking capacity                      | 36 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>40 kA Icu at 220/230 V AC 50/60 Hz conforming to IEC 60947-2<br>36 kA Icu at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>36 kA Icu at 400/415 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 18 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>20 kA at 220/230 V AC 50/60 Hz conforming to IEC 60947-2<br>18 kA at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>18 kA at 400/415 V AC 50/60 Hz conforming to IEC 60947-2                 |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                        |
| Utilisation category                   | Category A                                                                                                                                                                                                                                           |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                 |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                     |
| Trip unit rating                       | 400 A at 50 °C                                                                                                                                                                                                                                       |
| Protection type                        | Overload protection<br>Short-circuit protection                                                                                                                                                                                                      |
| Pollution degree                       | 3 conforming to IEC 60947                                                                                                                                                                                                                            |

### Complementary

|               |        |
|---------------|--------|
| control type  | Toggle |
| Mounting mode | Fixed  |

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# Product datasheet

Specifications



## circuit breaker EasyPact EZC630H - TMD - 500 A - 4 poles 3d

EZC630H4500N

### Main

|                                        |                                                                                                                                                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Range of product                       | EasyPact                                                                                                                                                                                                                                             |
| Product or component type              | Circuit breaker                                                                                                                                                                                                                                      |
| Device short name                      | EasyPact EZC630H                                                                                                                                                                                                                                     |
| Circuit breaker name                   | EasyPact EZC630H                                                                                                                                                                                                                                     |
| Device application                     | Distribution                                                                                                                                                                                                                                         |
| Poles description                      | 4P                                                                                                                                                                                                                                                   |
| Protected poles description            | 3t                                                                                                                                                                                                                                                   |
| Network type                           | AC                                                                                                                                                                                                                                                   |
| Network frequency                      | 50/60 Hz                                                                                                                                                                                                                                             |
| [In] rated current                     | 630 A at 40 °C                                                                                                                                                                                                                                       |
| [Ui] rated insulation voltage          | 690 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                          |
| [Uimp] rated impulse withstand voltage | 6 kV conforming to IEC 60947-2                                                                                                                                                                                                                       |
| [Ue] rated operational voltage         | 440 V AC 50/60 Hz conforming to IEC 60947-2                                                                                                                                                                                                          |
| Breaking capacity code                 | H                                                                                                                                                                                                                                                    |
| Breaking capacity                      | 50 kA Icu at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>50 kA Icu at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>70 kA Icu at 220/230 V AC 50/60 Hz conforming to IEC 60947-2<br>50 kA Icu at 400/415 V AC 50/60 Hz conforming to IEC 60947-2 |
| [Ics] rated service breaking capacity  | 25 kA at 440 V AC 50/60 Hz conforming to IEC 60947-2<br>70 kA at 220/230 V AC 50/60 Hz conforming to IEC 60947-2<br>50 kA at 380 V AC 50/60 Hz conforming to IEC 60947-2<br>50 kA at 400/415 V AC 50/60 Hz conforming to IEC 60947-2                 |
| Suitability for isolation              | Yes conforming to IEC 60947-2                                                                                                                                                                                                                        |
| Utilisation category                   | Category A                                                                                                                                                                                                                                           |
| Trip unit name                         | TM-D                                                                                                                                                                                                                                                 |
| Trip unit technology                   | Thermal-magnetic                                                                                                                                                                                                                                     |
| Trip unit rating                       | 500 A at 50 °C                                                                                                                                                                                                                                       |
| Protection type                        | Short-circuit protection<br>Overload protection                                                                                                                                                                                                      |
| Pollution degree                       | 3 conforming to IEC 60947                                                                                                                                                                                                                            |

### Complementary

|               |        |
|---------------|--------|
| control type  | Toggle |
| Mounting mode | Fixed  |

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# Product datasheet

Specifications



## Circuit breaker, EasyPact MVS, 4 Poles, 2000A, 50kA, EDO, ETA6, drawout, electrical

MVS20N4NW6A

### Main

|                                |                                             |
|--------------------------------|---------------------------------------------|
| Range                          | EasyPact                                    |
| product name                   | MVS20N                                      |
| Device short name              | MVS20N                                      |
| Product or component type      | Circuit breaker                             |
| Device application             | Distribution                                |
| Poles description              | 4P                                          |
| Protected poles description    | 4P 4d                                       |
| Neutral position               | Left                                        |
| [In] rated current             | 2000 A at 40 °C                             |
| Network type                   | AC                                          |
| [Ue] rated operational voltage | 690 V AC 50/60 Hz conforming to IEC 60947-2 |
| Breaking capacity code         | N                                           |
| Suitability for isolation      | Yes conforming to IEC 60947-2               |
| Utilisation category           | Category B                                  |
| Trip unit name                 | ETA6G                                       |
| Trip unit technology           | Electronic                                  |
| Trip unit rating               | 2000 A                                      |

### Complementary

|                                        |                                                                  |
|----------------------------------------|------------------------------------------------------------------|
| Network frequency                      | 50/60 Hz                                                         |
| control type                           | Electrically operated                                            |
| Mounting mode                          | Drawout                                                          |
| Mounting support                       | Rail<br>Base plate                                               |
| Connection position                    | Vertical<br>Horizontal                                           |
| Location of connection                 | Rear                                                             |
| [Ui] rated insulation voltage          | 1000 V AC 50/60 Hz conforming to IEC 60947-2                     |
| [Uimp] rated impulse withstand voltage | 12 kV conforming to IEC 60947-2                                  |
| Circuit breaker CT rating              | 2000 A                                                           |
| Breaking capacity                      | 50 kA Icu at 220...440 V AC 50/60 Hz conforming to IEC 60947-2   |
| [Ics] rated service breaking capacity  | 50 kA (Ics) at 220...440 V AC 50/60 Hz conforming to IEC 60947-2 |

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